SI-7000 官方资料包中文全文翻译（优化排版）

- 原始资料：../00_原始资料/SI7000官方资料包_含Modbus规范.pdf
- 正文来源：本目录内按官方资料逐章校订的中文译稿。
- 排版口径：新版 PDF 采用可读性优先的章节化排版，不强制与原 PDF 页数逐页对应；原图、图纸、照片和扫描页在正文或附录中保留。
- 术语口径：专业术语、协议名、功能码、寄存器地址、单位、部件英文名和界面英文按必要程度保留。

原资料包结构

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资料包封面与总目录

来源文件：../00_原始资料/SI7000官方资料包_含Modbus规范.pdf

抽取日期：2026-07-02

资料包共 101 页。该 PDF 是多个官方资料合并后的资料包，包含产品概览、认证页、DCS Modbus 接口规范、包装规格、外形和接线图、安装与调试手册、操作手册、维修手册和保修信息。

封面与总目录中文翻译

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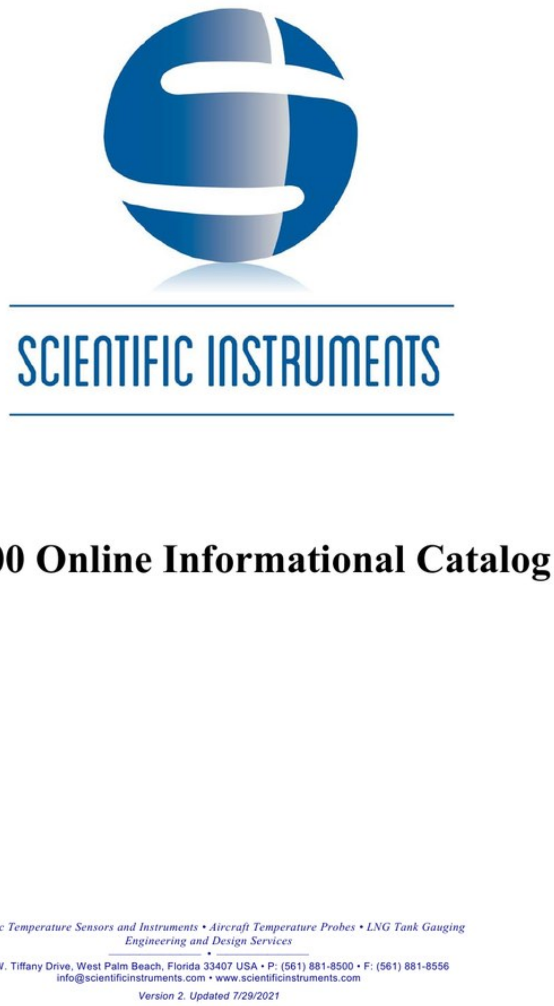
SCIENTIFIC INSTRUMENTS

SI-7000 Online Informational Catalog

Cryogenic Temperature Sensors and Instruments • Aircraft Temperature Probes • LNG Tank Gauging
Engineering and Design Services

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Version 2. Updated 7/29/2021

SI-7000 官方资料包封面

原文项目	中文译文
SI-7000 Online Informational Catalog	SI-7000 在线信息目录/资料包
Cryogenic Temperature Sensors and Instruments	低温温度传感器和仪表
Aircraft Temperature Probes	飞机温度探头
LNG Tank Gauging	LNG 储罐计量
Engineering and Design Services	工程和设计服务

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PDF 第 2 页：资料包总目录

SCIENTIFIC INSTRUMENTS

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DCS Modbus 接口规范中文翻译

来源文件：

- ../00_原始资料/SI7000官方资料包_含Modbus规范.pdf 第 10-24 页
- ../00_原始资料/SI7000_modbus_官方.pdf

原始标题：SI-7000 Tank Gauging System DCS Modbus Interface Specification

原始编号：020-701 Rev B

整理日期：2026-07-02

总体说明

本规范定义了 Scientific Instruments SI-7000 LTD 对 Modbus 协议的有限实现。它面向需要在上位机上开发和测试用户界面的程序人员，说明主机计算机如何读取或控制 SI-7000 的运行模式、探头运动、系统状态、报警、Profile 参数、传感器数据和 Profile 数据。

SI-7000 LTD 是用于大型低温液体储罐的罐计量系统，可测量液位、温度和密度，例如 LNG 储罐。系统通过驱动探头在罐内上下移动，按操作员设定采集温度和密度数据。采集到的数据存储在设备内存中，主机计算机可按需读取。

官方资料特别说明：程序人员应阅读并熟悉 SI-7000 操作手册，因为本 Modbus 规范不会重复介绍设备的基本运行逻辑。

Modbus 通信参数

通用 Modbus 规范中，帧格式和帧顺序等特性是固定的；接口类型、波特率、校验、停止位数量和传输模式等特性则可由具体实现决定。SI-7000 LTD 的 Modbus 实现采用以下参数：

项目	官方规定
接口类型	RS485
RS232 接入	如主机侧使用 RS232，需要外接转换器
波特率	9600 baud，当前不可修改
校验	Odd parity，奇校验
数据位	8 data bits
停止位	1 stop bit
传输模式	RTU
Device ID	使用每套系统的 Tank ID
主机请求最小间隔	Host 到 Device 的 Command/Request 最小时间应为 1 秒

当一台主机连接多个储罐时，必须逐个地址访问和扫描每一套 SI-7000。每台设备的 Device ID 使用该系统的 Tank ID。Tank ID 可在 SI-7000 触摸屏用户界面中设置。

支持的 Modbus 功能码总览

SI-7000 LTD Modbus 接口使用以下功能码：

功能码	官方用途	数据类型	读写方向
Function 1	读取系统模式、电机驱动方向和速度	1 bit 数字量	读
Function 5	设置系统模式、电机驱动方向和速度	1 bit 数字量	写
Function 2	读取单独的状态位和报警位	1 bit 数字量	读
Function 3	读取 Profile 和报警配置寄存器	16 bit 寄存器	读
Function 6	设置 Profile 和报警配置寄存器	16 bit 寄存器	写

功能码	官方用途	数据类型	读写方向
Function 4	读取传感器数据、当前罐况和 Profile 数据	16 bit 寄存器	读

官方说明中提到，液位、温度和密度输出在未来固件版本中将支持可配置缩放。当前 Rev B 文档给出的缩放参数为固定值。

模式和速度位

地址范围

模式和速度使用 1 bit 数字量。Function 1 用于读取，Function 5 用于设置。

项目	地址
Base Address	0001
Maximum Legal Address	0016

位定义

官方参数名	地址	中文含义
Manual	00001	手动模式
Calibrate	00002	校准模式
Auto	00003	自动模式
Profile	00004	Profile 模式
Reserved	00005	保留
Reserved	00006	保留
Reserved	00007	保留
Reserved	00008	保留
Stop	00009	停止探头运动
Up Slow	00010	向上慢速
Up Medium	00011	向上中速
Up Fast	00012	向上快速
Down Slow	00013	向下慢速
Down Medium	00014	向下中速
Down Fast	00015	向下快速
Reserved	00016	保留

模式位逻辑

地址 00001-00004 表示当前 LTD 运行模式。各模式互斥：如果某一个模式位置位，其他模式位会清零。

使用 Function 5 改变系统模式时，只需要对新模式对应的位写入置位命令。不需要也不能手动清除旧模式对应的位，设备会自动处理互斥关系。

速度和方向位逻辑

地址 00009-00015 表示探头运动方向和速度。各运动速度同样互斥：如果主机通过 Function 5 设置其中一个运动位，其他运动位会自动清零。

这些位可以随时读取，用于判断当前系统状态，读取动作不会影响设备运行模式。

需要重点注意：如果主机手动设置任何与探头方向或速度相关的位，会停止正在执行的 Profile 活动或液位跟踪，并把系统切换到 Manual drive mode。系统会无限期保持在该手动驱动状态，直到重新设置为 Auto、Cal 或 Profile。在

Manual mode 下，系统不能执行自动功能。

官方建议在上位机界面中突出显示 Manual mode，例如用红色提醒操作员“这不是正常自动运行状态”；而 Auto 模式可用绿色显示，因为这是持续跟踪液位的正常运行模式。

状态位和报警位

状态和报警使用 1 bit 数字量，通过 Function 2 读取。

项目	地址
Status Base Address	10001
Alarms Base Address	10017
Maximum Legal Address	10032

状态位地址表

状态位提供系统运行的重要信息，特别用于验证 LTD 上报的数据是否当前有效。

官方状态名	地址	中文含义
Bottom Reference	10001	底部参考开关触发，探头位于罐底
Lower Level Sensor	10002	下液位传感器处于液体中
Upper Level Sensor	10003	上液位传感器处于液体中
Interlock	10004	探头到达允许的最高位置，禁止继续向上
Profile Complete	10005	Profile 已完成
Metric	10006	使用公制单位
Reserved	10007	保留
Reel Alarm Disable	10008	卷盘报警禁用
Reel Alarm	10009	卷盘或物理驱动报警
Probe Un-calibrated	10010	探头未校准或当前位置可能不正确
Reserved	10011	保留
Interval Timer	10012	间隔定时 Profile 已启用
Probe At Liquid Level	10013	探头处于液气界面，液位读数为当前值
Reserved	10014	保留
Reserved	10015	保留
Reserved	10016	保留

建议上位机必须显示的状态

官方认为下列状态对操作员非常重要，应在自定义用户界面中明确报告：

- Bottom Reference
- Lower Level Sensor
- Upper Level Sensor
- Interlock
- Reel Alarm
- Probe Un-calibrated
- Interval Timer
- Probe At Liquid Level

其他状态可按需要显示。下面是各状态的含义整理。

状态含义

Bottom Reference 在底部参考开关闭合时置位，表示探头位于罐底。

Lower Level Sensor 和 Upper Level Sensor 在对应传感器处于液体中时置位。这两个状态对观察探头周围的实际物理状态、确认探头在罐内位置很重要。

Interlock 在探头到达允许的最高位置时置位；到达该位置后，探头不会继续向上运动。如果正常运行过程中探头触发 Interlock，通常表示某种异常，因为探头正常应处于液位或液位以下。当然，如果操作员手动向上驱动探头，也可能触发该状态。

Reel Alarm 置位表示发生物理驱动问题，需要维护人员检查。官方强调这是监控设备正常运行的关键状态。

Probe Un-calibrated 在当前探头位置可能不正确时置位。该状态也非常重要，因为它意味着上报的液位值也可能不正确。

Interval Timer 置位表示系统将按设定间隔自动执行 Profile；未置位时，Profile 可手动运行。

Probe At Liquid Level 在系统处于 Auto 且探头位于液气界面时置位。该状态表示当前上报的液位是最新值。

Profile Complete 对程序人员可能比对操作员更重要。该状态清零时，表示 LTD 正在执行 Profile；置位时，表示 Profile 已完成，数据可被采集和分析。

Reel Alarm Disable 正常情况下不应置位，因为它会覆盖设备中的安全功能。该安全功能可能在设备故障时防止损坏。官方建议可把该状态绑定到报警，防止它在操作员不知情时被设置。

Metric 置位表示数值以公制单位上报；清零表示以英制单位上报。该项一旦设定，通常不需要变更。

报警位地址表

报警位与罐内液体状态相关，对工厂运行很重要。

官方报警名	地址	中文含义
Low Density	10017	密度低报警
High Density	10018	密度高报警
Low Temperature	10019	温度低报警
High Temperature	10020	温度高报警
Low, Low Level	10021	低低液位报警
High, High Level	10022	高高液位报警
Low Level	10023	低液位报警
High Level	10024	高液位报警
Profile Temperature Deviation	10025	Profile 温度偏差报警
Profile Density Deviation	10026	Profile 密度偏差报警
Reserved	10027	保留
Reserved	10028	保留
Profile Low Temperature	10029	Profile 低温报警
Profile High Temperature	10030	Profile 高温报警
Profile Low Density	10031	Profile 低密度报警
Profile High Density	10032	Profile 高密度报警

Profile 和报警控制寄存器

Profile 和报警控制使用模拟输出或输入寄存器。Function 3 用于读取，Function 6 用于写入。通过这些寄存器，DCS 操作员可控制 Profile 参数并改变产品报警设定值。

地址范围

项目	地址
Base Address	40001
Maximum Address	40023

地址表

官方参数名	地址	中文含义
Profile First Point	40001	Profile 第一个可编程测点位置
Profile Increment	40002	Profile 测点间距增量
Profile Dwell Time	40003	Profile 每个测点停留时间，单位为秒
Reserved	40004	保留
Reserved	40005	保留
Reserved	40006	保留
Reserved	40007	保留
Reserved	40008	保留
Reserved	40009	保留
Automatic Profile Interval	40010	自动 Profile 间隔，单位为分钟
Automatic Profile Enable	40011	自动 Profile 使能
Automatic Profile Hour	40012	自动 Profile 首次启动小时
Automatic Profile Minute	40013	自动 Profile 首次启动分钟
Low Density Alarm Set Point	40014	低密度报警设定值
High Density Alarm Set Point	40015	高密度报警设定值
Low Temperature Alarm Set Point	40016	低温报警设定值
High Temperature Alarm Set Point	40017	高温报警设定值
Low Low Level Alarm Set Point	40018	低低液位报警设定值
High High Level Alarm Set Point	40019	高高液位报警设定值
Low Level Alarm Set Point	40020	低液位报警设定值
High Level Alarm Set Point	40021	高液位报警设定值
Temperature Deviation Alarm Set Point	40022	温度偏差报警设定值
Density Deviation Alarm Set Point	40023	密度偏差报警设定值

参数定义

当 Profile 启动时，探头先移动到罐底，并从罐底开始 Profile。Profile First Point 是探头上升过程中第一个可编程停靠测点，用于采集温度和密度。严格来说，它是 Profile 中的第二个点，因为第一个点总是在罐底。

Profile Increment 是用于计算后续测点位置的距离增量。设备从 Profile First Point 开始，每次加上该增量，决定返回液面过程中在哪些位置停下并采集读数。

Profile Dwell Time 是探头在每个测点暂停的时间，单位为秒，用于等待读数稳定后再记录。

Profile 可通过切换到 Profile Mode 手动启动，也可按每天固定时间或按重复间隔自动运行。

Automatic Profile Interval 表示连续两次 Profile 之间的分钟数。第一次 Profile 按 Automatic Profile Hour 和 Automatic Profile Minute 启动，之后按该间隔重复。

Automatic Profile Enable 表示是否启用自动 Profile：

值	含义
0	禁用自动 Profile
1	按设定时间和间隔自动启动 Profile

如果每天只需要一次 Profile，官方建议间隔设为 1440 分钟，这样第二天同一时间再次启动。启用该功能后的第一次 Profile 会在 Automatic Profile Hour 和 Automatic Profile Minute 定义的时间启动。

其余寄存器用于设置密度、温度、液位以及温度和密度偏差的报警限值。Temperature Deviation Limit 指 Profile 中相邻测点之间温度变化达到多少会触发报警；Density Deviation Limit 指相邻测点之间密度变化达到多少会触发报警。

缩放规则

总体口径

SI-7000 LTD 对 Function 3、Function 4 和 Function 6 的 Modbus 实现使用 16 bit 有符号或无符号整数与 DCS 通信。因此，每个整数计数代表的物理量需要通过缩放规则定义。

每个物理量都有低限、高限、低缩放值和高缩放值。Rev B 文档中这些限制当前为固定值。官方说明未来固件版本可能允许修改这些值，并在现场调试时为每个站点确定合适参数。

液位、温度和密度分别有独立缩放系数。

缩放表

物理量	低限	高限	低缩放值	高缩放值	LTD 上报示例值	发送到 DCS 的值	整理说明
Level, m	0 m	65.000 m	0	65000	15.000 m	15000	公制液位，1 count = 1 mm
Temperature, deg C	-327.68 deg C	327.67 deg C	-32768	32767	-160.00 deg C	-16000	有符号整数，1 count = 0.01 deg C
Density, kg/m3	0 kg/m3	655.35 kg/m3	0	65535	450.00 kg/m3	45000	1 count = 0.01 kg/m3
Level, ft	0 ft	200.00 ft	0	60000	100.00 ft	30000	英制液位
Temperature, deg F	-327.68 deg F	327.67 deg F	-32768	32767	-256.00 deg F	-25600	有符号整数，1 count = 0.01 deg F
Density, lb/ft3	0 lb/ft3	65.535 lb/ft3	0	65535	27.500 lb/ft3	27500	1 count = 0.001 lb/ft3

原文中的摄氏度和华氏度使用温度符号。为避免编码和工具兼容问题，本文表格中写为 deg C 和 deg F。

使用规则

公制液位的 Modbus 16 bit 整数值可直接按毫米读取，为无符号整数。也就是说，每 1 个 Modbus 计数表示 1 mm。

温度无论英制还是公制，每 1 个计数表示对应温标的 0.01 度，并作为有符号整数上报。

读取或写入液位相关量时，例如 Profile First Point、Profile Increment 和 High Level Alarm Set Point，应使用液位缩放系数。该规则同时适用于从主机接收的量和向主机发送的量。主机写入的量会按当前生效缩放值解释。

Dwell Time 使用秒，不按液位、温度或密度缩放。

传感器数据和 Profile 数据

传感器数据和 Profile 数据使用模拟输入寄存器，通过 Function 4 读取。它们表示探头当前位置的罐内当前状态，以及最近一次 Profile 期间采集的罐况信息。

地址范围

项目	地址
Sensor Data Base Address	30001
Maximum Legal Address	30620

地址表

官方参数名	地址	中文含义
Current Probe Position	30001	当前探头位置
Current Temperature	30002	当前温度
Current Density	30003	当前密度
Liquid Level	30004	最近一次液位读数
Reserved	30005	保留
Number of Points	30006	最近一次 Profile 已采集点数
Profile Month	30007	Profile 月
Profile Day	30008	Profile 日
Profile Hour	30009	Profile 小时
Profile Minute	30010	Profile 分钟
Current Hour	30011	当前系统小时
Current Minute	30012	当前系统分钟
Current Second	30013	当前系统秒
Reserved	30014	保留
Reserved	30015	保留
Reserved	30016	保留
Reserved	30017	保留
Reserved	30018	保留
Reserved	30019	保留
Reserved	30020	保留
Profile Probe Position Point 0	30021	Profile 第 0 点探头位置
Profile Temperature Point 0	30022	Profile 第 0 点温度
Profile Density Point 0	30023	Profile 第 0 点密度
Profile Probe Position Point 1	30024	Profile 第 1 点探头位置
Profile Temperature Point 1	30025	Profile 第 1 点温度
Profile Density Point 1	30026	Profile 第 1 点密度
Profile Probe Position Point n	30021 + 3 * n	Profile 第 n 点探头位置
Profile Temperature Point n	30022 + 3 * n	Profile 第 n 点温度
Profile Density Point n	30023 + 3 * n	Profile 第 n 点密度
Profile Probe Position Point 199	30618	Profile 第 199 点探头位置
Profile Temperature Point 199	30619	Profile 第 199 点温度
Profile Density Point 199	30620	Profile 第 199 点密度

官方备注：这里使用 Profile Probe Position，而不是 Profile Level，因为本文中的 Level 通常表示实际液位。常规 Profile 中只有最后一个点实际位于液位。

当前值定义

Current Position、Current Temperature 和 Current Density 是探头当前所在位置的值。无论探头在罐底还是液位处，这些当前值都表示探头当前位置。

Liquid Level 保存最近一次液位读数。如果探头处于液面以下，例如正在执行 Profile，液位读数不会更新，直到探头返回液面。前文的 Probe At Liquid Level 状态位用于指示当前上报液位是否为最新值。

Profile 点数和时间

Number of Points 表示最近一次 Profile 采集的点数。点数由以下因素决定：

- 可编程起始点位置；
- Profile 增量；
- 当前液位；
- 最大允许点数。

Profile 的第一个点始终是罐底。下一个点是可编程点，通常设为 1 m 这样的整数。之后每次加上可编程增量，增量通常也是 1 m。这样 Profile 点会落在 1 m、2 m、3 m 等整数高度。探头向上移动并采集数据，直到达到液位，或者达到最大点数而导致 Profile 提前结束。最后一个点通常位于液位。

官方文字说明最大点数为 500，即 0-499。如果想把点数限制到更低值，应设置 Profile increment，使最大液位下的点数不超过期望限制。

示例：如果最大液位为 30 m，Profile increment 为 1 m，则最大点数为 31 点，包括底部参考点。位置、温度和密度总共需要 93 个寄存器，即 31 * 3。

Profile Month、Profile Day、Profile Hour 和 Profile Minute 表示第一个 Profile 点被采集时的日期和时间。

Current Hour、Current Minute 和 Current Second 表示当前系统时间。官方建议可持续监控系统时间是否变化，用于验证与 LTD 的通信是否正常；如果时间停止变化，应产生报警。

每个 Profile 点的探头位置、温度和密度都可在地址表所列寄存器中读取。超过实际采集点数的寄存器数据为 0。

Function 4 缩放

Function 4 的缩放规则与 Function 3 和 Function 6 使用同一组参数。详见本文第 7 节。

硬件和通信连接

总体说明

主机计算机可通过 RS485 现场接线直接连接到 LTD。每个站点通常会提供现场接线图。为了建立通信，需要正确设置多个数据参数。

通信链路

在一种可能配置中，LTD 连接到 Tank Gauge Interface Module，即 TGIM。TGIM 作为罐计量仪表的电源开关、电源指示和现场接线连接点，通常安装在控制室或现场设备间。

通常有 2 条通信链路，可用于与主机计算机通信。它们称为 Host 1 和 Host 2。

LTD 通信链路使用 RS485。如果主机计算机使用 RS232，需要使用转换器。

默认波特率为 9600 baud，当前不可修改。其他参数为奇校验、8 个数据位、1 个停止位，传输模式为 RTU。

Modbus 协议要求使用地址。LTD 中的 Tank ID 可通过触摸屏界面设置，同时作为 Modbus 地址。如果同一数据总线上连接多个 LTD，必须为每台设备设置唯一地址。

编程建议

总体建议

要完整理解 LTD 的运行，程序人员应阅读操作手册。本节只给出对编程任务有帮助的相关细节。

LTD 的主要用途有两个：

1. 获取准确液位信息。

2. 获取温度和密度 Profile 信息，从而检测不安全的分层条件。

LTD 检测到的分层通过 Temperature Deviation 和 Density Deviation 报警指示。

各状态位的重要程度已在状态地址章节说明。开发用户界面时，应参考该章节决定哪些状态必须给操作员显示。

查找液位

LTD 上电后，只有在探头触碰罐底并返回液气界面后，准确液位信息才可用。

系统断电时，最后已知探头位置和液位会存入非易失存储器；系统重新上电时会恢复这些值。但是系统会显示 Un-cal 指示，因为这些值可能已经不再正确。

要清除 Un-cal 指示，必须让探头触碰罐底。触底会自动把探头位置设置为罐底应显示的值。

LTD 可配置为上电后保持 Manual Mode，也可直接进入 Cal Mode。

如果上电后保持 Manual，应手动启动校准。校准完成后，系统会保持 Auto Mode 并测量液位。对于已经在有液体储罐中完成投运的系统，这通常就是所需的全部操作。LTD 会自动执行查找液位的步骤，切换到 Auto mode，并监测液位。

Profile 信息

将控制器置于 Profile mode 可启动 Profile。启动方式包括：

- 触摸屏用户界面；
- 自动间隔定时器；
- 主机计算机发送的命令。

Point 0 始终是底部点，液位是最后一个点。采集点数由 Number of Points 变量给出。

探头在每个点会停留指定等待时间，即 dwell time，以便稳定。之后测量温度和密度，再移动到下一个点。Profile 完成后，设备返回 AUTO mode。

Profile 采集过程中，每增加一个新点，Number of Points 变量会递增。使用该变量可只导入有效数据点。

新的 Profile 开始时，探头首次触碰罐底后，内存中先前所有数据点和 Profile Complete 状态指示都会被清除。Profile 完成并且所有数据可用后，Profile Complete 状态指示会置位。

Profile 绘图

查看液体储罐中温度和密度分层的最简单方法是绘图。为了让图有意义，图形必须具备较高分辨率，因为预期数据变化很小，图形必须能显示这些小变化。

官方建议谨慎选择温度和密度的图形范围。一种缩放方式是：把罐底第一个点读取到的值作为图中心，然后随着液位增加绘制相对该值的偏差。

对于多数 LNG 储罐，温度或密度偏差图可采用以下量级作为中心偏差范围：

物理量	推荐偏差显示范围
密度，公制	中心值上下约 5 kg/m3
密度，英制	中心值上下约 0.3 lb/ft3

产品规格、认证与包装中文翻译

来源文件：../00_原始资料/SI7000官方资料包_含Modbus规范.pdf 第 3-9 页、第 25 页

整理日期：2026-07-02

SI-7000 LTD Tank Gauge System 封面

SI-7000 LTD Tank Gauge System 封面

图中文字：The New SI-7000 LTD Tank Gauge System

中文译文：新一代 SI-7000 LTD 罐计量系统。这里的 LTD 在官方资料中指 Level, Temperature and Density，即液位、温度和密度测量能力。

产品概览

SI-7000 产品概览

官方资料把 SI-7000 定位为 LNG 和 LPG 储罐中液位、温度和密度剖面测量的主力方案。第三代 LTD 用于检测储罐内分层现象，并向 DCS 和其他主机提供翻滚 rollover 预测、翻滚预防以及贸易计量或财务计算所需数据。

LTD 可提供实时测量数据和平均测量数据，用于监控罐内条件。这些数据还可给工厂操作员提供早期预警，使操作员在情况发展成危险紧急事件前做出判断并采取控制措施。

关键能力

官方栏目	原文要点	中文译文
Proven Technology	Accuracy & Reliability; Dedicated Sensors for Level, Temperature and Density embedded into a single housing / Probe	成熟技术。把液位、温度、密度专用传感器集成在一个壳体或探头中，强调准确性和可靠性。
Innovation	Smart Electronics; Single Control Module, easily accessible for service & maintenance	智能电子模块。单一控制模块，便于维修和维护接近。
Reliability	Reliable Mechanical Drive; Robust 316 SS Vertical-Gear-Drive mechanism (VGD) & 304 SS Chain	可靠机械驱动。使用坚固的 316 stainless steel 垂直齿轮驱动机构 VGD 和 304 stainless steel 链条。
Connectivity	Flexible Connectivity; Standard RS-485 MODBUS	灵活连接。标准通信为 RS-485 Modbus。

公司背景和选择理由

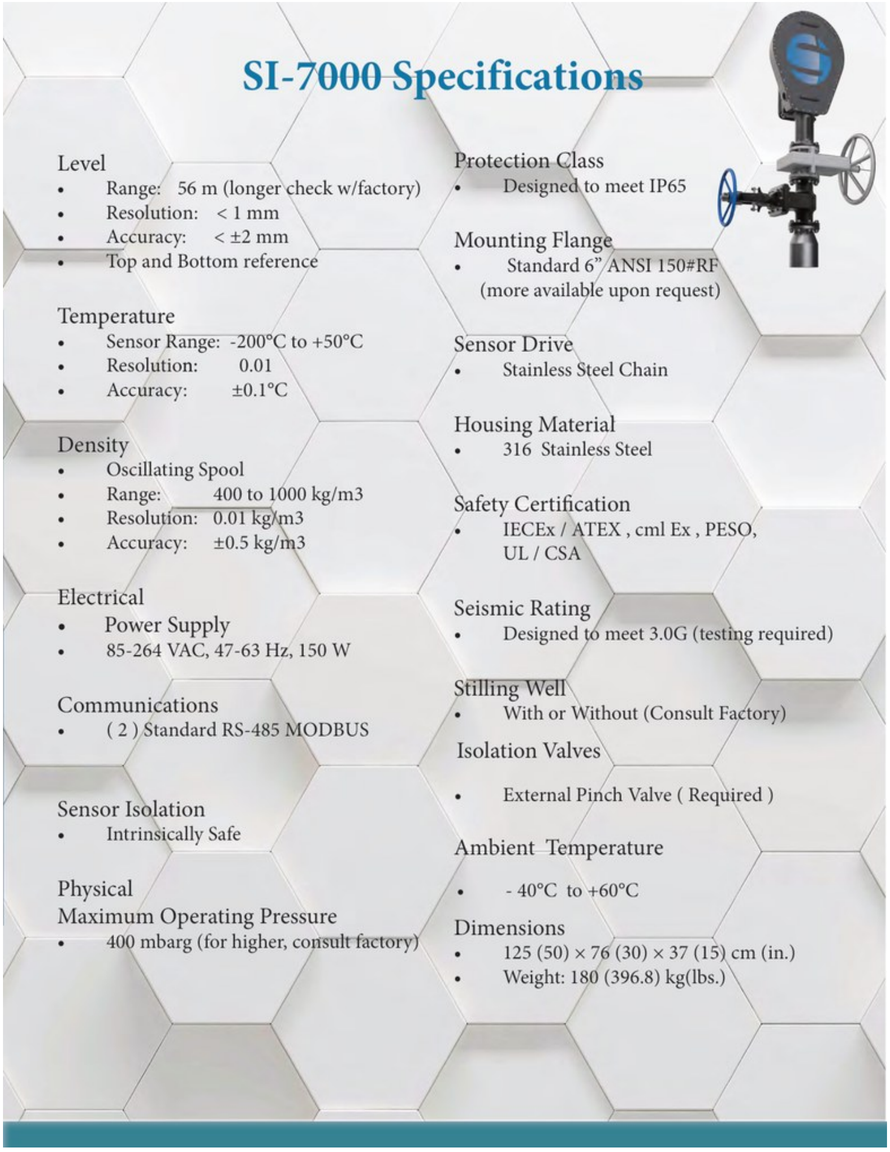
Scientific Instruments 创立于 1967 年，创始人包括航空工程师 Jack Hoey 和物理学家 Gib Halverson。官方资料称其核心能力集中在精密低温测温领域，并曾创造 NASA Apollo 任务使用的第一款 Germanium Resistance Thermometer GRT。

1975 年，Scientific Instruments 发明并取得第一套 LTD 系统专利。该系统把多种传感器嵌入单个探头，用于测量 LNG 储罐内液位、温度和密度。

官方给出的选择理由包括：

- 1975 年第一套 LTD 的原始发明者，全球识别为 SI Gauge。
- 自 1967 年以来聚焦 LNG 和低温测量方案。
- 在全球 LNG 储罐中安装超过 500 台。
- 提供全球销售和技术支持。
- 提供 Turn-Key Tank Gauge Solutions，即交钥匙罐计量解决方案。

SI-7000 规格



SI-7000 规格页

液位

项目	官方值	中文说明
Range	56 m, longer check w/factory	量程 56 m；更长量程需要向厂家确认。
Resolution	< 1 mm	分辨率小于 1 mm。
Accuracy	< +/-2 mm	精度小于 +/-2 mm。
Reference	Top and Bottom reference	顶部和底部参考。

温度

项目	官方值	中文说明
Sensor Range	-200 deg C to +50 deg C	传感器范围 -200 deg C 到 +50 deg C。
Resolution	0.01	分辨率 0.01，结合上下文为温度单位的 0.01 度。
Accuracy	+/-0.1 deg C	精度 +/-0.1 deg C。

原始页面使用温度符号。为避免 Markdown 和脚本环境中的编码差异，本文表格写为 deg C。

密度

项目	官方值	中文说明
Measurement principle	Oscillating Spool	振荡 spool 或振荡件密度测量。
Range	400 to 1000 kg/m3	范围 400 到 1000 kg/m3。
Resolution	0.01 kg/m3	分辨率 0.01 kg/m3。
Accuracy	+/-0.5 kg/m3	精度 +/-0.5 kg/m3。

电气和通信

项目	官方值	中文说明
Power Supply	85-264 VAC, 47-63 Hz, 150 W	电源输入 85-264 VAC，47-63 Hz，150 W。
Communications	(2) Standard RS-485 MODBUS	2 路标准 RS-485 Modbus。
Sensor Isolation	Intrinsically Safe	传感器隔离为本质安全设计。

机械和环境

项目	官方值	中文说明
Maximum Operating Pressure	400 mbarg, for higher consult factory	最大工作压力 400 mbarg；更高压力需咨询厂家。
Protection Class	Designed to meet IP65	防护等级设计满足 IP65。
Mounting Flange	Standard 6 inch ANSI 150#RF; more available upon request	标准 6 inch ANSI 150#RF 安装法兰；其他规格可按需提供。
Sensor Drive	Stainless Steel Chain	传感器驱动采用 stainless steel chain。
Housing Material	316 Stainless Steel	壳体材料为 316 stainless steel。
Safety Certification	IECEx / ATEX, cml Ex, PESO, UL / CSA	安全认证覆盖 IECEx / ATEX、cml Ex、PESO、UL / CSA。
Seismic Rating	Designed to meet 3.0G, testing required	抗震按满足 3.0G 设计，需测试确认。
Stilling Well	With or Without, Consult Factory	可带或不带静液井，需咨询厂家。
Isolation Valves	External Pinch Valve, Required	要求外部 Pinch Valve。
Ambient Temperature	-40 deg C to +60 deg C	环境温度 -40 deg C 到 +60 deg C。
Dimensions	125 (50) x 76 (30) x 37 (15) cm (in.)	尺寸约 125 cm x 76 cm x 37 cm，对应 50 in x 30 in x 15 in。

项目	官方值	中文说明
Weight	180 (396.8) kg (lbs.)	重量约 180 kg，即 396.8 lb。

译注：

- 56 m 液位量程与后续 Modbus 规范中的公制液位缩放高限 65.000 m 并不等同。前者是产品规格页给出的设备量程，后者是 Modbus 寄存器缩放范围。两者不能混用。
- External Pinch Valve (Required) 在规格页中标为必需，后续安装章节和机械布置图中应继续关注 Pinch Valve 相关安装要求。

全球安装基础

SI-7000 全球安装基础

官方页面标题为 OUR GLOBAL INSTALLED BASE SINCE 1976。页面以地图红点表示自 1976 年以来的全球安装基础，并注明包括新旧 LTD 版本，数量可能变化。

页面底部列出的业务范围：

- Cryogenic Temperature Sensors and Instruments，低温温度传感器和仪表。
- Aircraft Temperature Probes，航空温度探头。
- LNG Tank Gauging，LNG 储罐计量。

- Engineering Design Services, Integrated Solutions and Technical Services，工程设计服务、集成解决方案和技术服务。

IECEx Certificate of Conformity

第 7-9 页为 IECEx Certificate of Conformity，共 3 页。该部分属于证书扫描页，中文正文仅摘录清晰可读的关键字段，完整依据见原图。

证书第 1 页


Page 7

IECEx Certificate of Conformity

INTERNATIONAL ELECTROTECHNICAL COMMISSION
IEC Certification Scheme for Explosive Atmospheres
for rules and details of the IECEx Scheme visit www.iecex.com

Certificate No.: IECEx SEV 19.0035X
Issue No: 0
Certificate history: Issue No. 0 (2019-09-11)
Status: Current
Page 1 of 3
Date of Issue: 2019-09-11
Applicant: Scientific Instruments
4400 West Tiffany Drive
West Palm Beach
FL 33407
United States of America

Equipment: Tank Gauging System
Optional accessory:

Type of Protection: d, i, h
Marking:
Electrical Compartment: Ex db [ia IIC Ga] IIB T4 Gb
Mechanical Compartment: Ex h IIC T4 Ga
Probe Assembly: Ex ia IIC T4 Ga

Approved for issue on behalf of the IECEx Certification Body: Martin Plüss
Position: Manager Product Certification
Signature: (for printed version)
Date: 2019-09-11

1. This certificate and schedule may only be reproduced in full.
2. This certificate is not transferable and remains the property of the issuing body.
3. The Status and authenticity of this certificate may be verified by visiting the Official IECEx Website.

Certificate issued by:
Eurofins Electric & Electronic Product Testing AG
Luppmenstrasse 3
CH-8320 FEHRALTORF
Switzerland

eurofins | E&E

SWISS ACCREDITATION sas.admin.ch SCESp 0055

IECEx 证书第 1 页

字段	原文	中文译文
Certificate No.	IECEx SEV 19.0035X	证书号 IECEx SEV 19.0035X。
Issue No.	0	签发版本号 0。
Certificate history	Issue No. 0 (2019-09-11)	证书历史：2019-09-11 签发 Issue No. 0。
Status	Current	状态为当前有效。
Date of Issue	2019-09-11	签发日期 2019-09-11。
Applicant	Scientific Instruments, 4400 West Tiffany Drive, West Palm Beach, FL 33407, USA	申请方为 Scientific Instruments，地址位于美国佛罗里达州 West Palm Beach。

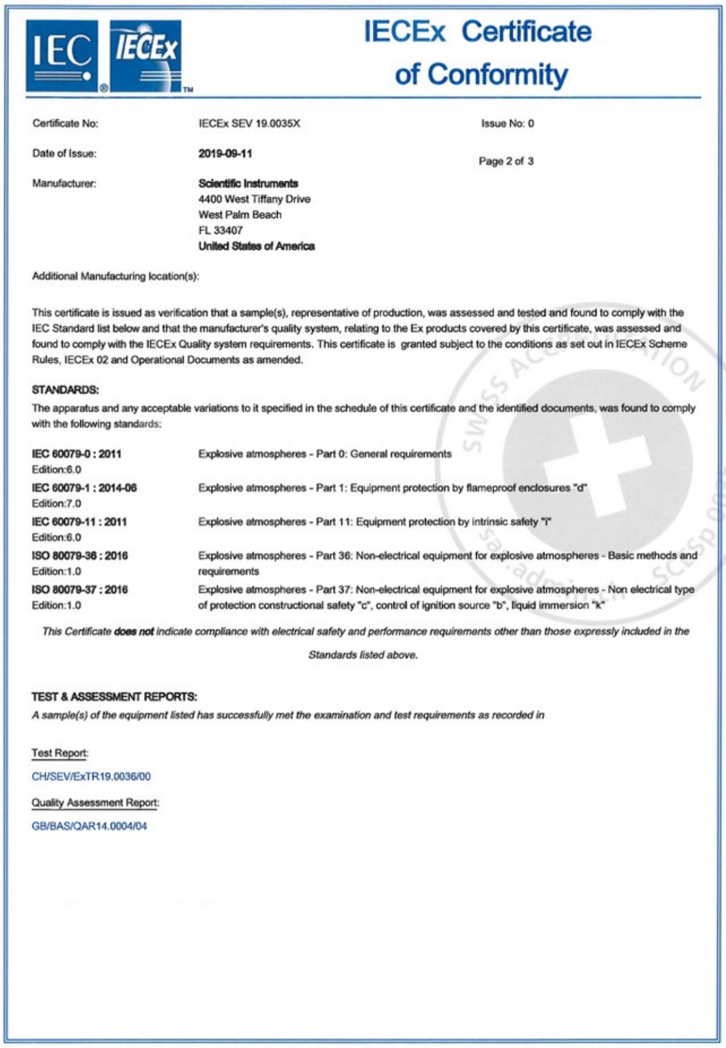
字段	原文	中文译文
Equipment	Tank Gauging System	设备为罐计量系统。
Type of Protection	d, i, h	防护类型包含 d、i、h。

防爆标志：

部位	原文标志	中文译文
Electrical Compartment	Ex db [ia IIC Ga] IIB T4 Gb	电气舱：隔爆 db，内部含本安关联保护 [ia IIC Ga]，气体组别 IIB，温度组别 T4，设备保护级别 Gb。
Mechanical Compartment	Ex h IIC T4 Ga	机械舱：非电气设备防爆保护 h，IIC，T4，Ga。
Probe Assembly	Ex ia IIC T4 Ga	探头组件：本质安全 ia，IIC，T4，Ga。

证书签发机构页面显示由 Eurofins Electric & Electronic Product Testing AG 签发。

证书第 2 页



IECEx Certificate of Conformity

Certificate No:	IECEx SEV 19.0035X	Issue No: 0
Date of Issue:	2019-09-11	Page 2 of 3
Manufacturer:	**Scientific Instruments** 4400 West Tiffany Drive West Palm Beach FL 33407 **United States of America**	

Additional Manufacturing location(s):

This certificate is issued as verification that a sample(s), representative of production, was assessed and tested and found to comply with the IEC Standard list below and that the manufacturer's quality system, relating to the Ex products covered by this certificate, was assessed and found to comply with the IECEx Quality system requirements. This certificate is granted subject to the conditions as set out in IECEx Scheme Rules, IECEx 02 and Operational Documents as amended.

STANDARDS:

The apparatus and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards:

IEC 60079-0 : 2011 Edition:6.0	Explosive atmospheres - Part 0: General requirements
IEC 60079-1 : 2014-06 Edition:7.0	Explosive atmospheres - Part 1: Equipment protection by flameproof enclosures "d"
IEC 60079-11 : 2011 Edition:6.0	Explosive atmospheres - Part 11: Equipment protection by intrinsic safety "i"
ISO 80079-36 : 2016 Edition:1.0	Explosive atmospheres - Part 36: Non-electrical equipment for explosive atmospheres - Basic methods and requirements
ISO 80079-37 : 2016 Edition:1.0	Explosive atmospheres - Part 37: Non-electrical equipment for explosive atmospheres - Non electrical type of protection constructional safety "c", control of ignition source "b", liquid immersion "k"

*This Certificate **does not** indicate compliance with electrical safety and performance requirements other than those expressly included in the Standards listed above.*

TEST & ASSESSMENT REPORTS:

A sample(s) of the equipment listed has successfully met the examination and test requirements as recorded in

Test Report:

CH/SEV/ExTR19.0036/00

Quality Assessment Report:

GB/BAS/QAR14.0004/04

IECEx 证书第 2 页

证书第 2 页说明：代表生产样品已评估和测试，符合下列 IEC 标准，且制造商与该证书覆盖的 Ex 产品相关的质量体系也已评估并符合 IECEx 质量体系要求。证书授予受 IECEx Scheme Rules、IECEx 02 和相关 Operational Documents 约束。

列出的标准：

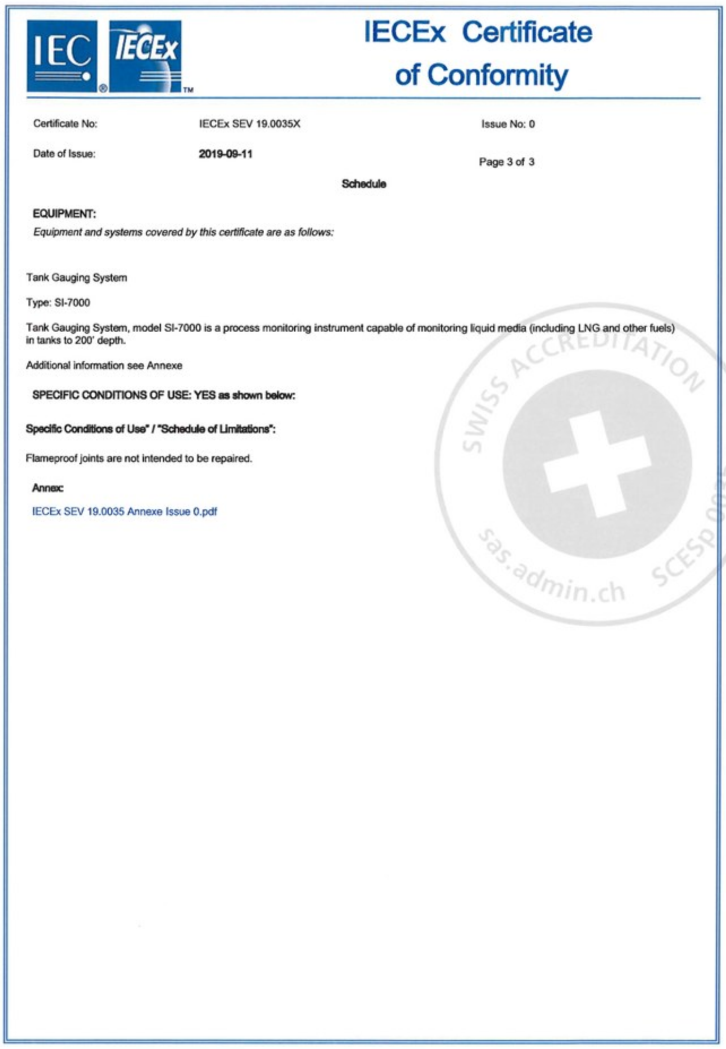
标准	版本	中文译文
IEC 60079-0:2011	Edition 6.0	爆炸性环境，第 0 部分：通用要求。
IEC 60079-1:2014-06	Edition 7.0	爆炸性环境，第 1 部分：隔爆外壳 d 设备保护。
IEC 60079-11:2011	Edition 6.0	爆炸性环境，第 11 部分：本质安全 i 设备保护。
ISO 80079-36:2016	Edition 1.0	爆炸性环境用非电气设备，基本方法和要求。
ISO 80079-37:2016	Edition 1.0	爆炸性环境用非电气设备，非电气保护型式，包括结构安全 c、点燃源控制 b、液浸 k。

页面备注：该证书不表示符合上述标准清单之外的电气安全和性能要求。

报告号：

项目	编号
Test Report	CH/SEV/ExTR19.0036/00
Quality Assessment Report	GB/BAS/QAR14.0004/04

证书第 3 页



IECEx Certificate of Conformity

Certificate No: IECEx SEV 19.0035X   Issue No: 0

Date of Issue: 2019-09-11   Page 3 of 3

Schedule

EQUIPMENT:
Equipment and systems covered by this certificate are as follows:

Tank Gauging System

Type: SI-7000

Tank Gauging System, model SI-7000 is a process monitoring instrument capable of monitoring liquid media (including LNG and other fuels) in tanks to 200' depth.

Additional information see Annexe

SPECIFIC CONDITIONS OF USE: YES as shown below:

Specific Conditions of Use" / "Schedule of Limitations":

Flameproof joints are not intended to be repaired.

Annex:

IECEx SEV 19.0035 Annexe Issue 0.pdf

IECEx 证书第 3 页

证书第 3 页为 Schedule。清晰可读的关键内容如下：

- Equipment：Tank Gauging System。
- Type：SI-7000。
- 设备描述：SI-7000 罐计量系统是一种过程监测仪表，可监测储罐内液体介质，包括 LNG 和其他燃料，储罐深度可到 200 ft。
- Additional information：见 Annex。
- Specific Conditions of Use：Yes。
- 限制条件：Flameproof joints are not intended to be repaired. 即隔爆接合面不 intended to be repaired，中文可理解为隔爆接合面不应由用户修理。
- Annex：IECEx SEV 19.0035 Annexe Issue 0.pdf。

译注：本资料包只包含证书正文 3 页，没有附带 Annex 文件内容。涉及具体限制条件、安装细节或维修边界时，应进一步查阅 Annex 和最新认证文件。

SI-7000 标准包装规格

SI-7000 标准包装规格

原始图纸信息：

字段	原文
Title	SI-7000 STANDARD PACKAGING SPECIFICATIONS
Drawing No.	025-119
Sheet	1 of 1
Scale	NTS
Drawn	GTN, 2016-06-22
Approved by	KN, 2017-04-28
Revision	-
Revision history	Initial Release, AS/WWJ, 2018-04-02, approved by WWJ

存储要求

包装设计用于室内存储，但可在有遮盖区域短期放置于室外。木箱上标有国际符号，用于指示正确搬运方式。

木箱结构

所有物品均使用木箱运输。木箱使用 5/8 inch 厚胶合板构造，并根据需要在所有角部和边缘使用 2 inch x 4 inch 木材支撑。根据内部支撑需要，还会额外使用 2 inch x 4 inch 和 2 inch x 12 inch 木材。木箱还使用钢带捆扎。

图中标注的典型支撑包括：

原文标注	中文译文
2 inch x 4 inch wooden supports	2 inch x 4 inch 木支撑
2 inch x 12 inch wooden support	2 inch x 12 inch 木支撑
4 inch x 4 inch wooden supports	4 inch x 4 inch 木支撑

包装要求

顶部罐体组件 Top of Tank Assemblies：

- 机械组件充氮并密封。
- 安装在机械组件上的电子外壳使用氮气吹扫，并在盖内放置干燥剂。

控制室电子设备和手册 Control Room Electronics/Manuals：

- 所有物品牢固装入纸箱，然后装入塑料袋中真空密封。
- 如果随系统订购备件，备件也装入塑料袋并真空密封。

近似重量和尺寸

项目	英制	公制	中文说明
SI-7000 ASSY crate size	40 L x 40 W x 60 H inch	1016 L x 1016 W x 1524 H mm	包装后外形尺寸约 40 x 40 x 60 inch。
Net Weight	393 lb	178 kg	净重约 393 lb，即 178 kg。
Gross Weight	600 lb	273 kg	毛重约 600 lb，即 273 kg。

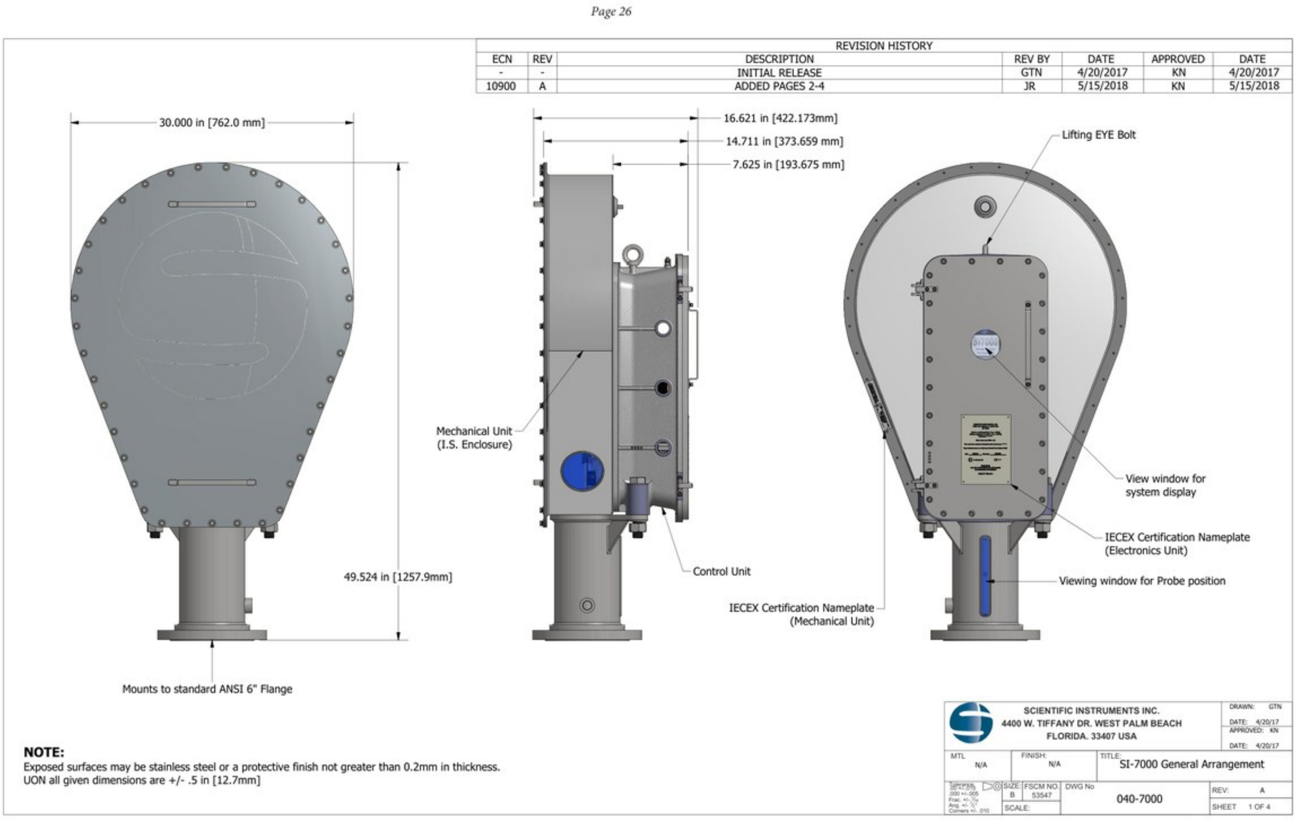
译注：规格页中设备重量为 180 kg，包装页净重为 178 kg，两者量级一致但不是完全相同数值。需要用于吊装、运输或现场工装设计时，应以项目图纸、装箱单和现场设备实物确认值为准。

机械布置与现场接线中文翻译

来源文件：../00_原始资料/SI7000官方资料包_含Modbus规范.pdf 第 26-31 页

整理日期：2026-07-02

SI-7000 General Arrangement, Sheet 1 of 4



SI-7000 总体布置图 Sheet 1

图纸信息：

字段	原文
Drawing No.	040-7000
Revision	A
Title	SI-7000 General Arrangement
Sheet	1 of 4
Drawn	GTN, 2017-04-20
Approved	KN, 2017-04-20
Revision history	Initial Release; ECN 10900 Rev A added pages 2-4, 2018-05-15

主要部件标注

原文标注	中文译文
Control Unit	控制单元
Mechanical Unit (I.S. Enclosure)	机械单元，本安外壳
Mounts to standard ANSI 6 inch Flange	安装到标准 ANSI 6 inch 法兰

原文标注	中文译文
Viewing window for Probe position	探头位置观察窗
View window for system display	系统显示观察窗
IECEX Certification Nameplate (Electronics Unit)	IECEx 认证铭牌，电子单元
IECEX Certification Nameplate (Mechanical Unit)	IECEx 认证铭牌，机械单元
Lifting EYE Bolt	吊装吊环螺栓

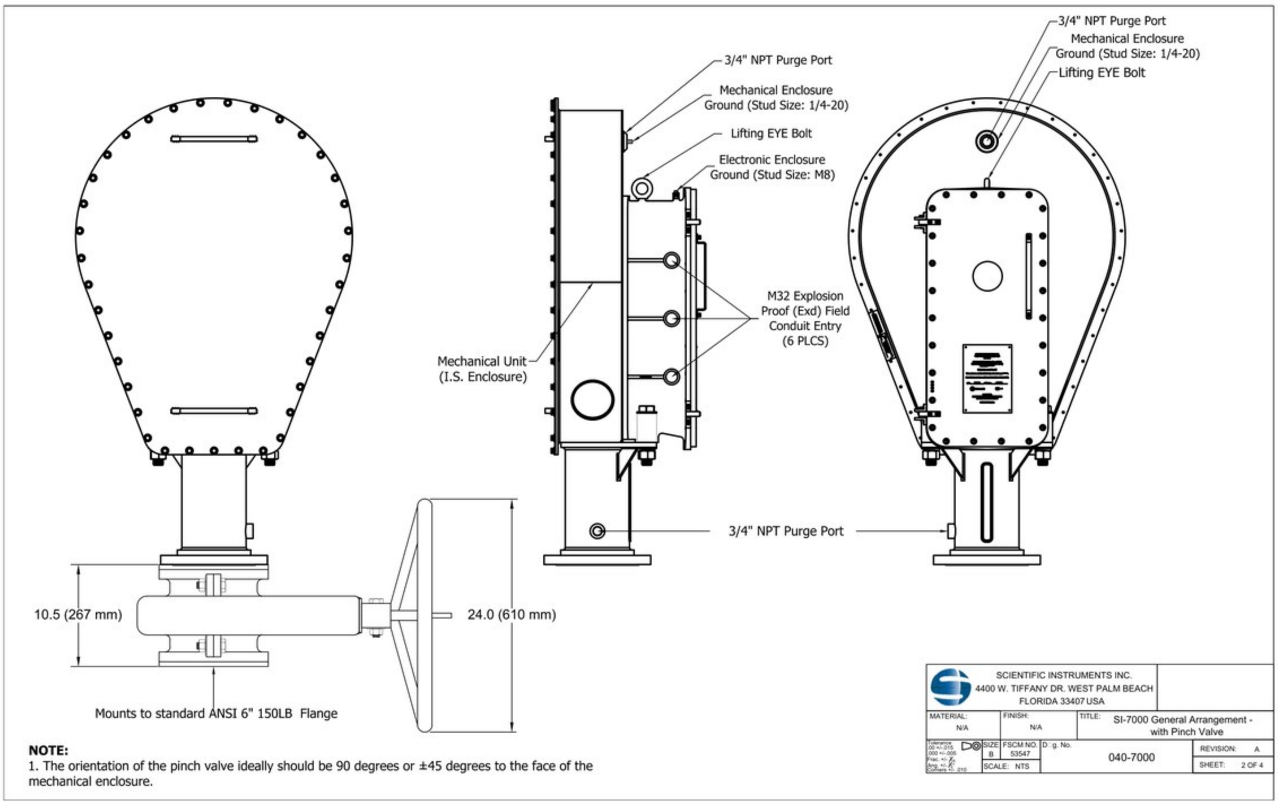
关键尺寸

尺寸	原文	中文说明
总宽	30.000 in [762.0 mm]	正视图宽度约 762.0 mm。
总高	49.524 in [1257.9 mm]	从安装法兰到顶部外形高度约 1257.9 mm。
侧向尺寸 1	16.621 in [422.173 mm]	侧视图上较大深度尺寸。
侧向尺寸 2	14.711 in [373.659 mm]	侧视图中间深度尺寸。
侧向尺寸 3	7.625 in [193.675 mm]	侧视图局部深度尺寸。

图纸备注：

- 外露表面可以是不锈钢，也可以是厚度不大于 0.2 mm 的保护涂层。
- 除非另有说明，所有给定尺寸公差为 +/-0.5 in，即 +/-12.7 mm。

SI-7000 General Arrangement with Pinch Valve, Sheet 2 of 4



SI-7000 带 Pinch Valve 布置图 Sheet 2

图纸信息：

字段	原文
Drawing No.	040-7000
Revision	A
Title	SI-7000 General Arrangement - with Pinch Valve
Sheet	2 of 4
Scale	NTS

主要部件和接口标注

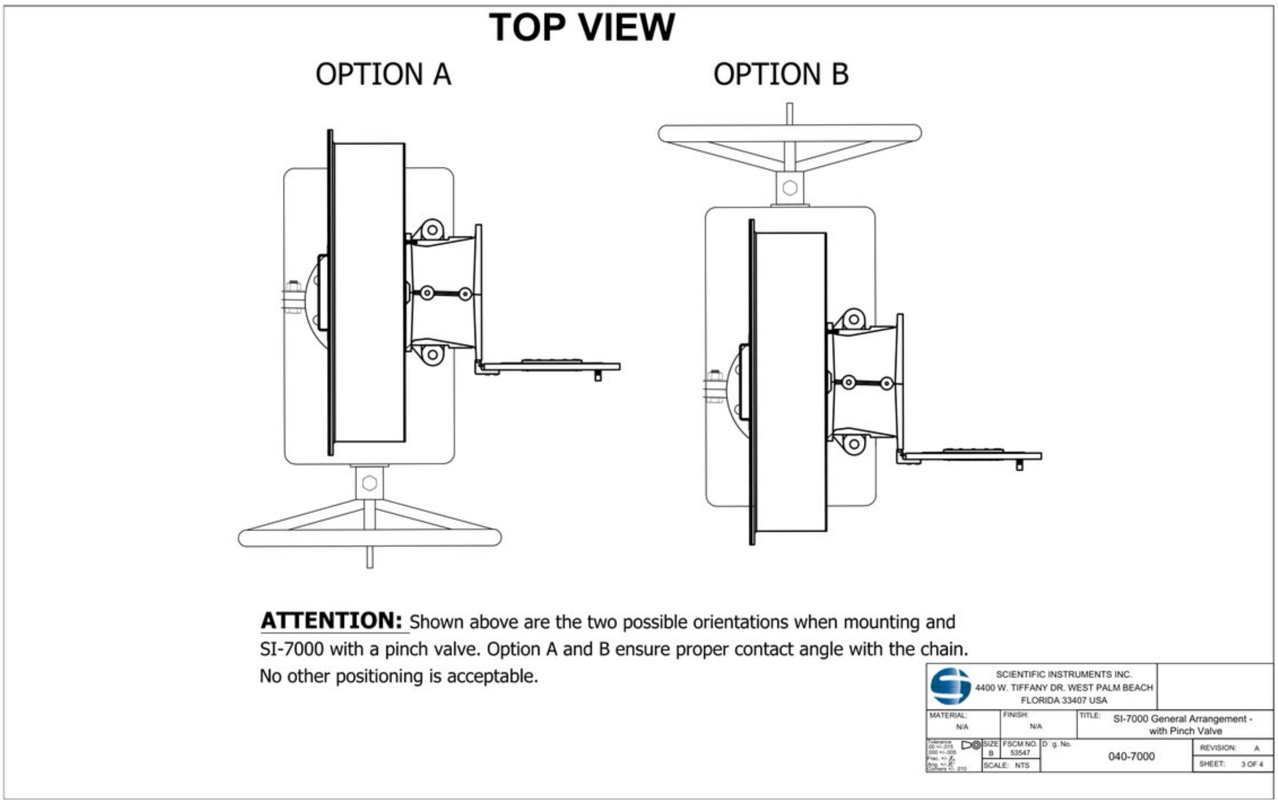
原文标注	中文译文
Mechanical Unit (I.S. Enclosure)	机械单元，本安外壳
3/4 inch NPT Purge Port	3/4 inch NPT 吹扫口
Mechanical Enclosure Ground (Stud Size: 1/4-20)	机械外壳接地点，螺柱尺寸 1/4-20
Electronic Enclosure Ground (Stud Size: M8)	电子外壳接地点，螺柱尺寸 M8
Lifting EYE Bolt	吊装吊环螺栓
M32 Explosion Proof (Exd) Field Conduit Entry (6 PLCS)	M32 隔爆 Exd 现场电缆管入口，共 6 处
Mounts to standard ANSI 6 inch 150LB Flange	安装到标准 ANSI 6 inch 150LB 法兰

Pinch Valve 相关尺寸和方向

原文	中文说明
10.5 [267 mm]	Pinch Valve 相关竖向尺寸，约 267 mm。
24.0 [610 mm]	Pinch Valve 手轮或侧向相关尺寸，约 610 mm。

图纸备注：Pinch Valve 的方向理想情况下应与机械外壳正面成 90 degrees 或 +/-45 degrees。

SI-7000 General Arrangement with Pinch Valve, Sheet 3 of 4



SI-7000 带 Pinch Valve 顶视方向图 Sheet 3

该页为顶视图 Top View，给出带 Pinch Valve 安装时允许的两个方向：

选项	中文说明
Option A	Pinch Valve 按图中 Option A 方向安装。
Option B	Pinch Valve 按图中 Option B 方向安装。

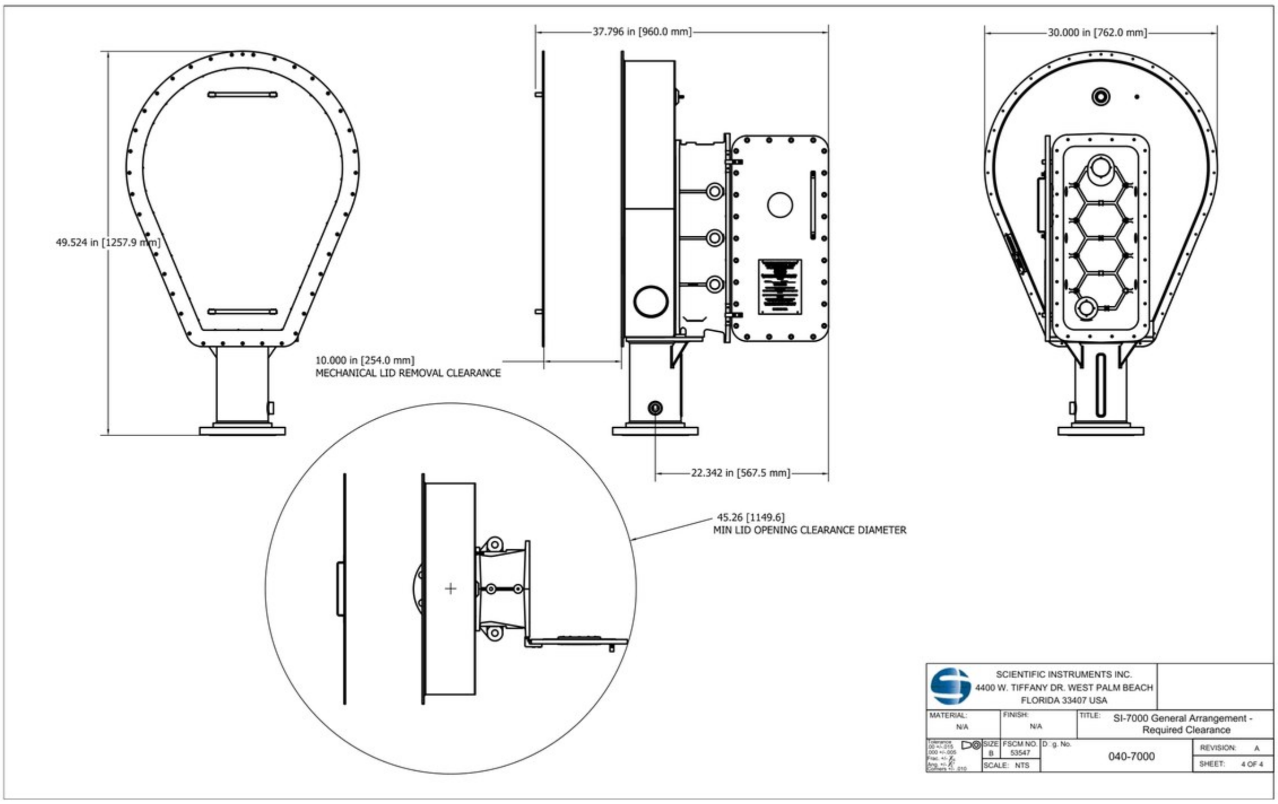
原文注意事项：

> Shown above are the two possible orientations when mounting an SI-7000 with a pinch valve. Option A and B ensure proper contact angle with the chain. No other positioning is acceptable.

中文译文：上图显示带 Pinch Valve 安装 SI-7000 时允许的两种方向。Option A 和 Option B 能确保链条具有正确接触角。其他安装定位均不可接受。

这条限制对现场安装非常关键。后续安装手册中涉及 Pinch Valve 和吊装定位时，应与该图互相核对。

SI-7000 General Arrangement Required Clearance, Sheet 4 of 4



SI-7000 所需间隙图 Sheet 4

图纸信息：

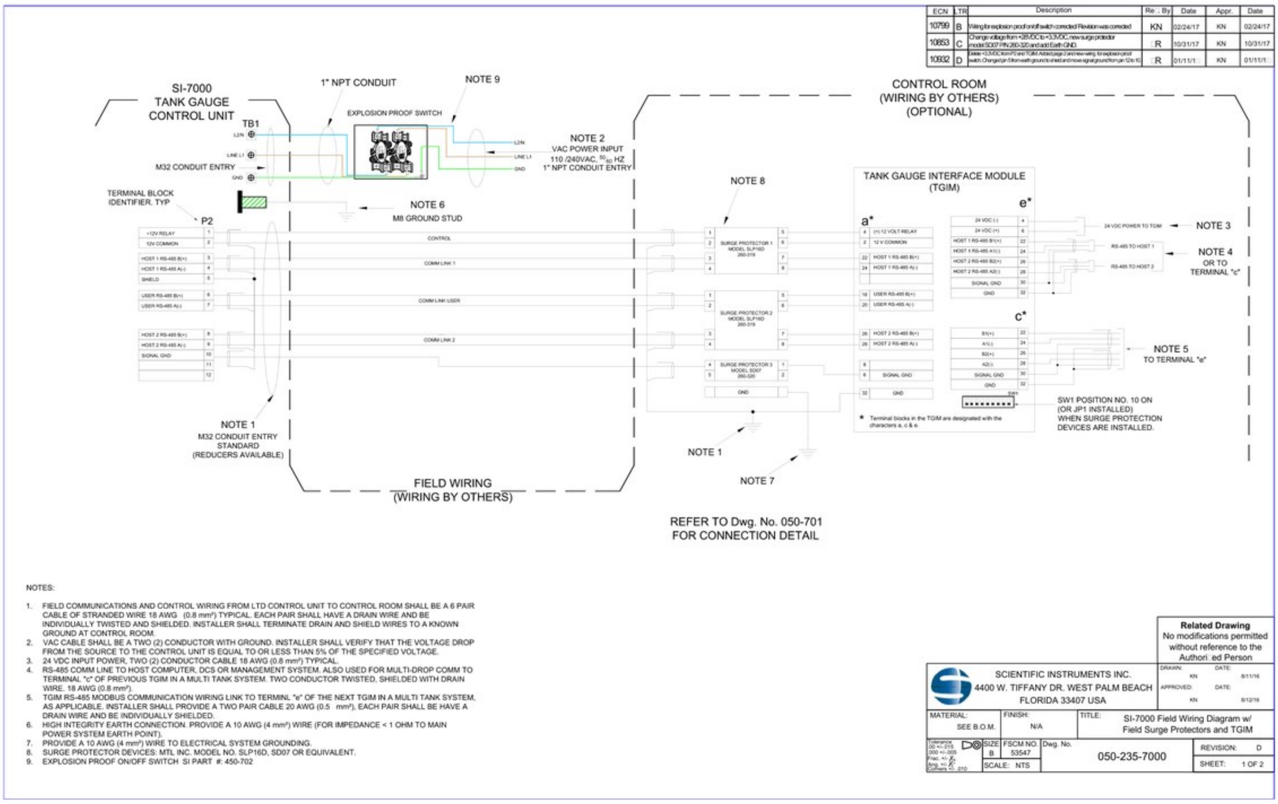
字段	原文
Drawing No.	040-7000
Revision	A
Title	SI-7000 General Arrangement - Required Clearance
Sheet	4 of 4

关键间隙和尺寸

原文尺寸	中文说明
49.524 in [1257.9 mm]	设备总高度参考，与 Sheet 1 一致。
37.796 in [960.0 mm]	侧向总占用尺寸。
30.000 in [762.0 mm]	正面宽度参考。
22.342 in [567.5 mm]	控制单元侧向突出或局部占用尺寸。
10.000 in [254.0 mm] Mechanical Lid Removal Clearance	机械盖拆卸所需间隙 254.0 mm。
45.26 [1149.6] Min Lid Opening Clearance Diameter	最小开盖间隙直径约 1149.6 mm。

译注：该页用于确认设备周边维护、开盖和拆盖空间。现场布置时不仅要满足安装法兰和设备外形尺寸，还要满足机械盖拆卸和开盖所需空间。

SI-7000 Field Wiring Diagram with Field Surge Protectors and TGIM, Sheet 1 of 2



SI-7000 带现场浪涌保护和 TGIM 的现场接线图

图纸信息：

字段	原文
Drawing No.	050-235-7000
Revision	D
Title	SI-7000 Field Wiring Diagram w/ Field Surge Protectors and TGIM
Sheet	1 of 2
Drawn	KN，2016-08-11
Approved	KN，2016-08-12
Related drawing note	No modifications permitted without reference to the Authorized Person

图中主要区域

原文区域	中文译文
SI-7000 Tank Gauge Control Unit	SI-7000 罐计量控制单元
Field Wiring (Wiring by Others)	现场接线，由其他方施工
Control Room (Wiring by Others) (Optional)	控制室接线，由其他方施工，可选
Tank Gauge Interface Module (TGIM)	罐计量接口模块 TGIM
Field Surge Protectors	现场浪涌保护器
Explosion Proof Switch	防爆开关

SI-7000 控制单元端子摘录

左侧 P2 端子在图中可辨识的信号包括：

端子或信号	中文译文
+12V Relay	+12 V 继电器相关端子
12V Common	12 V 公共端
Host 1 RS-485 B(+)	Host 1 RS-485 B(+)
Host 1 RS-485 A(-)	Host 1 RS-485 A(-)
User RS-485 B(+)	User RS-485 B(+)
User RS-485 A(-)	User RS-485 A(-)
Host 2 RS-485 B(+)	Host 2 RS-485 B(+)
Host 2 RS-485 A(-)	Host 2 RS-485 A(-)
Signal GND	信号地

图中还标出 M32 conduit entry、1 inch NPT conduit、M8 ground stud、110/240 VAC 电源输入和 50/60 Hz。

TGIM 相关连接

图中 TGIM 端子块以 a*、c*、e* 标识。可辨识连接包括：

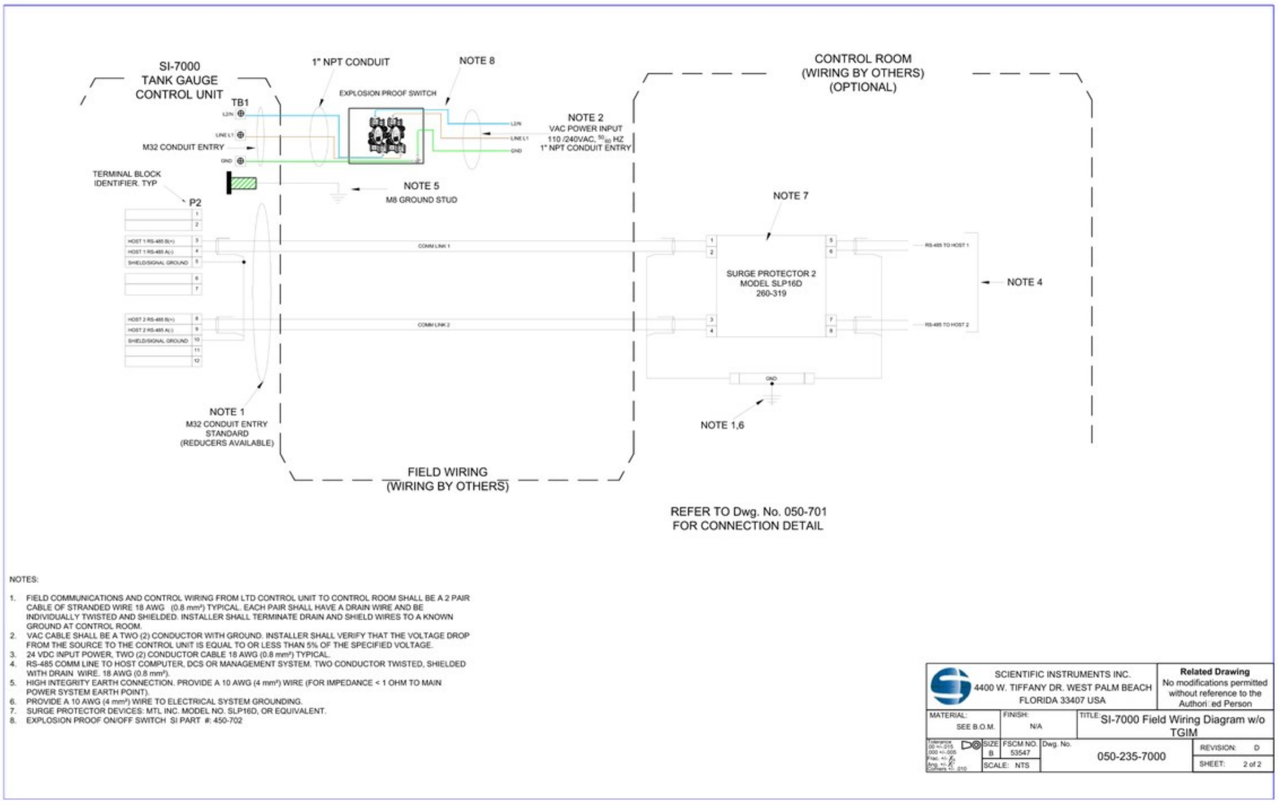
连接	中文译文
24 VDC power to TGIM	24 VDC 供电到 TGIM
RS-485 to Host 1	RS-485 到 Host 1
RS-485 to Host 2	RS-485 到 Host 2
To terminal c	到端子 c
To terminal e	到端子 e
SW1 position no. 10 on when surge protection devices are installed	安装浪涌保护器时，SW1 第 10 位拨到 ON，或安装 JP1。

图纸 notes 中文翻译

Note	中文译文
1	从 LTD 控制单元到控制室的现场通信和控制接线应使用 6 pair stranded wire，典型线径 18 AWG (0.8 mm2)。每对线都应有 drain wire，并且每对线应单独双绞和屏蔽。安装方应把 drain 和 shield wires 接到控制室已知接地点。
2	VAC 电缆应为两芯加接地。安装方应确认从电源到控制单元的压降不超过规定电压的 5%。
3	24 VDC 输入电源，典型为两芯 18 AWG (0.8 mm2) 电缆。
4	RS-485 通信线到主机计算机、DCS 或管理系统。也用于多点通信到终端 c 或上一台 TGIM。两芯双绞屏蔽线，带 drain wire，18 AWG (0.8 mm2)。
5	TGIM RS-485 Modbus 通信接线连接到多罐系统中下一台 TGIM 的 terminal e。安装方应提供 2 pair 20 AWG (0.5 mm2) 电缆，每对线应有 drain wire，并单独屏蔽。
6	高完整性接地连接。提供 10 AWG (4 mm2) 线缆，主电源系统接地点阻抗小于 1 ohm。
7	提供 10 AWG (4 mm2) 线缆连接到电气系统接地。
8	浪涌保护器：MTL Inc. model SLP16D、SD07 或等效产品。
9	防爆 On/Off 开关，SI part # 450-702。

译注：上表为图纸 notes 的中文阅读稿，实际接线、线径、接地和防爆器件选型必须以项目电气设计、最新图纸和当地规范为准。

SI-7000 Field Wiring Diagram without TGIM, Sheet 2 of 2



SI-7000 不带 TGIM 的现场接线图

图纸信息：

字段	原文
Drawing No.	050-235-7000
Revision	D
Title	SI-7000 Field Wiring Diagram w/o TGIM
Sheet	2 of 2

与 TGIM 版本的主要差异

不带 TGIM 的接线图保留 SI-7000 Tank Gauge Control Unit、现场接线、控制室可选接线、浪涌保护器和防爆开关，但控制室侧不再画出 TGIM 端子块。RS-485 通信线直接面向 Host 1 / Host 2。

图中可辨识要点：

- SI-7000 控制单元侧仍有 TB1 和 P2。
- Host 1 和 Host 2 各自有 RS-485 B(+) / A(-) 和 shield / signal ground。
- 现场通信线经浪涌保护器 Surge Protector 2 model SLP16D 260-319 后接往控制室主机。
- 图纸仍要求参考 Dwg. No. 050-701 获取 connection detail。

图纸 notes 中文翻译

不带 TGIM 图纸 notes 与带 TGIM 版本基本一致，但通信线数量和编号有所不同：

Note	中文译文
1	从 LTD 控制单元到控制室的现场通信和控制接线应使用 2 pair stranded wire，典型线径 18 AWG (0.8 mm2)。每对线都有 drain wire，并且每对线应单独双绞和屏蔽。安装方应把 drain 和 shield wires 接到控制室已知接地点。
2	VAC 电缆应为两芯加接地。安装方应确认从电源到控制单元的压降不超过规定电压的 5%。

Note	中文译文
3	24 VDC 输入电源，典型为两芯 18 AWG（0.8 mm2）电缆。
4	RS-485 通信线到主机计算机、DCS 或管理系统。两芯双绞屏蔽线，带 drain wire，18 AWG（0.8 mm2）。
5	高完整性接地连接。提供 10 AWG（4 mm2）线缆，主电源系统接地点阻抗小于 1 ohm。
6	提供 10 AWG（4 mm2）线缆连接到电气系统接地。
7	浪涌保护器：MTL Inc. model SLP16D 或等效产品。
8	防爆 On/Off 开关，SI part # 450-702。

安装与调试手册中文翻译

来源文件：../00_原始资料/SI7000官方资料包_含Modbus规范.pdf 第 32-51 页

原始标题：SI-7000 Tank Gauging System Installation & Commissioning Manual

原始编号：090-7000-1 Rev A

整理日期：2026-07-02

目录翻译

引言

SI-7000 Tank Gauging System 用于监测大型储罐内低温液体的液位、温度和密度，通常由控制室中的 PC 操作员界面和/或 DCS 用户界面控制和监测。

SI-7000 由机械部分和电气部分组成：

部分	原文说明	中文译文
Mechanical section	chain reel enclosure and probe enclosure	机械部分由链轮外壳和探头外壳组成。除驱动轴齿轮箱外，该部分包含在储罐内升降探头组件所需的全部机械部件。该部分应视为 Zone 0 环境。多数安装中不需要接触该部分。
Electrical section	electrical enclosure	电气部分由电气外壳组成，包含所有电子设备和驱动轴齿轮箱。电气部分外壳是防爆容器，应视为 Zone 1 环境。SI-7000 的输入电源和通信在该部分进行物理连接。

更详细的部件说明和功能介绍，官方要求参考 SI-7000 Operational Manual (090-7000-2)。

探头外壳安装在链轮外壳底部，链轮外壳和电气外壳背靠背安装。

Figure 1-6 的中文图注：

图号	原文图注	中文译文
Figure 1	Chain reel & probe enclosures	链轮和探头外壳
Figure 2	Chain reel enclosure w/o Cover	未安装盖板的链轮外壳
Figure 3	SI-7000 Side View	SI-7000 侧视图
Figure 4	Electrical Enclosure	电气外壳
Figure 5	Electrical Enclosure w/ door open	电气外壳门打开状态
Figure 6	SI-7000 transparent side view	SI-7000 透明侧视图

本手册提供在典型位置安装系统所需的信息。对于非典型位置，可能随本手册提供站点专用机械安装图和现场接线图，用于说明具体细节。

完整上电、系统测试和校准会在后续调试活动中完成，并且只能在厂家代表在场时进行。相关活动的更详细说明应参考 SI-7000 Operational Manual（090-7000-2）。不过，如果 SI 人员在场，本手册包含的初始上电和基本系统功能测试可以执行。

警告、注意和通知

Scientific Instruments, Inc. 不对 SI-7000 安装过程中造成的任何伤害、死亡或损坏承担责任。安装方或客户负责确保操作以安全方式执行，并符合所有政府、地方和主管机构法规。

Warning

原文类别	中文译文
WARNING	必须严格遵守所有适用的政府、地方和主管机构法规。否则可能导致死亡或严重伤害。
WARNING	雷电导电接地路径必须以合适方式实现，使升温、可点燃火花或喷射火花不会成为爆炸性环境的点火源。否则可能导致死亡、严重伤害或重大财产损失。
WARNING	必须确保用于起吊 SI-7000 的设备能够承受设备重量，以及起重机运动造成的突然冲击。否则可能导致死亡、严重伤害或重大财产损失。
WARNING	起吊和移动 SI-7000 到位时，必须确保结构、其他设备和人员之间保持足够间隙。否则可能导致死亡、严重伤害或重大财产损失。
WARNING	由于内部本质安全电路的位置，不得对左上入口端口进行任何连接。

Caution 和 Notice

原文类别	中文译文
CAUTION - PINCH POINT	将 SI-7000 定位到安装法兰上时必须小心，SI-7000 下放到位时所有人员必须让手远离夹点。否则手指可能被压碎或切断，造成严重伤害。
NOTICE	必须遵守本手册中的所有安装说明。否则可能损坏 SI-7000、导致保修失效或造成安装延误。
NOTICE	在 SI-7000 正在连接到储罐配对法兰之前，不要从探头外壳上移除保护泡棉。过早移除保护泡棉，会导致 SI-7000 移动和吊装到位时探头撞击探头外壳内壁。

安装要求

以下为 Scientific Instruments 建议的最低安装要求。如果这些最低要求未满足且存在安全风险，Scientific Instruments 人员不会参与安装活动。

要求	中文译文
法兰	可用的 6 inch 150 pound 法兰，安装在最小 6 inch 管道上，提供通向罐内的无阻碍通道。该法兰应能支撑 400 lb，即 181.4 kg，且不会损坏法兰或周边结构。

要求	中文译文
间隙	法兰周围至少 1 m 间隙。设备不应安装在难以操作 AC 电源断开开关的位置。
起重	起重机和合格操作员。
导向人员	两名人员引导 SI-7000 到位。
安装人员	一名具备机械和电气安装资质的人员。
安装人员能力	了解相关电气工程；理解并能够阅读、评估工程图；实际了解防爆原则和技术。
工作平台	稳定的临时或永久工作平台。

其他备注：

- 由于 SI-7000 始终安装在储罐外部，因此没有额外通风要求。
- 无声级相关要求。
- 安装 SI-7000 前不需要装配。
- 任何集成 SI-7000 的系统，其安全性由系统组装方负责。
- Scientific Instruments 当前不为 SI-7000 提供任何附件。

安装前检查清单

这些项目需要在 SI-7000 安装前确认。不满足要求会影响 SI-7000 运行、造成严重设备损坏或导致安装延误。

安装工具和硬件

安装 SI-7000 只需要少量基本工具。该检查清单应在登上罐顶前完成。

现场提供工具：

数量	原文	中文译文
2 ea	1 1/8 inch Wrench	1 1/8 inch 扳手 2 把
1 ea	6 mm Hex T-handle	6 mm 六角 T 型手柄 1 把
1 ea	Reversible Screwdriver Set (#2 Philips & 9/32 in. slotted)	可换向螺丝刀套装，#2 十字和 9/32 inch 一字
1 ea	Torque wrench (10 ft/lbs - 150 ft/lbs)	扭矩扳手，范围 10 ft-lb 到 150 ft-lb
1 ea	1 1/8 inch Socket, 3/8 inch drive	1 1/8 inch 套筒，3/8 inch 驱动
1 ea	1/2 inch to 3/8 inch drive adapter	1/2 inch 转 3/8 inch 驱动转换头
1 ea	1/2 inch Socket, 3/8 inch drive	1/2 inch 套筒，3/8 inch 驱动
1 ea	6 mm Hex bit socket, 3/8 inch drive	6 mm 六角批头套筒，3/8 inch 驱动

Scientific Instruments 提供硬件：

数量	原文	中文译文
1 ea	Explosion proof external AC power switch	防爆外部 AC 电源开关
16 ea	3/4-10 x 2.5 inch Stainless steel bolt	3/4-10 x 2.5 inch 不锈钢螺栓
4 ea	3/4 inch Stainless steel nut (Extra on Blind Flange)	3/4 inch 不锈钢螺母，盲法兰上额外提供
16 ea	3/4 inch Stainless steel lock washer	3/4 inch 不锈钢锁紧垫圈
16 ea	3/4 inch Stainless steel washer	3/4 inch 不锈钢平垫圈
4 ea	3/4-10 x 3 inch Stainless steel bolt (Extra on Blind Flange)	3/4-10 x 3 inch 不锈钢螺栓，盲法兰上额外提供

罐内要求

检查项	中文译文
罐清洁	储罐关闭和密封前是否已清洁？罐内残留的过量铁锈、尘粒、珍珠岩或其他异物，长期可能粘附到密度计上，开始影响密度读数。
探头垂直运动	从驱动机构一直到罐底或 stilling well，探头组件是否能够无阻碍垂直移动？任何障碍都可能使探头组件卡住，并损坏探头组件、链条和/或驱动机构。
避开泵和充装线	探头组件路径是否远离会产生高液流或湍流的泵或充装管线？高液流或湍流可能损坏探头组件，或导致密度和/或液位测量不准确。
Stilling well	如果使用 stilling well，它是否垂直且尺寸足够，确保探头不会接触 stilling well 内表面？与内表面接触可能导致不可预测运行，并给驱动系统增加额外应力，导致部件过早失效。

罐外要求

检查项	中文译文
工作平台	是否已有稳定工作平台？可以是永久或临时结构，但 SI 强烈建议安装永久工作平台。
安全设备	是否已有足够安全设备，且设备可用？
配对法兰	SI-7000 将安装到的配对法兰是否清洁、无损坏、无碎屑？
垫片	配对法兰和 SI-7000 之间使用的垫片是否清洁且可用？
隔离阀手轮	隔离阀手轮是否会妨碍接近机械或电气外壳？如会妨碍，机械外壳盖是否仍能拆下，电气外壳门是否仍能打开？
Pinch Valve	如果使用 Pinch Valve，是否安装为尽可能夹在链条平面侧，且手轮不会阻碍打开电气外壳门或拆除机械外壳盖？Pinch Valve 对不准可能导致密封不良，使维护时过量气体逸出，也可能阻碍人员接近 SI-7000 的一侧。

电气和通信

检查项	中文译文
外部电源开关位置	是否已在 SI-7000 可视范围内选择外部电源开关位置？
电源和通信电缆管	电源和通信电缆管是否准备好连接到 SI-7000？SI-7000 可以在电缆管尚未就位时安装，但这些连接完成前安装不视为完成。
上锁挂牌	SI-7000 电源是否已经 lock out / tag out？如果没有，在电源导体连接到 SI-7000 前必须完成。

SI-7000 开箱和检查

SI-7000 设计为坚固并可用于严苛条件，但移动或起吊设备时仍需小心。SI 设计了定制容器，使 SI-7000 到货后即可安装。不过仍需检查运输过程中可能发生的损坏。

步骤：

1. 确保运输木箱正立在平坦、稳定且相对水平的表面。

- 如果木箱来回摇晃，应加垫片使其稳定。

2. 确保冲击和倾斜指示器未被触发。

- 如果指示器已触发，应通知 SI，并尽可能提供照片。

3. 检查运输木箱是否有任何可见损坏迹象。

- 如有损坏，应通知 SI，并尽可能提供照片。

4. 移除木箱顶部、内部水平支撑和侧板。
5. 彻底检查 SI-7000 是否有任何损坏迹象。

- 如有损坏，应通知 SI，并尽可能提供照片。

6. 将吊带连接到电气外壳顶部吊眼，并施加轻微起吊力。
7. 使用 1/2 inch 扳手拆下将 SI-7000 固定在木箱底部的 4 个方头木螺钉。

相关图注：

图号	原文图注	中文译文
Figure 7	SI-7000 mounted on crate bottom	SI-7000 安装在木箱底板上
Figure 8	SI-7000 with lifting strap attached	已连接吊带的 SI-7000

安装

参考：Appendix A Bolt Tightening Worksheet 和其他随附站点专用图纸。

NOTICE：在 SI-7000 正在连接到储罐配对法兰之前，不要从探头外壳中移除保护泡棉。移除保护泡棉会导致 SI-7000 吊装到位时探头撞击探头外壳内壁。

步骤：

1. 确认 Pinch Valve 已按 4.3 节说明安装。
2. 将 SI-7000 吊到罐顶，注意不要让设备过度摆动。
3. 将 SI-7000 定位到安装法兰上方。
4. 移除 Blind Flange，并丢弃 3/4 inch 螺栓和螺母。保留平垫圈和锁紧垫圈，用于与 Pinch Valve 提供的 3/4-10 x 2.5 inch 螺栓配合安装。
5. 从探头外壳中移除保护泡棉。
6. 将 SI-7000 下放到 Pinch Valve 法兰上，确保孔位对齐，并让夹点区域保持无人员身体部位。
7. 使用随 Pinch Valve 提供的 8 组 3/4 inch stainless steel 硬件，把 SI-7000 安装到 Pinch Valve 法兰上。详细说明见 Appendix A。
8. 将等电位接地电缆连接到电气外壳顶部的接地凸耳，连接方式应能控制 30 m 雷击。原文为 in such a way that a bolt lightning of a 30m can be controlled，中文译文按雷电接地能力理解，现场应以电气设计和规范复核。

相关图注：

图号	原文图注	中文译文
Figure 9	Protective foam holding probe in place	固定探头位置的保护泡棉
Figure 10	无图题，仅 Figure 10	SI-7000 吊装到罐顶过程
Figure 11	Guiding SI-7000 into position	引导 SI-7000 到位
Figure 12	Remove probe protective foam	移除探头保护泡棉
Figure 13	Aligning SI-7000 with mating flange	将 SI-7000 与配对法兰对齐
Figure 14	Installing mounting hardware	安装固定硬件

连接系统接线

WARNING：所有用于 VAC 电源输入和/或通信的电气外壳螺纹入口，例如电缆密封套、导管接头、安装到电气外壳的堵头，都必须额定适用于爆炸性环境。

可选电缆密封套和螺纹入口种类很多，并具有覆盖国际和当地 wiring codes 的批准和证书。因此，安装方有责任确保电缆密封套、螺纹堵头和/或导管接头获得适用于 SI-7000 安装 Zone 和区域分类的 Ex 认证。更多信息应咨询制造商。

参考：Appendix B Drawing 050-7000-1 Typical Installation Wiring Diagram。

未使用的线缆和屏蔽层只能在控制室端接地。必须正确处理接地和屏蔽连接，以尽量降低通信线路噪声。

由于内部本质安全电路位置，不得对左上入口端口进行任何连接。连接该入口将使 SI-7000 的本质安全认证失效。

打开电气外壳门

在电气外壳内部作业时，必须特别小心，避免划伤或损伤 flame path 表面。装配和运输期间使用保护覆盖物，安装完成前应保持覆盖。任何 flame path 划伤或损坏都会使外壳防爆认证失效。

步骤：

- 松开所有固定外壳门关闭状态的螺栓。
- 完全打开门。
- 插入锁定销，将门保持在打开位置。

Figure 15 图注：Unused upper left entry port，中文为“未使用的左上入口端口”。

AC 电源输入和接地

AC 电源线应通过 SI-7000 的右下或左下入口端口连接，其中优先使用右下端口。SI-7000 设计用于 18 AWG 到 10 AWG 电源线，以及 8 AWG Earth ground。

Figure 16 和 Figure 17：

图号	原文图注	中文译文
Figure 16	Lower left-side AC entry port option	左下侧 AC 入口选项
Figure 17	Lower right-side AC entry port option	右下侧 AC 入口选项

步骤：

1. 确保 AC 电源已 lock out / tag out。
2. 确保外部 AC 电源开关处于 OFF 位置。
3. 将合适的 cable gland 和 conduit 连接到入口端口。
4. 拉入足够长度的电线，使 AC 电源线能够走线到电源端子排，Earth ground 线能够走线到外壳顶部接地螺柱，并且两者都有足够 service loop。
5. 在线缆端部压接合适的环形接线端子。
6. 拆下电源端子排盖板上的两颗螺钉，以接触连接端子。
7. 按接线图连接 AC 电源线。
8. 重新安装电源端子排盖板。
9. 按所有当地法规要求的 wiring practices，把 earth ground 端子连接到储罐接地。

备注：3/4 inch NPT conduit entries 为标准配置；其他批准尺寸可按需提供。

信号线连接

信号线可通过除左上端口和 AC 电源所用端口以外的任何入口端口连接到 SI-7000。为了便于内部走线，优先使用右侧中间端口。应注意不要让信号线靠近 AC 电源线走线。

SI-7000 设计用于 22 AWG 到 18 AWG 信号线，也可使用最高 14 AWG 的线规。但由于空间有限，建议使用尽可能小的线规。

所有信号线连接都通过电气外壳内主控制器组件 150-601 上的单个连接器 J2 完成。要连接到信号线的配对连接器 Phoenix Contact #1863259 已插在主控制器上，安装套件中还包含一个备用连接器。

Ferrules 不是必需的，但强烈建议使用。SI 推荐以下 Phoenix Contact 带套管 ferrules，也可使用类似 ferrules：

线规	Phoenix Contact 编号
22 AWG	3203024
20 AWG	3200687
18 AWG	3200690
16 AWG	3200755
14 AWG	3200522

步骤：

1. 在入口端口安装合适的 cable gland。

2. 拉入足够长度的线缆，使信号线能够走线到主控制器组件上的连接器，并留有足够 service loop。
3. 如果使用 ferrules，将 ferrules 压接到线端。
4. 从主控制器组件拆下配对连接器。
5. 按接线图把信号线连接到连接器。
6. 将连接器插回主控制器组件 J2。

关闭电气外壳门

在电气外壳内部作业时，必须特别小心，避免划伤或损伤 flame path 表面。任何 flame path 划伤或损坏都会使外壳防爆认证失效。

步骤：

1. 如果尚未完成 Initial Power Up，确保 AC 电源开关处于 OFF 位置。
2. 从 flame path 两侧移除保护覆盖物。
3. 使用套件中提供的酒精擦拭片清洁 flame path 表面，去除残留粘接剂。
4. 沿外壳侧 flame path 一周涂 1/8 inch 真空脂 bead，一条靠近外边缘，一条靠近中心。
5. 移除把门保持在打开位置的销。
6. 完全关闭门。
7. 使用提供的 Allen wrench 拧紧外壳门上的所有螺栓。
8. 将螺栓扭矩拧至 20 ft-lb。
9. 擦除从 flame path 中挤出的任何硅润滑剂。

译注：证书页明确 Flameproof joints are not intended to be repaired.，因此 flame path 清洁、保护和关门紧固步骤必须严格按厂家资料和现场防爆要求执行。

调试

NOTE：如果 SI-7000 未按 Scientific Instruments 规定的方式使用，设备提供的保护可能受损。

初始上电和调试只能由 Scientific Instruments 人员执行，或在其陪同下执行。在 Scientific Instruments 人员检查安装之前给 SI-7000 上电或操作 SI-7000，会导致保修失效，并且 Scientific Instruments 不对设备损坏承担责任。

初始上电

步骤：

1. 确保 SI-7000 电源开关处于 OFF 位置。
2. 移除 AC 电源 lock out / tag out。
3. 将 SI-7000 电源开关切到 ON 位置。
4. 检查主控制器上的以下项目：

- 红色 A/C 电源 LED 亮。如果不亮，检查：
- 确保 AC 电源已供给控制器组件。
- 确保电源插头完全插入插座。
- 绿色 FUSE LED 亮。如果不亮，检查：
- 检查 AC 插头模块中的保险丝，套件中含备用保险丝。
- 绿色 PWR LED 亮。如果不亮，最可能原因是：
- DC 电源不工作，更换控制器组件。
- LCD 点亮并显示主菜单。如果没有：
- 拆下控制器组件，确认 LCD 已完全连接。
- 如果仍不成功，则 LCD 或 CPU 不工作，更换控制器组件。

5. 继续下一项测试，或将 SI-7000 电源开关切到 OFF 位置。

电机驱动测试

步骤：

1. 确保 SI-7000 电源开关处于 ON 位置。
2. 在触摸屏 LCD 上进入 Motor menu。
3. 触摸 down arrow 一次。

- 电机应以最低速度开始降低探头组件。

4. 第二次触摸 down arrow。

- 电机应增加到中速。

5. 第三次触摸 down arrow。

- 电机应增加到最高速度。

6. 第四次触摸 down arrow。

- 电机应回到中速。

7. 触摸 stop button。

- 电机应停止，并将探头保持在当前位置。

8. 触摸 up arrow 一次。

- 电机应以最低速度开始提升探头组件。

9. 第二次触摸 up arrow。

- 电机应增加到中速。

10. 第三次触摸 up arrow。

- 电机应增加到最高速度。

11. 第四次触摸 up arrow。

- 电机应回到中速。

12. 允许电机把探头提升到 parked position。

- 电机应自动停止，并把探头保持在 parked position。

Bench Marking

步骤：

1. 确保 SI-7000 电源开关处于 ON 位置。
2. 在触摸屏 LCD 上进入 Motor menu。
3. 触摸 down arrow 一次。

- 电机应以最低速度开始降低探头组件。

4. 通过探头外壳窗口观察链条运动，当链条上的指定标记与参考点对齐时停止电机。记录 step count。

- 电机应停止，并将探头保持在当前位置。

5. 触摸 down arrow 三次。

- 电机应开始降低探头组件。

6. 通过探头外壳窗口观察链条运动，当链条上的第二个标记与参考点对齐时停止电机。记录第二个 step count。
7. 计算 level scale。详细信息咨询厂家。
8. 触摸 up arrow 三次。

- 电机应开始提升探头组件。

9. 允许电机把探头提升到 parked position。

- 电机应自动停止，并把探头保持在 parked position。

通信

步骤：

1. 确保 SI-7000 电源开关处于 ON 位置。
2. 确保绿色 COM LED 亮。
3. 从控制室发送降低探头命令。

- 电机应开始降低探头组件。

4. 从控制室发送停止电机命令。

- 电机应停止，并将探头保持在当前位置。

5. 从控制室发送提升探头命令。

- 电机应开始提升探头组件。

6. 允许电机把探头提升到 parked position。

- 电机应自动停止，并把探头保持在 parked position。

Appendix A - Bolt Tightening Work Sheet

该附录是螺栓紧固工作表，用于记录位置、螺栓规格、法兰表面处理、润滑剂、每步签字和最终签名。

需要填写字段：

原文	中文译文
Location/Identification	位置/标识
Nominal Bolt Size	公称螺栓尺寸
Surface Finish on Flange	法兰表面处理
Lubricant Used	使用的润滑剂
Name	姓名
Signature	签名
Date	日期

备注翻译：螺栓润滑会显著影响安装垫片时使用的扭矩值。为了实现相同的垫片压缩量，相比使用有效润滑剂如二硫化钼，干螺栓需要高得多的扭矩。润滑剂不应作为脱模剂涂到垫片或法兰面上。

每一步需要签名或首字母确认：

1. 确保系统处于环境温度并已泄压。
2. 目视检查并清洁法兰、螺栓、螺母和垫圈。
3. 润滑螺栓、螺母以及螺母表面的法兰接触面。
4. 从安装法兰底部插入螺栓并保持到位。
5. 将平垫圈和锁紧垫圈依次放到螺栓端部。
6. 将螺母安装到螺栓端部并手紧。
7. 对剩余七组硬件重复步骤 3 到 6。
8. 使用下列扭矩值和交叉顺序紧固硬件。每个 1 到 8 的紧固序列构成一轮 round。
9. 在每一轮之间检查圆周间隙，每隔一个螺栓测量一次。如果圆周间隙不够均匀，应通过选择性拧紧螺栓做适当调整后再继续。

扭矩轮次：

轮次	原文	中文译文
Round 1	Torque to 30 ft-lbs	拧紧到 30 ft-lb。
Round 2	Torque to 60 ft-lbs	拧紧到 60 ft-lb。

轮次	原文	中文译文
Final Rotational Round	100% of Final Torque	最终旋转轮，达到最终扭矩 100%，与上文 Round 3 相同。按顺时针旋转顺序，从 Bolt No. 1 开始完成一整轮，并继续直到任何螺母在最终扭矩 100% 下不再发生进一步旋转。

附录图给出的交叉紧固顺序为：1、5、3、7、8、4、6、2。

NOTE：初次紧固后 4 到 24 小时之间，由于螺栓松弛，可能发生短期螺栓预紧力损失。SI 建议在初次 Final Rotation Round 至少 24 小时后和/或即将调试前重复步骤 10。

操作手册中文翻译

来源文件：../00_原始资料/SI7000官方资料包_含Modbus规范.pdf 第 52-85 页

原始标题：

- SI-7000 Tank Gauging System Operational Manual Part A，090-7000-2A Rev -，PDF 第 52-59 页
- SI-7000 Tank Gauging System Operational Manual Part B，090-7000-2B Rev A，PDF 第 60-85 页

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-	Initial Release	KN	2019-08-21	KN	2019-08-21

Part A 目录：

Part A 引言

总体说明

Scientific Instruments, Inc. SI-7000 是先进的机电系统，用于测量低温液体中的液位、温度和密度，包括 LNG、丙烷和丁烷。系统设计重点包括可靠性、模块化、易维护和易操作。SI-7000 通过计算机控制运行，使操作员能够获取工厂智能、安全运行所需的关键数据。

除提供准确储罐液位测量外，SI-7000 还可通过把多传感器探头组件定位到储罐任意高度，实现整个储罐内温度和密度的精确测量。由此可获取 Profile，准确呈现罐内当前状态。对于低温液体的安全储存和处理，该信息很关键，因为低温液体可能发生分层，并随时间形成潜在的 rollover 条件。

应用

LNG 等低温液体储存中的重要安全问题之一是 rollover。如果发生 rollover，储罐内部压力可能升至过高水平。储罐配有安全放空阀，用于防止压力上升到可能造成结构损坏的水平。然而，当这些阀动作时，气体会以不受控速率排向大气，

这本身也是额外安全风险。

在特定条件下，分层 LNG 趋于平衡时会发生 rollover。分层是指储罐中产品形成具有不同密度和不同温度的层。随着两个或多个层的密度接近相等，任何 LNG 储罐中都可能发生突然混合。混合期间，系统内滞留的热量迅速释放，生成蒸汽，可能超过储罐的排放能力。

Scientific Instruments SI-7000 通过周期性获取储罐 Profile，可检测分层和潜在 rollover 条件，并产生报警提示需要操作员干预。操作员随后可以采取预防措施，例如混合液体或将液体转移到另一储罐。

系统配置

集成触摸屏 LCD 是设置所有系统参数的主要界面。系统配置也可通过 Modbus 串行通信接口远程修改。

Part A 系统规格

认证标准

官方列出的认证标准：

- IEC 60079-0:2011
- IEC 60079-1:2014
- IEC 60079-11:2011
- EN 60079-0:2012/A11:2013
- EN 60079-1:2014
- EN 60079-11:2012
- ISO 80079-36
- ISO 80079-37
- IEC/EN61010-1:2010

设备标志

部位	原文标志	中文译文
Electrical Compartment	II (1) 2 G Ex db [ia IIC Ga] IIB T4 Gb (-20 deg C <= Ta <= +50 deg C)	电气舱：隔爆 db，关联本安 [ia IIC Ga]，IIB T4 Gb，环境温度 -20 deg C 到 +50 deg C。
Mechanical Compartment	II 1 G Ex h IIC T4 Ga (-20 deg C <= Ta <= +50 deg C)	机械舱：Ex h IIC T4 Ga，环境温度 -20 deg C 到 +50 deg C。
Probe Assembly	II 1 G Ex ia IIC T4 Ga (-200 deg C <= Ta <= +50 deg C)	探头组件：Ex ia IIC T4 Ga，温度范围 -200 deg C 到 +50 deg C。

测量能力

类别	项目	官方值	中文译文
Level	Range	56 m	液位量程 56 m。
Level	Resolution	1 mm	液位分辨率 1 mm。
Level	Accuracy	+/-2 mm	液位精度 +/-2 mm。
Temperature	Range	-200 deg C to +50 deg C	温度范围 -200 deg C 到 +50 deg C。
Temperature	Resolution	0.01 deg C	温度分辨率 0.01 deg C。
Temperature	Accuracy	+/-0.1 deg C	温度精度 +/-0.1 deg C。
Density	Range	400 to 1000 kg/m3	密度范围 400 到 1000 kg/m3。
Density	Resolution	0.01 kg/m3	密度分辨率 0.01 kg/m3。
Density	Accuracy	+/-0.5 kg/m3	密度精度 +/-0.5 kg/m3。
Profile	Profile Points	250+	Profile 点数 250+。

类别	项目	官方值	中文译文
Sensor Isolation	Internal I.S. Barriers	内部本安栅。	

电气、通信、机械

类别	项目	官方值
Electrical	Operational Voltage	85-240 VAC
Electrical	Certified Operational Voltage	120-240 VAC
Electrical	Frequency	50-60 Hz
Electrical	Power Draw	Imax 3.15 A
Electrical	Overvoltage	Category III
Electrical	Pollution	Degree 1
Communications	Standard	(2) Standard RS485 / MODBUS Protocol
Communications	Optional	4-20 mA, Fiber Optic, HART & Ethernet
Mechanical	Explosion Group of Ex Atmosphere	IIC
Mechanical	Housing Material	Stainless Steel
Mechanical	Dimensions	125 cm x 76 cm x 37 cm
Mechanical	Weight	180 kg
Mechanical	Ambient Temp	-40 deg C to +60 deg C
Mechanical	Safety Certs	IECEx
Mechanical	Protection class	Designed to meet IP65
Mechanical	Seismic Rating	Designed to withstand 3.0 g
Mechanical	Drive Chain	SS304
Mechanical	Mounting	6 inch ANSI 150#RF Flange
Mechanical	Op. Pressure	400 mbar
Mechanical	Stilling Well	Not Required
Mechanical	Isolation Valve	Required
Mechanical	Pinch Valve	Required

内部连接

所有内部连接器的阻燃等级为 V-0。跳过的 designators 已从设计中移除。并非所有连接都会使用，有些连接可能不存在于组件上。

SI-7000 Controller Assembly:

连接器	中文译文
J1	Mains Power Input，通用 AC 输入
J2	Field Wiring Input，RS485
J6	Reel Encoder Input
J8	Motor Power Output
J10	Probe Interface Module，PIM
J14	JTAG Programming，厂家使用
J22	Alternative 28 VDC - 48 VDC Power Input

Probe Interface Module PIM:

连接器	中文译文
J1	SI-7000 Controller Assembly
J2	Slip-ring
P1	Primary Earth Ground
P2	Optional Earth Ground

Part A 警告、注意和通知

Scientific Instruments, Inc. 不对 SI-7000 操作期间造成的任何伤害、死亡或损坏承担责任。客户负责确保所有操作和维护以安全方式进行，并符合政府、地方和主管机构法规。

Warning

原文类别	中文译文
WARNING	必须严格遵守所有适用的政府、地方和主管机构法规。否则可能导致死亡或严重伤害。
WARNING	打开电子外壳盖或拆除机械外壳盖之前，必须通过开关或断路器断开外部电源。否则可能导致死亡、严重伤害或重大财产损失。
WARNING	起吊和移动 SI-7000 到位时，必须确保结构、其他设备和人员之间保持足够间隙。否则可能导致死亡、严重伤害或重大财产损失。
WARNING	存在低温灼伤风险。如果 SI-7000 安装在含 LNG 等低温液体的储罐上，维护时提升探头后，必须给探头足够时间达到环境温度，避免处理探头时发生低温灼伤。通常需要超过 4 小时达到环境温度。当触摸屏主显示上的温度达到环境温度时，打开机械单元应是安全的。
WARNING	SI-7000 不用于氧气服务。
WARNING	存在爆炸性环境时不得打开。气体/空气混合物最大实验安全间隙 MESG >= 0.05 mm。

Caution、特殊使用条件和 Notice

原文类别	中文译文
CAUTION - PINCH POINT	关闭 SI-7000 电子外壳盖时必须小心，所有人员必须让手远离。否则手指可能被夹伤。
Special Conditions of Use	Flameproof joints are not intended to be repaired. 隔爆接合面不应修理。
NOTICE	使用 SI-7000 不涉及消耗性材料。
NOTICE	正常运行期间没有会因高温导致烫伤的表面。如果设备以服务手册未覆盖的方式失效，请在触碰设备前咨询 Scientific Instruments。
NOTICE	探头清洁说明见服务手册。探头是唯一可能需要周期性清洁的 wetted part。驱动链不需要清洁，非 wetted parts 不需要周期性清洁。在仪表设计适用范围内，不需要去污。
NOTICE	必须遵守本手册中的所有操作说明。否则可能损坏 SI-7000 并导致保修失效。

Part B 修订记录和目录

Part B 原始编号为 090-7000-2B Rev A。

修订记录：

Rev	Description	Rev by	Date	Appr by	Date
-	Initial Release	KN	2019-08-21	KN	2019-08-21
A	Updated Photos	JR	2020-11-16	KN	2020-11-16

Part B 目录覆盖：

- Theory of Operation，工作原理；
- Operator Interface，操作员界面；

- Home Screen，主界面；
- System Info，系统信息；
- System Configuration Menu，系统配置菜单；
- Level、Temperature、Density、System、Com Settings、Profile & Zoom Scan Settings、Start Mode；
- Command Menu，命令菜单；
- Motor，电机；
- Technical Assistance，技术支持；
- Disclaimer，免责声明。

工作原理

概述

SI-7000 控制器接收机械单元中传感器的输入，并根据当前运行模式发送信号控制电机，从而控制悬挂在储罐内的多传感器探头组件运动。

探头内两个液位传感器向控制器指示探头是在液体中、蒸气中，还是在液体和蒸气界面。控制器监测温度传感器和密度计信号，持续计算更新信息并传输到主机计算机。

系统有多种运行模式，用于适应维护活动和日常运行。系统可按客户要求设定为定期间隔执行 Profile，以提供罐内状态更新信息。未执行 Profile 时，系统通常设置为跟踪液位，提供最新液位读数。

液位

液位传感器根据其处于液体还是蒸气中产生不同电压。控制器利用这些传感器电压差异判断其状态。通过把液位传感器在竖直方向上相隔一小段距离布置，当下液位传感器处于液体中、上液位传感器处于蒸气中时，控制器可识别该点为液位。

温度

温度传感器是四引线铂元件，在 0 deg C 时电阻为 100 ohm。使用 1 mA 激励电流时，传感器两端产生与其电阻成比例的电压。该电压由模数转换器数字化，并与控制器中的数据表比较。控制器计算相应温度，并随着温度变化持续更新。

密度

密度计及其 associated maintaining amplifier 向控制器输出频率与密度成比例的信号。控制器确定该信号频率，并使用密度转换公式以及特定密度计的校准参数，计算密度计浸没介质的密度。

Bottom Reference Switch

系统使用机械开关组件，在探头接触罐底时向控制器发信号。当控制器感知到 Probe at Reference 条件时，会停止探头向下运动，并把探头位置设置为预编程的底部参考值。底部参考值会计入液位传感器距离探头底部的距离。

Top Reference Switch

第二个机械开关组件用于在探头到达探头腔顶部时通知控制器。当控制器感知到 Probe at Top 条件时，会停止探头向上运动。该功能为维护活动前将探头停放在顶部提供了简便方法。

运行模式

系统可通过触摸屏界面在五种主要模式中运行：Automatic Auto、Calibration、Profile、Zoom Scan 和 Manual。日常运行最常用模式为 Auto 和 Profile。Manual 通常只用于维护活动。Calibration 可在维护活动后使用，或在断电后用于快速重新查找底部参考，再回到液面处 Auto mode，也可在需要重新建立液位时使用。

Automatic Mode，Auto

系统进入 Auto 后，控制器使探头定位并跟踪液体/蒸气界面。当下传感器在液体中、上传感器在蒸气中时，系统认为探头处于界面。Auto 是正常运行模式；所有报警都会报告，如果已设定，Profile run 可自动开始。Profile run 或 calibration run 完成后，系统会返回 Auto。

Calibration Mode，Cal

选择 Cal mode 后，控制器驱动探头到罐底以建立 bottom reference，然后返回液位界面，随后系统回到 Auto。校准最常用于断电后或某些维护活动后。断电时系统保存最后探头位置，但假设断电期间探头可能被手动移动，例如维护活动中，因此上电时系统会标记为未校准。

Profile Mode

Profile mode 从罐底到液位采集温度和密度读数，为操作员提供罐内当前状态的准确表示。

启动 Profile mode 后，探头被驱动到罐底并重新建立 bottom reference。在罐底，探头按编程延迟时间暂停，使探头周围罐内条件稳定。规定延迟后，系统采集位置、温度和密度读数，然后驱动探头向上，按编程增量停靠并采集温度和密度数据。每个测点都有相同延迟，以便读数稳定。Profile 在采集到最大点数或找到液位时结束。

控制 Profile 的参数可通过 LCD 界面输入，也可通过 Modbus 串行接口输入。

采集所有读数后，系统分析数据并按适用条件生成报警。Profile 完成后系统返回 Auto。这样，系统采集并存储完整储罐 Profile，可输出到主机计算机。

NOTE：SI-7000 设计为在液体完全静止和稳定时执行 Profile。不要在储罐装液或卸液时尝试执行 Profile。顶部或底部装载 LNG 都会产生湍流、旋涡和液体不稳定。卸载期间使用大功率压缩机，也会产生低液位湍流和不稳定。为了获得最佳精度和结果，应在装载或卸载过程前一天或后一天安排 Profile。更多信息咨询厂家。

Zoom Scan Mode

该功能当前未实现。

Manual Mode

当需要控制探头并给出特定上升或下降命令时，使用 Manual mode。它最常用于维护活动。探头可以停止而不跟踪液位，也可以按快速、中速或慢速向上或向下驱动。

NOTE：Manual mode 下不会报告任何报警，液位读数也不会更新。必须认识到，系统会无限期保持 Manual mode，直到操作员或维护人员把它返回 Auto。

Power Up Condition

SI-7000 以罐底为参考，因此上电后重新建立 bottom reference 很重要。如果该参考尚未建立，当前探头位置会标记为 Uncal。

系统可设置为断电后自动重新建立 bottom reference，也可设置为上电进入 Manual 并停止电机。该设置位于 LCD 界面 Configuration 区域，显示为 Change to Start in Auto 或 Change to Start in Manual。如果设置为从 Auto mode 启动，系统会先找到罐底，然后返回液位。

如果系统未设置为自动重新建立 bottom reference，只需启动 Cal run，即进入 Calibration Mode。探头会移动到罐底，重新建立 bottom reference，然后返回液位并进入 Auto mode。

操作员界面

系统状态信息通常通过 Modbus 协议传输到控制室中的主机计算机或 DCS。可用信息包括当前运行模式、探头运动方向和速度、当前位置、液位、温度和密度、Profile 数据，以及各种状态指示和报警。由于每个站点可选 DCS 操作员界面的编程和实现不同，以下描述的某些功能可能不会出现。以下章节描述 SI-7000 LCD 屏幕上的可用内容。

Home Screen

Home screen 是仪表上电后的默认显示。该屏显示液位、温度、密度和其他关键参数。

当前位置、温度和密度值是探头当前所在位置测得的值。如果密度值非常低，例如为 0 或等于 0，探头可能处于蒸气中而不是液体中。

如果上电后已经建立液位，当前液位值也应显示。如果尚未找到液位，该值会为 0 或最后记录值。注意，只有探头实际位于液位时，才能保证液位读数正确。这由下文的 At Liquid Level 指示。除此之外，要保证准确液位，还不应有 Probe Uncal 状态指示。

基本状态包括上、下液位传感器、底部参考开关和顶部限位开关。底部开关跨接在下液位传感器上，顶部限位开关跨接在上传感器上。因此每个传感器有三种可能状态：

状态	中文译文
Closed	对应开关闭合，传感器短接，因此无法指示液体或蒸气。
Liquid	对应开关打开且传感器在液体中。下传感器显示为 L-LIQ，上传感器显示为 U-LIQ。
Vapor	对应开关打开且传感器在蒸气中。下传感器显示为 L-VAP，上传感器显示为 U-VAP。

当探头完全浸没时，两个传感器都应在液体中。如果探头完全离开液体，两个传感器都应显示蒸气。如果探头位于界面，即液位处，下传感器应在液体中，上传感器应在蒸气中。

SI-7000 相对于罐底测量液位，因此要报告准确液位，必须先建立罐底位置。如果该参考尚未建立，当前探头位置会标记为 Uncal。系统断电时保存最后位置；但由于维护人员可能在断电期间物理移动探头及其 cable，因此系统首次上电时探头位置总是标记为 Uncal。必须重新建立 bottom reference 才能清除该条件。

如果探头位置下降到先前找到 bottom reference 的位置以下，而没有在那里再次找到 bottom reference，Probe Uncal 指示也会点亮。

Home Screen 报警说明

SI-7000 提供多个报警，用于提醒操作员异常或潜在危险条件。有些报警与当前条件有关，有些与 Profile run 期间发现的罐内条件有关。

Reel Alarm：如果 Reel Alarm 指示亮，表示机械单元 reel 未转动，可能指示驱动机构机械故障或罐内障碍。维护人员应调查。

Probe at Liquid Level：系统处于 Auto mode 并位于液位时，该指示应点亮。如果该指示不亮，操作员无法确认当前液位读数正确。如果探头在液面以下或在蒸气区，就无法知道液位是否正在变化。

Level Alarms：与当前条件有关的最重要报警参数可能是液位。液位报警包括 low level、low low level、high level 和 high high level。每个报警设定值应由运行人员确定并相应编程。

Temperature and Density Alarms：其他当前条件报警包括高低温报警和高低密度报警。在多数环境中，这些报警不是关键报警，因为通常用来检测储罐 Profile 中温度和密度异常变化的是 deviation alarms，而不是高低报警。

Deviation Alarms：Profile run 期间，在每个编程停靠点采集温度和密度数据。运行结束后分析数据点。如果相邻两个点的温度偏差大于编程报警值，则设置 temperature deviation alarm。类似地，如果相邻点密度变化大于报警阈值，则设置 density deviation alarm。这些报警在下一次 Profile run 执行并验证报警条件不再存在之前不会清除。

Deviation alarms 非常重要，因为密度或温度变化可能表示 layering 或 stratification；如果不检查，可能导致潜在 rollover 条件。

Alarm Functionality in Manual Mode：必须理解，当系统处于 Manual Mode 时，即使存在报警条件，也不会报告报警。系统必须处于 Auto mode 才会报告报警。Profile 或 Cal run 后系统会立即切换到 Auto mode。

System Info

System Info 屏显示软件版本和 Tank ID。

System Configuration Menu

在主界面按 CONFIG 会打开 System Configuration Menu。该菜单允许访问可通过 LCD 触摸屏显示修改的参数。

配置菜单

Level

LTD 的主要功能之一是测量储罐液位。如果液位低于低限或高于高限，系统会生成报警。这些报警的设定值在 Level 菜单中设置。

修改数值时，按 Edit Parameter 按钮。新屏幕会打开并允许修改数值。触摸蓝色方块，然后输入期望 reference。所有液位输入后，点击 enter 并保存。使用右箭头访问第四个参数。

Temperature

SI-7000 中的温度传感器安装在密度计内部。每个传感器理论上是 100 ohm platinum resistance thermometer PRT，但电阻可能与理论值略有差异。每个传感器发货前都会执行校准流程，并在冰点和氮气温度下测量准确值。结果值包含在每个密度计的校准证书中。

SI-7000 电子电路经过校准，可准确测量电阻，因此输入所用传感器的正确值后，仪表可在系统规格规定公差内显示温度。

项目	官方值
Temperature Measurement Range	-200 deg C to +50 deg C
Error over-range	999.99 deg C
Error under-range	-555.55 deg C

Low Temperature 和 High Temperature 报警可由客户确定设定值并触发，设定值以 Kelvin 输入。

Deviation alarm 是非常重要的报警，用于判断储罐是否分层。在 Profile 中，如果相邻两点之间差异大于 Deviation alarm limit，会触发 Temperature Deviation TD alarm。该报警保持有效，直到执行另一次 Profile 并清除条件。

Density

密度计是经过校准的设备。每台发货前都会执行校准流程，并确定三个系数，这些系数包含在每个密度计的校准证书中。

SI-7000 电子电路设计为可准确测量频率，因此输入密度计的正确值后，仪表可在规格公差内显示密度。

项目	官方值
Density Measurement Range	400 kg/m3 to 600 kg/m3
Error over-range	9999.99 kg/m3

Low Density 和 High Density 报警可由客户确定设定值并触发，设定值以 kg/m3 输入。

Deviation alarm 用于判断储罐是否分层。在 Profile 中，如果相邻两点差异大于 Deviation alarm limit，会触发 Density Deviation DD alarm。该报警保持有效，直到执行另一次 Profile 并清除条件。

System

System Menu 提供多个子菜单。

Bottom Reference：SI-7000 相对于罐底测量液位，每次 Cal run 和 Profile 开始时都会触底。当探头触碰罐底时，电机 pulse count 设为 0。下液位传感器距离探头底部的高度会加到 pulse count 上，以建立下液位传感器的准确位置。修改 bottom reference 时，触摸蓝色方块，输入期望 reference，点击 enter 并保存。

Tank ID：对于多罐安装，Tank ID 用于区分不同储罐。它也是 Modbus Address。修改 Tank ID 或 Address 时，触摸蓝色方块，输入期望 ID 或 Address，点击 enter 并保存。

Time/Date：SI-7000 保存日期和时间记录，并在断电后维持时间最长一个月。如果控制器断电超过一个月，可能需要重新设置当前日期和时间。修改日期和时间时，按对应按钮；新屏幕打开后设置日期，按 Next 后设置当前时间，完成后按 Done。

内部使用且需要密码的项目：

- Level Sensor Cal Closed
- Microcontroller Cal
- Level Sensor Cal Liquid

Units of Measure (Metric/English)：该按钮可切换计量单位。如果系统当前为公制，按钮显示切换到 English 的选项；如果系统当前为 English units，按钮显示切换到 metric 的选项。

100 OHM CAL：选择该菜单时会显示确认信息 Calibrate at 100 OHM?。这是内部使用项目，只应在技术支持部门指示时执行。按 No 可退出菜单而不校准。

Nitrogen Temperature Cal：选择该菜单时会显示确认信息 Calibrate at Nitrogen temperature?。这是内部使用项目，只应在技术支持部门指示时执行。按 No 可退出菜单而不校准。

Level Scale：液位公式如下：

```
Level = Motor Pulses * Level Scale + Bottom Reference
```

因此，Level Scale 是把 pulse 与实际探头位置关联起来的校准值。该值在厂家执行校准后确定，并记录在站点专用 Configuration Worksheet 上。此处显示值应与 Configuration Worksheet 中的值匹配。

Interlock：Interlock 是探头运动高限。通常设置为使探头保持在储罐冷区。如果 DCS 或 PC 用户界面中的 Interlock 指示亮，表示探头已到达允许最高位置。除非探头被手动提升到该位置，否则这是异常情况，维护人员应调查。修改 interlock 时，触摸蓝色方块，输入期望 interlock，点击 enter 并保存。

Com Settings

该菜单当前不支持。

Profile & Zoom Scan Settings

该菜单有两个子菜单。

Profile Setup：SI-7000 最重要功能之一是在罐内不同点采集数据。Profile 从罐底开始，沿向上方向采点，直到达到最大点数或找到液位。该菜单允许自定义要采集的 Profile。

Profile 的实际第一个点在罐底采集。第一个可编程点由 First Point 字段指定。探头离开罐底后，会先在该点停止，然后加上 Increment 计算下一个停靠点。Dwell Time 是探头在每个点等待后再采集读数的时间，单位为 milliseconds。Interval 是下一次 Profile 采集前的时间，单位为 minutes。

时间设置屏幕可选择 Profile 开始的准确时间。注意，如果编程时间到达时系统处于 Manual，Profile run 不会开始；如果系统处于 Auto、Calibrate 或 Profile，则会开始。

Zoom Scan Setup：该菜单当前未实现。

Start Mode，Auto/Manual

系统上电时的状态可编程。它可以保持 Manual Mode 等待指令，也可以直接进入 Auto Mode 并测量液位。维护期间通常设置为保持 Manual；正常运行时可改为以 Auto 启动。

如果当前设置为 Manual 启动，按钮会显示改为 Auto 启动的选项。如果设置为 Auto 启动，按钮会显示改为 Manual 启动的选项。

Command Menu

Command Menu 允许改变当前 LTD 模式。

按钮	中文译文
Manual Mode	使探头进入 Manual Stop。系统会无限期保持在该模式，等待操作员命令。
Level Calibration	探头移动到罐底以重新建立 bottom reference，然后返回液位。
Auto Mode	使 LTD 返回正常 Auto mode，持续跟踪液位。
Start Profile	启动新 Profile。探头移动到罐底，然后返回液位，在编程停靠点采集数据。
Zoom Scan	当前未实现。
Back	返回上一菜单。

Manual Mode

Manual mode 下探头由操作员控制。按 Up arrow 一次，探头开始慢速向上移动；再次按下变为中速；第三次按下变为高速。

同样，按 down 一次、两次或三次，使探头分别以慢速、中速或快速向下移动。

改变方向前应按 Stop 按钮，以避免链条和探头受到冲击。

Level Calibration

按 Level Calibration 后，探头开始向下移动到罐底。到达罐底后，会返回液位。按下 calibration 按钮后，显示返回正常显示。

Auto Mode

按 Auto Mode 会使 LTD 进入 Auto，探头会从当前位置返回液位，并持续跟踪液位。按下 Auto Mode 按钮后，显示返回正常显示。

Auto 中，探头通常以慢速驱动以跟踪液位，但可按需要加速。

Start Profile

按 Start Profile 后，探头开始向下移动到罐底。到达罐底并在第一个点采集数据后，探头开始向上移动，并沿途在编程停靠点停止采集数据。到达液位时采集最后一个点，Profile 完成。按下 Start Profile 按钮后，显示返回正常显示。

Motor

按 MOTOR 会打开一个屏幕，显示与电机驱动功能和其他系统参数相关的详细诊断信息。

技术支持和免责声明

技术支持联系方式：

项目	内容
Company	Scientific Instruments
Address	4400 W. Tiffany Dr., West Palm Beach, FL 33407, U.S.A.
Telephone	+1 (561) 881-8500
Facsimile	+1 (561) 881-8556
Email	support@scientificinstruments.com
Website	www.scientificinstruments.com

免责声明翻译：

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维修手册与保修信息中文翻译

来源文件：../00_原始资料/SI7000官方资料包_含Modbus规范.pdf 第 86-101 页

原始标题：

- SI-7000 Tank Gauging System Service Manual, 090-7000-3 Rev -, PDF 第 86-100 页
- Systems Warranty Policy, CSF0030 Rev -, PDF 第 101 页

整理日期：2026-07-02

维修手册封面和目录

维修手册封面信息：

项目	内容
系统	SI-7000 Tank Gauging System
文件	Service Manual
公司	Scientific Instruments, Inc.
地址	4400 W. Tiffany Dr., West Palm Beach, FL 33407 U.S.A.
文件号	090-7000-3 Rev -

维修手册目录翻译：

原章节	中文译文
1 Introduction	1 引言
2 Warnings, Cautions, and Notices	2 警告、注意和通知
3 Overview	3 概述
4 Guidelines for Maintenance	4 维护指南
4.1 Preparation	4.1 准备
4.2 Possible Materials Needed	4.2 可能需要的材料
4.3 Calibration of Density Meter	4.3 密度计校准
4.4 Calibration of Temperature Measurement	4.4 温度测量校准
4.5 Opening Electrical Enclosure Door	4.5 打开电气外壳门
4.6 Closing Electrical Enclosure Door	4.6 关闭电气外壳门
4.7 Mechanical Enclosure	4.7 机械外壳
5 Periodic Maintenance Procedure	5 周期性维护流程
6 Basic Functionality Testing	6 基本功能测试
6.1 Power Up	6.1 上电
6.2 Motor Driver Test	6.2 电机驱动测试
6.3 Checking Level Calibration	6.3 液位校准检查
6.4 Communications Test	6.4 通信测试

引言和系统分区

SI-7000 Tank Gauging System 用于监测大型储罐内低温液体的液位、温度和密度，通常由控制室中的 PC 操作员界面和/或 DCS 用户界面控制和监视。

系统由机械部分和电气部分组成：

部分	官方说明	中文译文
Mechanical section	Chain reel enclosure and probe enclosure	机械部分包括链轮卷筒外壳和探头外壳。除 drive shaft gear box 外，升降储罐内探头组件所需的机械部件都在该部分内。官方要求该部分视为 Zone 0 环境。基础维修通常不需要进入该部分。
Electrical section	Electrical enclosure	电气部分包括电气外壳，内含电子部件和 drive shaft gear box。该外壳是 explosion-proof container，官方要求视为 Zone 1 环境。SI-7000 的输入电源和通信线缆在这里进行物理连接。

维修手册说明，更详细的组件功能解释应参考 SI-7000 Operational Manual，文件号 090-7000-2。探头外壳安装在链轮卷筒外壳底部，链轮卷筒外壳与电气外壳背靠背连接。

图示翻译：

图号	原图标题	中文译文
Figure 1	Chain reel & probe enclosures	链轮卷筒和探头外壳
Figure 2	Chain reel enclosure w/o Cover	未安装盖板的链轮卷筒外壳
Figure 3	SI-7000 Side View	SI-7000 侧视图
Figure 4	Electrical Enclosure	电气外壳
Figure 5	Electrical Enclosure w/ door open	电气外壳门打开状态
Figure 6	SI-7000 transparent side view	SI-7000 透明侧视图

本维修手册提供在典型现场维修系统所需的信息。随手册可能同时提供现场专用机械安装图和现场接线图，用于说明典型现场细节。其他活动的详细说明应参考 SI-7000 Operational Manual。

警告、注意和通知

Scientific Instruments, Inc. 不对 SI-7000 操作和维修期间造成的任何伤害、死亡或损坏承担责任。客户负责确保所有操作和维护以安全方式进行，并符合政府、地方和主管机构法规。

Warning

原文类别	中文译文
WARNING	必须严格遵守所有适用的政府、地方和主管机构法规。否则可能导致严重伤害和/或死亡。
WARNING	打开电子外壳盖或拆除机械外壳盖之前，必须通过开关或断路器断开外部电源。否则可能导致严重伤害、死亡或重大财产损失。
WARNING	存在低温灼伤风险。如果 SI-7000 安装在含 LNG 等低温液体的储罐上，维护时提升探头后，必须给探头足够时间达到环境温度，避免处理探头时发生低温灼伤。通常需要超过 4 小时达到环境温度。当触摸屏主显示上的温度达到环境温度时，打开机械单元应是安全的。
WARNING	存在爆炸性环境时不得打开。气体/空气混合物最大实验安全间隙 MESG >= 0.05 mm。

Caution 和 Notice

原文类别	中文译文
CAUTION - PINCH POINT	关闭 SI-7000 电子外壳盖时必须小心，所有人员必须让手远离。否则手指可能被夹伤。
NOTICE	使用 SI-7000 不涉及消耗性材料。
NOTICE	正常运行期间没有会因高温导致烫伤的表面。如果设备以服务手册未覆盖的方式失效，请在触碰设备前咨询 Scientific Instruments。
NOTICE	探头清洁说明见本维修手册。探头是唯一可能需要周期性清洁的 wetted part。驱动链不需要清洁，非 wetted parts 不需要周期性清洁。在仪表设计适用范围内，不需要去污。原文使用 whetted / un-whetted，此处按 wetted / unwetted 理解。
NOTICE	必须遵守本手册中的所有操作说明。否则可能损坏 SI-7000 并导致保修失效。
NOTICE	必须遵守本手册中的所有维修说明。否则可能损坏 SI-7000 和/或附近其他设备。

原文类别	中文译文
NOTICE	所有维修工作必须由合格人员执行。
NOTICE	保险丝必须由合格人员更换。
NOTICE	维修期间必须执行 Lock-out Tag-out 程序。

维护概述和维护指南

概述

一般情况下，系统检查和例行维护会从控制室开始，此时探头仍在液体中。这样可以在探头离开液体前观察系统性能和现有问题，便于探头离开液体后采取纠正措施。

完成该检查后，将探头提升到探头外壳并等待升温。在系统升温期间，Mechanical Unit 不能打开，但 Electronic Unit 可在此期间检查。当探头达到环境温度后，可以打开机械单元并检查内部组件。

现场图纸

维护现场接线和安装细节应参考 site specific drawings。

准备

开始维护流程前，建议检查是否需要固件更新。如果需要固件更新，可能需要先完成固件更新。

可能需要的材料

- Vacuum grease，真空润滑脂。
- Grease for pinch valve，Pinch Valve 用润滑脂。
- Can of compressed air or clean instrument air，压缩空气罐或洁净仪表空气。

校准和外壳操作指导

密度计校准

部分客户关注密度测量的绝对精度。对这类客户，官方建议每 3-5 年将密度计返厂校准一次，具体周期取决于客户对绝对精度的关注程度。

用于检测分层 stratification 时并不要求绝对精度。为检测分层，系统只需要观察储罐内相对密度差异。

密度计校准不包含在正常例行维护流程中。如需细节，应联系厂家。

温度测量校准

温度测量校准由两部分组成：

1. 校准电子电路。
2. 校准温度传感器。

由于温度传感器位于密度计内部，当密度计返回 Scientific Instruments 时，可同时请求校准温度传感器。

只有在具备正确测试设备时，才可在现场执行温度测量电子电路校准。所需设备包括 decade box、precision DVM 和连接电缆。具体细节应联系厂家。

温度校准也可每 3-5 年执行一次，具体取决于客户对温度测量绝对精度的关注程度。通常情况下绝对精度不是关键；分层检测依靠观察温度的相对变化。

温度测量校准不包含在正常例行维护流程中。如需细节，应联系厂家。

打开电气外壳门

执行维护活动时，需要打开电气外壳门。进入电气外壳工作时必须极其小心，避免划伤或损伤 flame path 表面。任何划伤或损伤都会使外壳的 explosion proof certification 失效。

步骤：

1. 使用随附 Allen wrench 松开固定外壳门的所有螺栓。
2. 完全打开门。
3. 插入 pin，将门固定在打开位置。

关闭电气外壳门

维护活动完成后，应关闭电气外壳门。关闭前同样必须避免划伤或损伤 flame path 表面。

关闭电气外壳前应执行基本功能测试。如果与控制室通信可用，也应同时测试通信。这样可以在盖板关闭前发现潜在问题。

步骤：

1. 沿外壳侧 flame path 一周涂一圈薄的 silicone vacuum grease。
2. 取下用于保持门打开的 pin。
3. 完全关闭门并插入所有螺栓。
4. 使用随附 Allen wrench 拧紧外壳门上的所有螺栓。
5. 螺栓扭矩为 18-20 lb-ft，即 24.5-27 N*m。
6. 擦掉从 flame path 挤出的任何硅脂。

机械外壳

本节提供机械外壳相关工作指导：

原文类别	中文译文
WARNING	Shafts and bushings 属于 flameproof joints，不应修理。如需替换这些部件，应联系 Scientific Instruments。
NOTICE	轴承应在使用 10 年后更换，该时间对应额定寿命的 90%。预期使用工况为每天连续机械运动 2 小时。

周期性维护流程

以下表格按官方 Periodic Maintenance Procedure 翻译。该流程是维修手册的核心内容，执行时应同时遵守现场安全规程和厂家要求。

On Tank，在罐上检查

活动	备注
将当前可编程参数值与 Configuration Worksheet 比较。	记录并解决任何差异。
探头仍在液体中时，观察系统在 Auto 且位于液位处的运行情况。	除非储罐内因泵送动作产生湍流，否则探头应稳定在液位处：下传感器在液体中，上传感器在蒸气中。持续 hunting 表示需要把液位传感器间距调大。
检查 Motor 屏幕并确认所有参数正常。观察是否存在报警。	无额外备注。
检查主界面上的温度和密度值是否合理。	对 LNG，温度通常在 -157 deg C（-251 deg F）到 -163 deg C（-261 deg F）之间，密度通常在 425 kg/m3（26.5 lb/ft3）到 470 kg/m3（29.3 lb/ft3）之间。这只是功能检查。温度重新校准和密度重新校准更复杂，不包含在此流程中。
在 Manual 下将探头驱动到储罐底部。	无额外备注。
稍微提升探头，并确认两个传感器都指示液体。	只需要稍微提升探头。Bottom reference switch 不得闭合，因为它会短接下液位传感器。
启动 level calibration，并检查系统是否正确找到液位。	探头到达液位时应快速停止。
驱动探头上升，直到在 upper limit switch 处停止。探头到达顶部后禁用驱动机构。	禁用驱动机构可防止探头因误操作再次向下进入液体。
打开机械单元前，等待探头升温到环境温度。期间执行电子单元检查流程。	温暖天气下，探头约 4 小时达到环境温度。机械单元在此期间不能打开，但电子单元可以打开。探头升温期间不必关闭 Pinch Valve。

Pinch Valve

活动	备注
确认探头位于 top limit switch 后，检查 Pinch Valve 是否动作顺畅。此时不要关闭 Pinch Valve。	关闭 Pinch Valve 前必须确认探头顶部位于 upper limit switch，否则可能损坏探头。将手轮顺时针旋转 5 圈，再逆时针旋转 5 圈。如果配有 grease fitting，应使用优质润滑脂润滑。

Electronic Unit Checks，电子单元检查

活动	备注
检查驱动到顶部时记录的数值。牢固关闭 Pinch Valve。	Top stop value 应与 Configuration Worksheet 中的值基本一致。
驱动探头向下接触 Pinch Valve。	该步骤验证环境温度下的 bottom reference 动作。
让探头停靠在 Pinch Valve 上。	这可确保打开系统时 cable 有足够松弛量。
断电。	无额外备注。
检查电子单元外壳内端子是否无腐蚀、无异物且固定可靠。	检查连接器和端子螺钉是否紧固，并确认无腐蚀。
检查是否有部件劣化迹象。	无额外备注。
检查外露硬件，确保紧固。	无额外备注。
检查电机连接，确保连接紧固。	无额外备注。

Mechanical Unit，升温后的机械单元检查

活动	备注
松开机械单元盖板螺栓，使系统泄压。	如果 15 秒后气流仍未停止，将 Pinch Valve 再关紧一点。
拆下盖板并对内部组件进行目视检查。	无额外备注。
拆下适用硬件，将探头从外壳中提起，放到合适工作台面上。	无额外备注。
拆下 probe shroud、switch assembly 和 screen。	无额外备注。
检查密度计，确保清洁、干燥并牢固固定在 probe cone 上。	如有必要，用压缩空气清洁。如果必须使用液体清洗剂，探头上的任何水分都必须在探头返回液体前干燥。
检查液位传感器，确保无碎屑和异物。	无额外备注。
记录密度计 spool number。	检查它是否与 Configuration Worksheet 匹配。
重新安装 probe shroud、switch assembly 和 screen。	无额外备注。
检查所有部件动作是否正常，尤其是 bottom switch。	Bottom switch 正常工作非常重要，必须仔细确认。
将探头放回探头外壳并重新安装硬件。	无额外备注。
检查链条和链轮是否过度磨损。	如果出现严重磨蚀迹象且曾报告驱动问题，应对整条驱动链进行完整检查。注意：链条必须正确处理，否则检查期间可能受到不利影响。
清洁驱动机构内部并清除异物。	无额外备注。
检查 decoder 的 reel feedback signal。	无额外备注。

Closing Mechanical Unit，关闭机械单元

活动	备注
在 gasket 上涂少量 vacuum grease，在盖板螺栓上涂 anti-seize compound。重新安装盖板。	无额外备注。
如果系统已上电，则断电。	下一步涉及放空气体。该步骤期间不得有外露带电电子部件。
按现场规程对仪表进行 purge。	仪表有 lower purge port 和 upper purge port。
Purge 完成后，确保所有 purge openings 固定且拧紧。	检查盖板接合处周围是否无明显气体泄漏。如果泄漏无法停止，可能需要更换机械单元 gasket。

活动	备注
当区域内无残留气体时，打开 Electrical Unit，或重新上电。	无额外备注。
完全打开 Pinch Valve，并在 Manual 下驱动系统向下。	使用高速。无需在液体界面处停止。驱动期间监测系统，确认运行平稳。
稍微提升探头，并确认传感器读取为液体。	无额外备注。
启动 level calibration。	这是让探头返回液位的最快方式。等待 Cal Run 完成。
探头到达液位后，确认它快速停止。	无额外备注。

Closing Electronic Unit，关闭电子单元

活动	备注
在电子单元密封面涂少量 silicon grease，以防环境影响。	无额外备注。
在 socket bolts 上涂 grease 或 anti-seize compound，并关闭电子单元。	无额外备注。
执行一次 Profile。如果用户界面允许，可从控制室启动。	这将向主机计算机提供密度和温度 Profile 值。
更新 Configuration Worksheet。	确保所有数值都是最新的。

Optional PC Interface，可选 PC 接口

活动	备注
如果 SI 提供了 PC，检查通信电缆是否固定可靠。	无额外备注。
执行备份等工作。	无额外备注。
如有历史 Profile，检查过去的 Profile。	等待探头完成运动期间，是检查 Profile 历史的合适时间。查看 Profile 数据中是否有异常。

基本功能测试

Power Up，上电

1. 确保 SI-7000 电源开关处于 OFF 位置。
2. 解除 AC power 的 lock out/tag out。
3. 将 SI-7000 电源开关切换到 ON 位置。
4. 检查主控制器上的以下项目：

- 红色 A/C power LEDs 应点亮。如果未点亮，确认 AC power 已供给 controller assembly，并确认电源插头已完全插入插座。
- 绿色 FUSE LEDs 应点亮。如果未点亮，检查 AC plug module 中的保险丝，套件中包含备用保险丝。
- 绿色 PWR LED 应点亮。如果未点亮，最可能是 DC power supply 不工作，需要更换 controller assembly。
- LCD 应点亮并显示主菜单。如果未点亮，拆下 controller assembly 并确认 LCD 完全连接；如果仍无法成功，则 LCD 或 CPU 不工作，需要更换 controller assembly。

5. 继续下一项测试，或将 SI-7000 电源开关切换到 OFF 位置。

Motor Driver Test，电机驱动测试

1. 确保 SI-7000 电源开关处于 ON 位置。
2. 在触摸屏 LCD 上进入 Motor 菜单。
3. 触摸 down arrow 一次，电机应开始以最低速度降低探头组件。
4. 第二次触摸 down arrow，电机应提高到中速。
5. 第三次触摸 down arrow，电机应提高到最高速度。
6. 第四次触摸 down arrow，电机应回到中速。

7. 触摸 stop button，电机应停止并保持探头在当前位置。
8. 触摸 up arrow 一次，电机应开始以最低速度提升探头组件。
9. 第二次触摸 up arrow，电机应提高到中速。
10. 第三次触摸 up arrow，电机应提高到最高速度。
11. 第四次触摸 up arrow，电机应回到中速。
12. 允许电机将探头提升到 parked position，电机应自动停止并保持探头在 parked position。

Checking Level Calibration，液位校准检查

1. 确保 SI-7000 电源开关处于 ON 位置。
2. 在触摸屏 LCD 上进入 Motor 菜单。
3. 触摸 down arrow 一次，电机应开始以最低速度降低探头组件。
4. 通过探头外壳窗口观察链条运动，当链条上指定标记与参考点对齐时停止电机，并记录 step count。电机应停止并保持探头在当前位置。
5. 触摸 down arrow 三次，电机应开始降低探头组件。
6. 通过探头外壳窗口观察链条运动，当链条上第二个标记与参考点对齐时停止电机，并记录第二个 step count。
7. 计算 level scale。细节咨询厂家。
8. 触摸 up arrow 三次，电机应开始提升探头组件。
9. 允许电机将探头提升到 parked position，电机应自动停止并保持探头在 parked position。

Communications Test，通信测试

1. 确保 SI-7000 电源开关处于 ON 位置。
2. 确保绿色 COM LED 点亮。
3. 从控制室发送降低探头的命令，电机应开始降低探头组件。
4. 从控制室发送停止电机的命令，电机应停止并保持探头在当前位置。
5. 从控制室发送提升探头的命令，电机应开始提升探头组件。
6. 允许电机将探头提升到 parked position，电机应自动停止并保持探头在 parked position。

系统保修政策

50 YEARS of SCIENTIFIC INSTRUMENTS 1967-2017

Systems Warranty Policy

Scientific Instruments, Inc. (SII) warrants its LTD tank gauging system against all defects in material and workmanship for a period of (24) months from the date of shipment of goods from SII dock, at 4400 West Tiffany Drive, West Palm Beach, Florida.

SII shall not be liable for damages caused by raw materials, designs of work methods instructed by the Purchaser or incompatibility of the Purchaser's equipment with SII products. SII shall not be liable for damages caused by negligence of a contracting party or a third party and not for damages caused by the use of SII product for other purposes than it is intended.

Sub-assemblies not of SII manufacture, but as part of an SII system, such as software, printers, terminals, modems, etc., are warranted in accordance with the terms of sales as extended by the product supplier.

SII shall not be liable for indirect losses and pure financial losses, lost profit or other consequential economic losses.

Should a defect become apparent within this warranty period, SII or properly trained and certified personnel will perform field repair at the customer's facility. Transportation costs and living expenses for the required field service personnel shall be borne by the customer, at cost.

In the event of failure of sub-assemblies or components which can be physically returned to SII, all warranty repairs/replacements will be made at no charge, provided that the customer pays all freight costs, foreign duties, etc., incurred in returning such sub-assembly or components to SII. Any U.S.A. duties or fees imposed upon products entering this country will be paid by SII.

This warranty policy is considered null and void if the installation and commissioning of the system is not performed by SII personnel, or personnel formally trained and certified by SII. A Certificate of Training will be issued by SII when independent contractors or consultants have successfully completed such training.

Authorized by: Leigh Ann Hoey

Name/Title: Leigh Ann Hoey/President

4400 West Tiffany Drive, West Palm Beach, Florida, USA
www.scientificinstruments.com
Ph: 561-881-8500 or 800-466-6031

CSF0030 Rev. - 7/01/18

系统保修政策

PDF 第 101 页为 Systems Warranty Policy，文件标识 CSF0030 Rev -，页面日期 7/01/18。该页为扫描件，已保留原图：

保修条款中文译文如下：

原文主题	中文译文
保修范围和期限	Scientific Instruments, Inc. SII 对其 LTD tank gauging system 的材料和工艺缺陷提供保修。保修期为货物从 SII 位于 4400 West Tiffany Drive, West Palm Beach, Florida 的 dock 发运之日起 24 个月。
不承担责任的情形	SII 不对因买方指定的原材料、工作方法设计、买方设备与 SII 产品不兼容造成的损害负责。SII 也不对合同方或第三方的疏忽造成的损害负责，也不对把 SII 产品用于非预期用途造成的损害负责。
非 SII 制造的子组件	作为 SII 系统一部分但非 SII 制造的子组件，例如软件、打印机、终端、调制解调器等，按产品供应商延伸出的销售条款提供保修。
间接损失	SII 不对间接损失、纯经济损失、利润损失或其他 consequential economic losses 负责。
现场维修	如果缺陷在保修期内显现，SII 或经过适当培训和认证的人员将在客户现场执行现场维修。所需现场服务人员的交通费用和生活费用由客户按成本承担。
可返修子组件	如果故障子组件或部件可以实物返回 SII，且客户承担将该子组件或部件返回 SII 产生的所有运费、外国关税等费用，则所有保修维修/更换免费完成。产品进入美国时产生的任何美国关税或费用由 SII 支付。
保修失效条件	如果系统安装和调试不是由 SII 人员执行，或不是由 SII 正式培训和认证的人员执行，则本保修政策视为无效。独立承包商或顾问成功完成培训后，SII 会签发培训证书。
授权签署	Authorized by: Leigh Ann Hoey，Name/Title: Leigh Ann Hoey/President。

原文主题	中文译文
联系方式	4400 West Tiffany Drive, West Palm Beach, Florida, USA；www.scientificinstruments.com；Ph: 561-881-8500 or 800-466-6031。

译注：保修政策与操作手册中的免责声明一致，都强调安装、调试和维护必须由 SII 或经 SII 培训认证的人员执行，否则可能导致保修失效。

附录 A 原始页面图片对照

以下页面图像用于对照原始图纸、照片、证书、扫描页和表格版式。

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SCIENTIFIC INSTRUMENTS

SI-7000 Online Informational Catalog

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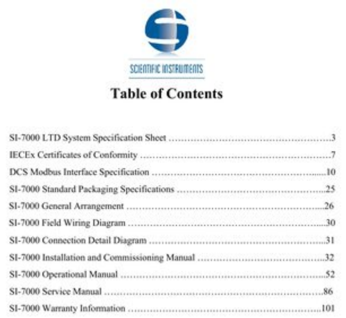

SCIENTIFIC INSTRUMENTS

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SI-7000 Specifications

Level
- Range: 56 m (longer check w/factory)
- Resolution: < 1 mm
- Accuracy: < ±2 mm
- Top and Bottom reference

Temperature
- Sensor Range: -200°C to +50°C
- Resolution: 0.01
- Accuracy: ±0.1°C

Density
- Oscillating Spool
- Range: 400 to 1000 kg/m3
- Resolution: 0.01 kg/m3
- Accuracy: ±0.5 kg/m3

Electrical
- Power Supply
- 85-264 VAC, 47-63 Hz, 150 W

Communications
- (2) Standard RS-485 MODBUS

Sensor Isolation
- Intrinsically Safe

Physical
Maximum Operating Pressure
- 400 mbarg (for higher, consult factory)

Protection Class
- Designed to meet IP65

Mounting Flange
- Standard 6" ANSI 150#RF (more available upon request)

Sensor Drive
- Stainless Steel Chain

Housing Material
- 316 Stainless Steel

Safety Certification
- IECEx / ATEX , cml Ex , PESO, UL / CSA

Seismic Rating
- Designed to meet 3.0G (testing required)

Stilling Well
- With or Without (Consult Factory)

Isolation Valves
- External Pinch Valve (Required)

Ambient Temperature
- - 40°C to +60°C

Dimensions
- 125 (50) × 76 (30) × 37 (15) cm (in.)
- Weight: 180 (396.8) kg(lbs.)

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IECEx Certificate of Conformity

INTERNATIONAL ELECTROTECHNICAL COMMISSION
IEC Certification Scheme for Explosive Atmospheres

eurofins | E&E

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IECEx Certificate of Conformity

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IECEx Certificate of Conformity

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SCIENTIFIC INSTRUMENTS

SI-7000
Tank Gauging System

DCS Modbus
Interface Specification

SCIENTIFIC INSTRUMENTS, INC.
4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
020-701 Rev B

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SCIENTIFIC INSTRUMENTS, INC.
4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
020-701 Rev B

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1 General Description

This document defines a limited implementation of the Modbus protocol for the Scientific Instruments SI-7000 LTD. This document should provide programmers with the details necessary to prepare and test the user interface on a host computer.

The LTD is a tank gauging system designed to measure level, temperature and density in large tanks of cryogenic liquids (e.g. liquid natural gas). A probe is driven up and down in a tank to collect temperature and density data, as specified by the operator. The data is stored in memory for output to the host computer on demand. The programmer should read and be familiar with the Operation Manual for the LTD, as the basic system operation information is not duplicated here.

In the general Modbus specification, certain characteristics are fixed, such as frame format and frame sequence. Other characteristics are variable, including the type of interface (e.g. RS232, RS485), baud rate, parity, number of stop bits and transmission mode (ASCII or RTU). In the LTD implementation of the Modbus interface, the interface type is RS485. (To interface with the host computer using RS232, a converter must be used.) The baud rate is fixed at 9600 baud, and the other parameters are odd parity, 8 data bits, and 1 stop bit. The transmission mode used is RTU.

The Modbus interface allows the host computer access to the LTD's current mode of operation, the direction and speed of the probe, individual system status and alarm bits, profile data, and current tank conditions. The mode of operation can be controlled via Modbus, and the probe can be driven manually at several different speeds.

Output of level, temperature, and density will be scalable in future firmware versions.

In the last section, programming suggestions are given to assist the programmer in handling important information and status bits appropriately.

Note:

The minimum time for the Command/Request from Host (PC) to Device (LTD) should be 1 Second.

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2 Data Addressing & Data Definition

A number of Modbus functions are utilized in the LTD Modbus interface. Function 1 is used to read system mode, motor drive direction, and speed, while Function 5 is used to set them. Function 2 is used to read individual status and alarm bits. Function 3 is used to read profile and alarm configuration registers and Function 6 makes it possible to set them. Function 4 allows the host computer to read sensor data from the LTD.

When more than one tank is connected to a host computer, each tank must be addressed and scanned individually. The Device ID is the Tank ID of each system, as explained below in the Communication Settings section of the Hardware Specifications.

2.1 Mode, Speed (Digital Output or Input, 1 Bit) (Functions 1, 5)

Mode, Speed Base Address = 0001
Maximum Legal Address = 0016

The bits are defined as follows:

Parameter	Address
Manual	00001
Calibrate	00002
Auto	00003
Profile	00004
Reserved	00005
Reserved	00006
Reserved	00007
Reserved	00008
Stop	00009
Up Slow	00010
Up Medium	00011
Up Fast	00012
Down Slow	00013
Down Medium	00014
Down Fast	00015
Reserved	00016

The information shown above gives the current LTD mode of operation and the probe's movement and can be accessed using Function 1. These are individual bits, not registers. Any of these quantities may be set using Function 5 with the same address.

Bits 1 – 5 indicate the current mode of operation. Each mode is mutually exclusive, so that if one bit is set, the others are cleared. When using Function 5 to change the system mode, it is only necessary to set the bit corresponding to the new mode. It is not necessary (or possible) to manually clear the bit corresponding to the old mode.

Bits 9 – 15 indicate the probe's direction and speed of travel. Again, each drive speed is mutually exclusive, so if the host computer sets one bit using Function 5, the others are automatically cleared.

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These bits may be read to determine current system status at any time without any effect on the system mode of operation, but it should be noted that manually setting any of the bits relating to the probe's direction and speed of travel will halt any profile activity or liquid level tracking that is taking place and put the system in the Manual drive mode. The system will continue indefinitely in this mode until it is set back to "Auto", "Cal", or "Profile." The system will not be able to perform any of its automatic functions while it remains in the Manual mode. One suggestion to bring attention to this condition would be to indicate the manual mode in red to alert the operator that this is not the normal mode of operation. The "Auto" mode, on the other hand, could be indicated in green, since this is the normal mode of operation where liquid level will be tracked continuously.

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2.2 Status, Alarms (Digital Input, 1 Bit) (Function 2)

The main purpose of this document is to provide the information necessary for a programmer to develop a user interface for the LTD on a host computer.

Status Base Address = 10001
Alarms Base Address = 10017
Maximum Legal Address = 10032

2.2.1 Status Bits

The status bits provide important information about system operation and are essential for verifying that data reported by the LTD is current and valid. Each status bit (or status indicator) is given along with its address. They are defined as follows:

Status Indicator	Address	
Bottom Reference	10001	
Lower Level Sensor	10002	
Upper Level Sensor	10003	
Interlock	10004	
Profile Complete	10005	
Metric	10006	
Reserved	10007	
Reel Alarm Disable	10008	
Reel Alarm	10009	
Probe Un-calibrated	10010	
Reserved	10011	
Interval Timer	10012	
Probe At Liquid Level	10013	
Reserved	10014	
Reserved	10015	
Reserved	10016	

Some of these status indicators provide information that is more important than others for plant operators to see. The ones that are essential and should definitely be reported in any custom user interface are Bottom Reference, Lower Level Sensor, Upper Level Sensor, Interlock, Reel Alarm, Probe Un-calibrated, Interval Timer, and Probe At Liquid Level. The others are optional, as explained below.

Bottom Reference is activated whenever the bottom reference switch is closed, indicating that the probe is on the bottom of the tank. **Lower Level Sensor** and **Upper Level Sensor** are set whenever the sensors are in liquid. These are important for observing the physical conditions around the probe and verifying where it is in the tank.

Interlock is set whenever the probe is at the maximum allowed probe position; the probe will not drive up any more when it reaches Interlock. It usually indicates some kind of error condition if the probe reaches Interlock during normal operation, since the probe should be at liquid level or below. This could also happen if an operator drives the probe up manually.

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When **Reel Alarm** is set, it shows that a physical drive problem has occurred. This indicates the need for investigation by maintenance personnel and is absolutely essential for monitoring proper operation of the equipment.

Probe Un-calibrated is activated whenever it is possible that the current probe position is not correct. This status indicator is also important, since it means that the liquid level value reported may be incorrect as well.

When **Interval Timer** is set, it indicates that the system will perform profiles automatically at the programmed interval. If not set, profiles can be run manually.

Probe At Liquid Level is set whenever the system is in Auto and the probe is at the liquid/vapor interface. This means that the level reported is current.

The status indicator, **Profile Complete**, may be of more importance to the programmer than to the operator. When this indicator is cleared, the LTD is in the process of running a profile. When this indicator is set, it means that the profile is complete and the data can be collected and analyzed. The remaining status indicators are not normally needed by operators, but they are useful for someone checking the complete status of the equipment. **Reel Alarm Disable** should not normally be set, because this overrides a safety feature in the equipment that could prevent possible damage in the event of equipment failure. Having an alarm tied to this indicator would help prevent it from being set without operator knowledge. The status indicator, **Metric**, will be set if values are reported in metric units, and cleared if values are reported in English units. Once this is set, it should not need to be changed.

2.2.2 Alarms

The alarm bits relate to conditions of liquid in the tank and are important with respect to plant operation. Each alarm is given along with its address. The alarms are defined as follows:

Alarm	Address	
Low Density	10017	
High Density	10018	
Low Temperature	10019	
High Temperature	10020	
Low, Low Level	10021	
High, High Level	10022	
Low Level	10023	
High Level	10024	
Profile Temperature Deviation	10025	
Profile Density Deviation	10026	
Reserved	10027	
Reserved	10028	
Profile Low Temperature	10029	
Profile High Temperature	10030	
Profile Low Density	10031	
Profile High Density	10032	

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2.3 Profile & Alarm Control (Analog Output or Input) (Functions 3, 6)

Functions 3 and 6 are used to read and control profile settings and product alarm settings. This makes it possible for operators to completely control the profiles and change alarm settings from the DCS.

2.3.1 Addressing

Base Address = 40001
Maximum Address = 40023

Parameter	Address
Profile First Point	40001
Profile Increment	40002
Profile Dwell Time	40003
Reserved	40004
Reserved	40005
Reserved	40006
Reserved	40007
Reserved	40008
Reserved	40009
Automatic Profile Interval	40010
Automatic Profile Enable	40011
Automatic Profile Hour	40012
Automatic Profile Minute	40013
Low Density Alarm Set Point	40014
High Density Alarm Set Point	40015
Low Temperature Alarm Set Point	40016
High Temperature Alarm Set Point	40017
Low Low Level Alarm Set Point	40018
High High Level Alarm Set Point	40019
Low Level Alarm Set Point	40020
High Level Alarm Set Point	40021
Temperature Deviation Alarm Set Point	40022
Density Deviation Alarm Set Point	40023

2.3.2 Definition

When a profile is initiated, the probe drives to the tank bottom and starts the profile there. **Profile First Point** is the first programmed point at which the probe will stop on the way up to take temperature and density measurements. (Actually it is the second point in the profile, since the first point is always at the bottom of the tank.)

Profile Increment is the distance that is added to the first point and each successive point to determine where to stop and take readings on the way back to the surface of the liquid.

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Profile Dwell Time is the time in seconds that the probe will pause at each point to allow readings to stabilize before recording the value.

A profile can be started manually by switching the mode to Profile Mode. Alternatively, a profile can be run automatically at either a fixed time each day or at regularly repeated intervals.

Automatic Profile Interval designates the number of minutes between each successive profile run (after the first profile starting at **Automatic Profile Hour** and **Automatic Profile Minute**).

Automatic Profile Enable designates whether profiles will be run automatically or not. A value of 0 disables automatic profiles. A value of 1 causes profiles to be initiated at the programmed time each day and at the interval chosen. If only one profile is desired per day, the interval should be 1440 minutes (to start it the next day at the same time). The first profile (after enabling the feature) will be at the time defined in **Automatic Profile Hour** and **Automatic Profile Minute**.

The remaining quantities set the alarm limits for low and high density values, temperature values, liquid level, and temperature and density deviations. **Temperature Deviation Limit** is the amount of change in temperature from one point to the next in a profile that will cause an alarm. **Density Deviation Limit** is the amount of change in density from one point to the next in a profile that will cause an alarm.

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2.3.3 Scaling

In the LTD implementation of the Modbus specification for Functions 3, 4, and 6 the values are communicated to the DCS using 16 bit signed or unsigned integers, so it must be decided what each integer will represent. Each quantity is scaled between high and low limits for the value being reported, along with high and low scale values. These limits are currently fixed. (In a future firmware version it will be possible to change them, and the proper values for each site will be established during commissioning if not previously set.)

Separate scaling factors are set for level, temperature, and density. The following table defines the scaling.

	Low Limit	High Limit	Low Scale	High Scale	Value to be reported from LTD	Value Sent to DCS
Level (m)	0 m	65.000 m	0	65000	15.000 m	15000
Temperature	-327.68	327.67 °C	-32768	32767	-160.00	-16000
Density (kg/m^3)	0	655.35 kg/m^3	0	65535	450.00	45000
Level (ft)	0 ft	200.00 ft	0	60000	100.00 ft	30000
Temperature	-327.68	327.67 °F	-32768	32767	-256.00 °F	-25600
Density (lb/ft^3)	0	65.535 lb/ft^3	0	65535	27.500	27500

For Level (in metric units), the 16-bit integer value in the Modbus register would read directly in millimeters of liquid level as an unsigned integer. Each count of the Modbus integer would represent 1 mm. For temperature (both English and metric), each count would represent 100th's of a degree in each respective scale, and would be reported as a signed integer.

When reading or writing quantities relating to level (such as Profile First Point, Profile Increment, and High Level Alarm Set Point), use the scaling factors for Level. This applies both to quantities received and transmitted. Quantities received from the host computer are interpreted according to the current scaling values in effect. Dwell times are in seconds.

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2.4 Sensor Data, Profile Data (Analog Input) (Function 4)

The current conditions in the tank at the location of the probe and information about the condition of the tank during the last profile are given by the sensor data and profile data quantities.

2.4.1 Addressing

Base Address for Sensor Data = 30001
Maximum Legal Address = 30620

Parameter	Address
Current Probe Position	30001
Current Temperature	30002
Current Density	30003
Liquid Level	30004
Reserved	30005
Number of Points	30006
Profile Month	30007
Profile Day	30008
Profile Hour	30009
Profile Minute	30010
Current Hour	30011
Current Minute	30012
Current Second	30013
Reserved	30014
Reserved	30015
Reserved	30016
Reserved	30017
Reserved	30018
Reserved	30019
Reserved	30020
Profile Probe Position Point 0	30021
Profile Temperature Point 0	30022
Profile Density Point 0	30023
Profile Probe Position Point 1	30024
Profile Temperature Point 1	30025
Profile Density Point 1	30026
Profile Probe Position Point n	30021 + (3 * n)
Profile Temperature Point n	30022 + (3 * n)
Profile Density Point n	30023 + (3 * n)
Profile Probe Position Point 199	30618
Profile Temperature Point 199	30619
Profile Density Point 199	30620

Note: The term "profile probe position" is used instead of "profile level," because in this document, level is usually used to indicate the actual liquid level. Only the last point in a regular profile is actually at "liquid level."

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2.4.2 Definition

Current Position, Current Temperature, and **Current Density** are the current values where the probe is now, whether at the bottom of the tank or at liquid level. **Liquid Level** holds the most recent liquid level reading. If the probe is beneath the surface, such as during a profile, the liquid level reading will not be updated until the probe returns to the surface. The status indicator "At Liquid Level" (described in a previous section) is used to show when the value reported is up to date.

Number of Points contains the number of profile points collected during the last profile. The number of points is determined by the location of the programmed starting point, the increment added, and liquid level. The first point in a profile is always the bottom of the tank. The next point is programmable, and is usually set to an whole number such as 1 m, and then a programmable increment is added (also normally 1 m). This puts profile points at whole numbers, such as 1 m, 2 m, 3 m, etc. The probe moves up, collecting data until liquid level is reached (or the maximum number of points is reached, causing the profile to terminate prematurely). The last point is normally at liquid level.

The maximum number of points allowed is 500 (0 to 499). To limit the number of points to a lower value, set the profile increment so that at maximum liquid level, the number of points will not exceed the desired limit. For example, if maximum liquid level is 30 m and the profile increment is 1 m, the maximum number of points would be 31 (including the bottom reference point) and the total number of registers needed for position, temperature and density would be 93 (31 x 3).

Profile Month, Profile Day, Profile Hour, and **Profile Minute** show the date and time when the first profile point is collected. **Current Hour, Current Minute, and Current Second** show the current system time. One way to constantly verify communications with the LTD is to monitor the system time and verify that it is always changing. If it stops changing, an alarm should be generated.

The probe position, temperature, and density for each point collected in a profile are available at the addresses shown. Data in registers beyond the number of points collected will be zero.

2.4.3 Scaling

The scaling for Function 4 values is controlled by the same parameters that are used for values in Functions 3 & 6. Please refer to that section for more information about scaling.

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3 Hardware Specifications

3.1 General

A host computer may be connected directly to the LTD using the RS485 field wiring. A site-wiring diagram is generally provided for each site. In order to establish communications, there are a number of data parameters that must be set properly.

3.2 Communication Settings

In one possible configuration, the LTD is connected to the Tank Gauge Interface Module (TGIM). The TGIM serves as a power switch and power indicator for the gauge and a connection point for all field wiring. It is usually located in the control room or in a field equipment room. There are normally 2 communication links, and these may be used for communication with a host computer. They are referred to as Host 1 and Host 2.

The LTD communication links use RS485. To interface with the host computer using RS232, a converter must be used. The default baud rate is 9600 baud, and this cannot be changed at this time. The other parameters are odd parity, 8 data bits, and 1 stop bit. The transmission mode used is RTU.

The Modbus protocol requires the use of an address. The Tank ID in the LTD may be set using the touch screen user interface, and also serves as the Modbus address. If more than one LTD is connected on the same data bus, it is important to set a unique address for each unit.

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4 Programming Suggestions

4.1 Introduction

To completely understand the operation of the LTD, the programmer should read the Operation Manual. While not all of the information is duplicated here, many relevant details are given to assist in the programming task. Below are some suggestions on how to use the information available from the LTD.

The primary uses of the LTD are 1) to obtain accurate liquid level information, and 2) to obtain temperature and density profile information that will enable the detection of unsafe layering conditions. Layers detected by the LTD are indicated by the Temperature and Density Deviation alarms.

The relative importance of the various status bits has been covered in the section above where the addresses of the status bits are given. Refer to that section for a better understanding of which indicators are most important for the operators to see.

4.2 Finding Liquid Level

After powering on the LTD, accurate liquid level information is available only after the probe has touched the bottom of the tank and been driven back to the liquid/vapor interface. When the system is powered off, the last known probe position and liquid level are stored in non-volatile memory, and the same values are recalled when the system is powered on, but the "Un-cal" indication appears, since it is possible that these values are no longer correct.

To clear the "Un-cal" indicator, it is necessary to touch the bottom of the tank. This will automatically set the probe position to the value that should be displayed at the bottom of the tank.

The LTD can be set to remain in Manual Mode or go directly into Cal Mode on power up. If it remains in Manual, a calibration should be initiated manually and after that it will remain in Auto Mode, measuring liquid level. With a system commissioned in a tank with liquid, that should be all that is necessary. The LTD will automatically go through the steps to find liquid level, switch to Auto mode and monitor the liquid level.

4.3 Profile Information

A profile is initiated by placing the controller into the Profile mode. This may be done via the touch screen user interface, the automatic interval timer, or a command sent from a host computer. Point 0 is always the bottom point, and the liquid level is the last point. The number of points collected is given in the "Number of Points" variable. The probe is stopped at each point for a specified wait time (called "dwell time") for stabilization. Temperature and density are measured, and then the probe is driven to the next point. The unit is returned to the AUTO mode when the profile is completed.

As a profile is being collected, the "Number of Points" variable is incremented after each new point. Using this variable makes it possible to import only valid data points. When the probe first touches the bottom of the tank at the beginning of a new profile all the previous data points in memory and the "Profile Complete" status indicator are cleared. When the profile is complete and all data is available, the "Profile Complete" status indicator is set.

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4.4 Profile Graphing

The easiest way to see stratification of temperature and density in a tank of liquid is by graphing the data. In order for this graph to be meaningful, it must have a high resolution. The expected changes in the data are small, and the graph must be able to show those changes. It is recommended that the range of temperature and density be chosen carefully. One way to achieve the correct scaling is to make the values read at the first point at the bottom of the tank as the center of the graph, and then plot a deviation from those values as the level increases. On most LNG tanks, it should be suitable to show a deviation from the center of 5 kg/m^3 or 0.3 lb/ft^3.

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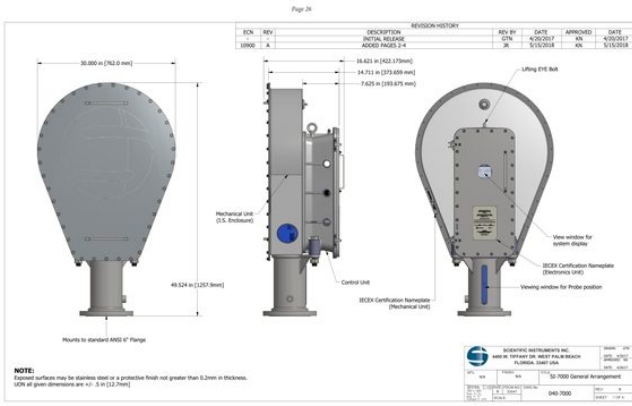

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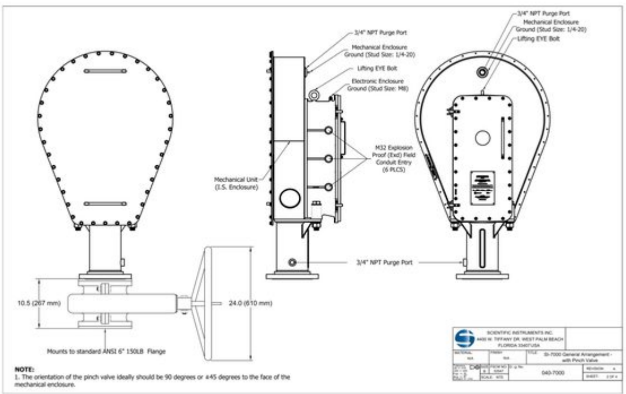

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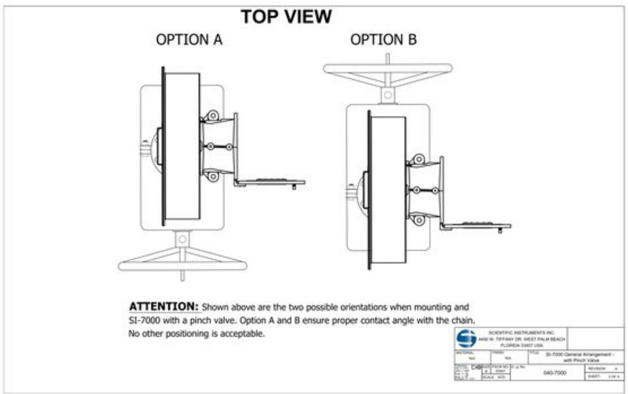

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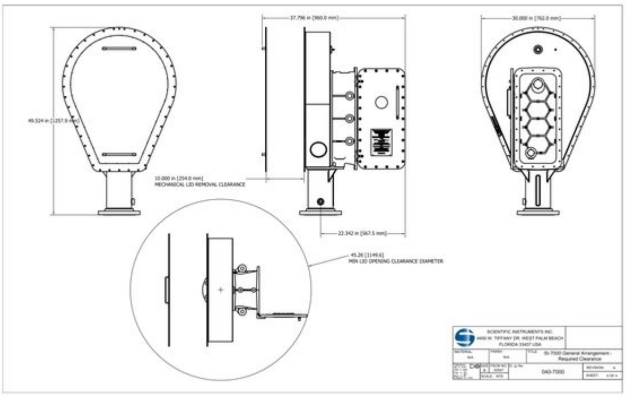

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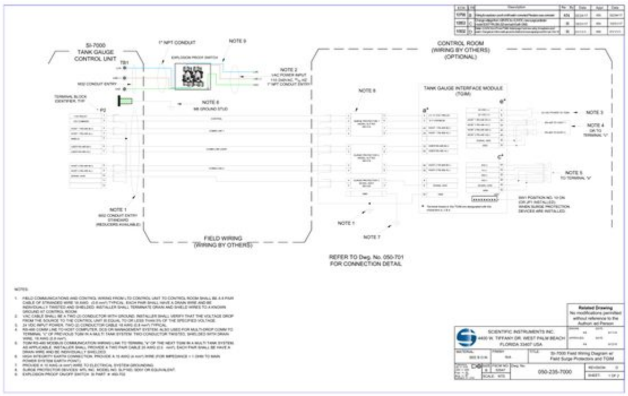

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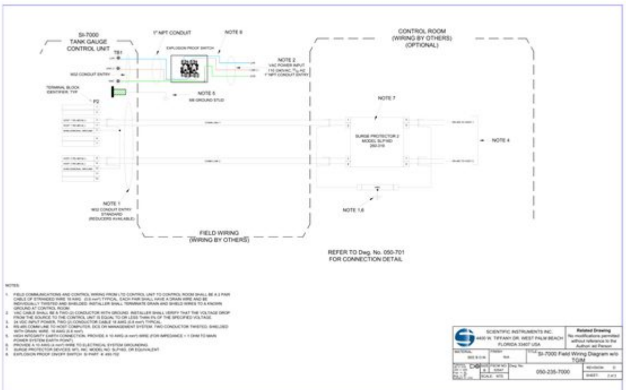

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SCIENTIFIC INSTRUMENTS

SI-7000

Tank Gauging System

Installation & Commissioning Manual

SCIENTIFIC INSTRUMENTS, INC.
4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
090-7000-1 Rev A

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1 INTRODUCTION

The SI-7000 Tank Gauging System monitors the level, temperature, and density of the cryogenic liquid in a large storage tank and is typically controlled and monitored by a PC based operator interface and/or DCS user interface in the control room. The SI-7000 consists of two sections, mechanical and electrical. The mechanical section consists of the chain reel enclosure and probe enclosure. This section contains all of the mechanical components, except the drive shaft gear box, required to lift and lower the probe assembly in the tank and shall be considered a Zone 0 environment. For most installations this section will not need to be accessed. The electrical section consists of the electrical enclosure and contains all the electronics and drive shaft gear box. The electrical section housing is an explosion-proof container and shall be considered a Zone 1 environment. This section is where the input power and communications are physically connected to the SI-7000. For a more detailed explanation of the components in these sections and their function please refer to the "SI-7000 Operational Manual" (090-7000-2). The probe enclosure is mated to the bottom of the chain reel enclosure, and the chain reel enclosure and electrical enclosure are mated back to back.

Figure 1 Chain reel & probe enclosures

Figure 2 Chain reel enclosure w/o Cover

Figure 3 SI-7000 Side View

Figure 4 Electrical Enclosure

Figure 5 Electrical Enclosure w/ door open

Figure 6 SI-7000 transparent side view

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This manual provides information necessary to install the system at a typical location. A site-specific mechanical installation drawing and field wiring diagram may be provided along with this manual for clarification of details at an atypical location.

The complete power-up, system testing, and calibration are accomplished during commissioning activities at a later time and only with the presence of a factory representative. Please refer to the "SI-7000 Operational Manual" (090-7000-2) for more detailed instructions on these activities. However, initial power-up and basic system functionality testing are included in this manual and may be performed if SI personnel are present.

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2 WARNINGS, CAUTIONS, AND NOTICES

Scientific Instruments, Inc. is not liable for any injuries, death, or damages caused during the installation of the SI-7000. It shall be the installer's or customer's responsibility to ensure the operation is performed in a safe manner and complies with all government, local, and governing agency regulations.

WARNING: All applicable government, local, and governing agency regulations shall be strictly adhered to. Failure to do so could result in death or serious injury.

WARNING: The conductive grounding paths for lightning shall be achieved in such a way, that warming up, ignitable sparks alternatively spray sparks cannot become the ignition source of the explosive atmosphere. Failure to do so could result in death, serious injury, or extensive property damage.

WARNING: Care must be taken to ensure equipment used to lift the SI-7000 is adequate for the weight as well as any sudden jolts from crane movement. Failure to do so could result in death, serious injury, or extensive property damage.

WARNING: Care must be taken to ensure adequate clearances are maintained between structures, other equipment, and personnel when lifting and moving the SI-7000 into position. Failure to do so could result in death, serious injury, or extensive property damage.

WARNING: No connections shall be made to the upper left entry port due to the location of the internal intrinsic safety circuitry.

CAUTION - PINCH POINT: Care must be taken when positioning the SI-7000 on the mounting flange and all personnel shall keep their hands clear as the SI-7000 is lowered into place. Failure to do so may result in serious injuries as fingers may be crushed or severed.

NOTICE: All installation instructions in this manual shall be followed. Failure to do so may result in damage to the SI-7000, void the warranty, or cause installation delays.

NOTICE: Do not remove the protective foam from the probe enclosure until the SI-7000 is being attached to the tank's mating flange. Remove the protective foam will allow the probe to impact the walls of the probe enclosure while the SI-7000 is being moved and lifted into place.

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3 INSTALLATION REQUIREMENTS

These are the minimum requirements for installation that Scientific Instruments suggests. If these minimum requirements are not met and there is a safety risk, Scientific Instruments personnel shall not engage in installation activities.

- An available 6" 150 pound flange mounted on a 6" minimum pipe providing unobstructed access to the tank interior. This flange should be capable of supporting 400 pounds (181.4 kg) without causing damage to the flange or surrounding structures.
- A minimum of 1m of clearance around the flange. The equipment should not be mounted where it is difficult to operate the A/C power disconnect switch.
- Crane and qualified operator.
- Two personnel to guide SI-7000 into position.
- One qualified personnel for mechanical and electrical installation.
    - General understanding of relevant electrical engineering.
    - Understanding of and ability to read and assess engineering drawings.
    - Practical understanding of explosion protection principles and techniques.
- Stable temporary or permanent working platform.

Other Notes:

- Since the SI-7000 is always mounted outside on the tank, there are no additional ventilation requirements.
- There are no requirements relating to sound levels.
- No assembly is required before installing the SI-7000.
- The safety of any system incorporating the SI-7000 is the responsibility of the assembler of the system.
- Scientific Instruments does not currently provide any accessories for the SI-7000.

4 PRE-INSTALLATION CHECK LISTS

These items need to be verified prior to the SI-7000 being installed. Failure to meet these requirements can adversely affect the operation of the SI-7000, cause severe damage to equipment, or create installation delays.

4.1 SI-7000 INSTALLATION TOOL AND HARDWARE

Only a few basic tools are required to install the SI-7000. This checklist should be completed prior to climbing to the top of tank.

- Tools, Site supplied
    - ☐ 2ea 1⅛" Wrench
    - ☐ 1ea 6mm Hex T-handle
    - ☐ 1ea Reversible Screwdriver Set (#2 Philips & 9/32 In. slotted)
    - ☐ Torque wrench (10ft/Lbs.-150ft/Lbs.)
    - ☐ 1ea 1⅛" Socket, ⅜" drive
    - ☐ 1ea ½" to ⅜" drive adapter
    - ☐ 1ea ½" Socket, ⅜" drive
    - ☐ 1ea 6mm Hex bit socket, ⅜" drive

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- Hardware, Scientific Instruments supplied
    - ☐ 1ea Explosion proof external AC power switch
    - ☐ 16ea ¾-10x2.5" Stainless steel bolt
    - ☐ 4ea ¾" Stainless steel nut (Extra on Blind Flange)
    - ☐ 16ea ¾" Stainless steel lock washer
    - ☐ 16ea ¾" Stainless steel washer
    - ☐ 4ea ¾-10x3" Stainless steel bolt (Extra on Blind Flange)

4.2 INTERNAL TANK REQUIREMENTS

- ☐ Was the tank cleaned before it was closed and sealed? Excess rust, dust particles, perlite, or other foreign particles that are left in the tank could stick to the density meter over time and begin to affect density readings.
- ☐ Can the probe assembly move vertically in an unrestricted manner from the drive mechanism all the way to the bottom of the tank or stilling well? Any obstructions may cause the probe assembly to become stuck and result in damage to the probe assembly, chain, and/or drive mechanism.
- ☐ Is the probe assembly's path away from any pumps or filling lines that would produce high liquid flow or turbulence? High liquid flow or turbulence may result in damage to the probe assembly or inaccurate density and/or level measurements.
- ☐ If a stilling well is used, is it plumb and of adequate size to ensure that the probe does not contact the stilling well's inner surface? Contact with the inner surface may cause unpredictable operation and place additional stress on the drive system resulting in premature failure of components.

4.3 EXTERNAL TANK REQUIREMENTS

- ☐ Is a stable work platform in place? This may be a permanent or temporary structure; however SI strongly suggests that a permanent work platform be installed.
- ☐ Is adequate safety equipment in place and serviceable?
- ☐ Is the mating flange where the SI-7000 will be mounted clean and free of damage and debris?
- ☐ Is the gasket that will be used between the mating flange and SI-7000 clean and serviceable?
- ☐ Will the hand-wheel of the isolation valve obstruct access to the mechanical or electrical enclosures? If it will obstruct access, can the mechanical enclosure cover still be removed and the electrical enclosure door opened?
- ☐ If a pinch valve is used, is it installed so that it closes on the flat sides of the chain as much as possible and that the hand wheel does not prevent opening the electrical enclosure door or the ability to remove the mechanical enclosure cover? Misalignment of the pinch valve may prevent it from sealing properly and allow excessive gas to escape during maintenance and may also hinder personnel from having access to one side of the SI-7000.

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4.4 ELECTRICAL AND COMMUNICATIONS

- ☐ Has a place been chosen for the external power switch within sight of the SI-7000?
- ☐ Are the electrical power and communication cable conduits ready to be connected to the SI-7000? The SI-7000 may be mounted without the cable conduits in place. However, the installation will not be considered complete until these connections are made.
- ☐ Is the electrical power supply to the SI-7000 locked out/tagged out? If not, this shall be done prior to the power conductor being attached to the SI-7000.

5 SI-7000 UNPACKING AND INSPECTION

The SI-7000 is designed for ruggedness and use in severe conditions, but care should still be used when moving or lifting the unit. SI has designed a custom container so that the SI-7000 will arrive ready to be installed. However, the unit should still be inspected for any damage that may have occurred during shipping.

1. Ensure shipping crate is upright on a flat, stable, and relatively level surface.
    a. If the crate rocks back and forth then it shall be shimmed to hold it steady.
2. Ensure the shock and tilt indicators have not been activated.


    a. Notify SI if the indicators have been activated and provide photos if possible.
3. Inspect shipping crate for any visible signs of damage.
    a. Notify SI of any damage and provide photos if possible.
4. Remove the crate top, internal horizontal supports, and side panels.

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Figure 7 SI-7000 mounted on crate bottom

5. Thoroughly inspect the SI-7000 for any signs of damage.
    a. Notify SI of any damage and provide photos if possible.
6. Attach lifting straps to eyelet on top of the electrical enclosure and apply a slight lifting force.

Figure 8 SI-7000 with lifting strap attached

7. Use a 1/2" wrench to remove the 4 lag screws holding the SI-7000 to the crate bottom.

6 INSTALLATION

REF: Appendix A, Bolt Tightening Worksheet and other site specific drawings supplied.

NOTICE: DO NOT remove the protective foam from the probe enclosure until the SI-7000 is being attached to the tank's mating flange. Removing the protective foam will allow the probe to impact the walls of the probe enclosure while the SI-7000 is being lifted into place

1. Verify that the pinch valve has been installed following the instructions in section 4.3.

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Figure 9 Protective foam holding probe in place

2. Lift SI-7000 to top of tank taking without causing it to swing excessively.

Figure 10

3. Position SI-7000 over mounting flange.

Figure 11 Guiding SI-7000 into position

4. Remove Blind Flange & disregard the 3/4bolts & nuts. Keep flat & lock washers to use w/the ¾-10x2.5" bolts provided w/the Pinch Valve for mounting.
5. Remove protective foam from the probe enclosure.

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Figure 12 Remove probe protective foam

6. Lower SI-7000 onto the Pinch Valve flange ensuring the holes are aligned and pinch points are kept clear.

Figure 13 Aligning SI-7000 with mating flange

7. Use 8 sets of the ¾" SS hardware included w/the Pinch Valve to mount the SI-7000 to the Pinch Valve flange. See Appendix A for detailed instruction.

Figure 14 Installing mounting hardware

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8. Connect an earth bonding cable to the ground lug on top of the electrical enclosure in such a way that a bolt lightning of a 30m can be controlled.

7 CONNECTING SYSTEM WIRING

WARNING: ALL THREADED ENTRIES TO ELECTRICAL ENCLOSURE FOR VAC POWER INPUT AND/OR COMMUNICATIONS, SUCH AS CABLE GLANDS, CONDUIT FITTINGS, PLUGS FITTED TO ELECTRICAL ENCLOSURE SHALL BE RATED FOR USE IN EXPLOSIVE ATMOSPHERES.

THERE IS A WIDE SELECTION OF CABLE GLANDS, AND THREADED ENTRIES AVAILABLE WITH APPROVALS AND CERTIFICATES COVERING INTERNATIONAL AND LOCAL WIRING CODES. THEREFORE IT IS THE RESPONSIBILITY OF THE INSTALLER TO ENSURE THAT CABLE GLANDS, THREADED PLUGS AND/OR CONDUIT FITTINGS ARE Ex APPROVED FOR THE ZONE AND AREA CLASSIFICATION WHERE THE SI-7000 WILL BE INSTALLED. CONSULT THE MANUFACTURER FOR ADDITIONAL INFORMATION.

REF: Appendix B Drawing 050-7000-1 Typical Installation Wiring Diagram
All unused wires and shields should be terminated to ground in the control room only. Proper grounding and shield connections must be made to minimize noise levels on the communication lines. No connections shall be made to the upper left entry port due to the location of the internal intrinsic safety circuitry. Connecting to this will void the SI-7000's intrinsic safety certification.

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Figure 15 Unused upper left entry port

7.1 OPENING ELECTRICAL ENCLOSURE DOOR

Extreme care must be taken not to scratch or mar the flame path surfaces while working inside the electrical enclosure. A protective covering is used during assembly and shipping and should remain in place until the installation is completed. Any scratches or damage to the flame path void the enclosure's explosion proof certification.

- Loosen all bolts holding the enclosure's door shut.
- Open door fully.
- Insert latching pin to hold the door in the open position.

7.2 AC POWER INPUT & GROUND

AC power lines shall be connected to the SI-7000 using either the lower right or lower left entry port with the lower right port being the preferred option. The SI-7000 has been designed for use with 18 AWG to 10 AWG power lines and an 8 AWG Earth ground.

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Figure 16 Lower left-side AC entry port option

Figure 17 Lower right-side AC entry port option

1) Ensure AC power source is locked out / tagged out.
2) Ensure external AC power switch is in the off position.
3) Connect appropriate cable gland and conduit to entry port.
4) Pull in enough wire so that the AC power lines can be routed to the power terminal strip and the Earth ground wire can be routed to the grounding stud at the top of the enclosure with adequate service loops for both.
5) Crimp the appropriate ring lug connectors to the ends of the wires.
6) Remove the two screws on the power terminal strip cover to access the connection terminals.
7) Connect the AC power lines as shown in the wiring diagram.
8) Reinstall the power terminal strip cover.
9) Connect the earth ground terminal to the tank ground using wiring practices in accordance with all local regulations.

Note:

¾" NPT conduit entries are standard. Other approved sizes are available upon request

7.3 SIGNAL WIRING CONNECTIONS

Signal lines can be connected to the SI-7000 using any entry port other than the upper left port and the entry port being used for the AC power. For ease of internal routing, the center right port is the preferred option. Care should be taken not to route the signal wires in close proximity to the AC power lines.

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The SI-7000 has been designed for use with 22 AWG to 18 AWG signal lines, and wire gages up to 14 AWG may be used. However, due to limited space it is recommended that smallest possible gage wire be used. All signal wire connections are made via a single connector (J2) on the main controller assembly (150-601) inside the electrical enclosure. The mating connector (Phoenix Contact #1863259) that will be attached to the signal wires has been inserted on the main controller. A spare connector has also been included in the installation kit. Ferrules are not required, but it is strongly suggested that they be used. SI recommends the following Phoenix Contact sleeved ferrules, but any similar ferrules may be used: 22AWG – 3203024, 20 AWG – 3200687, 18 AWG – 3200690, 16 AWG – 3200755, 14 AWG – 3200522.

3 Left-side signal entry port options

4 Right-side signal entry port options

1. Install appropriate cable gland into entry port.
2. Pull in enough wire so that the signal lines can be routed to the connector on the main controller assembly with an adequate service loop.
3. If used, crimp ferrules to the ends of the wires.
4. Remove mating connector from the main controller assembly.
5. Connect the signal lines to the connector as shown in the wiring diagram.
6. Plug connector into main controller assembly (J2).

7.4 CLOSING ELECTRICAL ENCLOSURE DOOR

Extreme care must be taken not to scratch or mar the flame path surfaces while working inside the electrical enclosure. Any scratches or damage to the flame path void the enclosure's explosion proof certification.

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- If the "Initial Power Up" (Step 8.1) has not been completed, ensure the AC power switch in the "OFF" position.
- Remove the protective covering from both sides of the flame path.
- Clean flame path surfaces with alcohol wipes provided in kit to remove any residual adhesive.
- Apply a 1/8" bead of vacuum grease all the way around the enclosure side flame path, one near the outer edge and one near the center.
- Remove pin holding the door in the open position.
- Close door fully.
- Tighten all bolts on the enclosure door with the Allen wrench provided .
- Torque bolts to 20ft/lbs.
- Wipe off any silicone lubricant that has been squeezed out from the flame path.

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8 COMMISSIONING

NOTE: If the SI-7000 is used in a manner not specified by Scientific Instruments, the protection provided by the equipment may be impaired.

Initial power up and commissioning shall only be performed by or with Scientific Instruments personnel. Applying power to or operating the SI-7000 prior to Scientific Instruments personnel inspecting the installation will void the warranty and Scientific Instruments will not be liable for any damage to the equipment.

8.1 INITIAL POWER UP

1. Ensure the SI-7000 power switch is in the "OFF" position.
2. Remove AC power lock out/tag out.
3. Move the SI-7000 power switch to the "ON" position.
4. Check the following on the main controller.
    - Red "A/C" power LEDs are on. If not, try the following:
        - Ensure AC power is being supplied to the controller assembly.
        - Ensure power plug is full inserted into socket .
    - Green "FUSE" LEDs are on. If not, try the following:
        - Check fuses in AC plug module (spare fuses are included in kit).
    - Green "PWR" LED is on. If not, it is most likely the following:
        - DC power supply is not functioning, replace controller assembly.
    - LCD is on and displaying the main menu. If not,
        - Remove controller assembly and ensure LCD is fully connected.
        - If still no success, the LCD or CPU is not functioning, replace controller assembly.
5. Continue to the next test or move the SI-7000 power switch to the "OFF" position.

8.2 MOTOR DRIVER TEST

1. Ensure the SI-7000 power switch is in the "ON" position.
2. On the touch screen LCD navigate to the Motor menu.
3. Touch the down arrow once.
    - The motor should begin to lower the probe assembly at its slowest speed.
4. Touch the down arrow a second time.
    - The motor should increase to its medium speed.
5. Touch the down arrow a third time.

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- The motor should increase to its maximum speed.

6. Touch the down arrow a fourth time.
    - The motor should return to its medium speed.
7. Touch the stop button.
    - The motor should stop and hold the probe at its current position.
8. Touch the up arrow once.
    - The motor should begin to lift the probe assembly at its slowest speed.
9. Touch the up arrow a second time.
    - The motor should increase to its medium speed.
10. Touch the up arrow a third time.
    - The motor should increase to its maximum speed.
11. Touch the up arrow a fourth time.
    - The motor should return to its medium speed.
12. Allow the motor to lift the probe to the parked position.
    - The motor should automatically stop and hold the probe in the parked position.

8.3 BENCH MARKING

1. Ensure the SI-7000 power switch is in the "ON" position.
2. On the touch screen LCD navigate to the Motor menu.
3. Touch the down arrow once.
    - The motor should begin to lower the probe assembly at its slowest speed
4. Observe the chain movement through the probe enclosure window and stop the motor when the designated marking on the chain is aligned with a reference point. Record the step count.
    - Motor should stop and hold the probe at the current location
5. Touch the down arrow three times.
    - The motor should begin to lower the probe assembly
6. Observe the chain movement through the probe enclosure window and stop the motor when the second marking on the chain is aligned with the reference point. Record the second step count.
7. Calculate the level scale. Consult with factory for details.
8. Touch the up arrow three times.

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- The motor should begin to lift the probe assembly.

9. Allow the motor to lift the probe to the parked position.
    - The motor should automatically stop and hold the probe in the parked position.

8.4 COMMUNICATIONS

1. Ensure the SI-7000 power switch is in the "ON" position.
2. Ensure the green "COM" LED(s) are on.
3. From the control room send a command to lower the probe.
    - The motor should begin to lower the probe assembly .
4. From the control room send a command to stop the motor.
    - The motor should stop and hold the probe at its current position.
5. From the control room send a command to lift the probe.
    - The motor should begin to lift the probe assembly.
6. Allow the motor to lift the probe to the parked position.
    - The motor should automatically stop and hold the probe in the parked position.

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APPENDIX A

BOLT TIGHTENING WORK SHEET

Location/Identification:______________________________ Nominal Bolt Size: ________

Surface Finish on Flange:_______________ Lubricant Used:______________________

Note: Bolt lubrication greatly affects the torque values used when installing gaskets. To achieve the same gasket compression, a much higher torque value is required for a dry bolt versus using an effective lubricant such as molybdenum disulfide. Lubricant should not be applied to the gasket or flange faces as a release agent.

(Initial each step)

_____1. Ensure system is at ambient temperature and depressurized.
_____2. Visually examine and clean flanges, bolts, nuts and washers.
_____3. Lubricate bolt, nut, and flange facings on the nuts surfaces.
_____4. Insert bolt from the bottom of the mounting flange and hold it in place.
_____5. Place the washer and then the lock washer on the end of the bolt.
_____6. Install nut on end of bolt and hand tighten.
_____7. Repeat steps 3 to 6 for remaining seven hardware sets
_____8. Use the torque values and cross-pattern sequence below to tighten hardware (each tightening sequence 1 through 8 constitutes a "round").
_____9. Check gap around the circumference between each of these rounds, measured at every other bolt. If the gap is not reasonably uniform around the circumference, make the appropriate adjustments by selective bolt tightening before proceeding.

- **Round 1** - Torque to 30 ft-lbs.
- **Round 2** - Torque to 60 ft-lbs.

_____10. Final Rotational Round - 100% of Final Torque (same as Round 3 above). Use ROTATIONAL, clockwise tightening sequence, starting with Bolt No. 1, for one complete round and continue until no further nut rotation occurs at 100% of the Final Torque value for any nut.

NOTE: Short-term bolt preload loss can occur between four to twenty-four hours after initial tightening due to bolt relaxation. SI suggests repeating step 10 at least twenty-four hours after the initial "Final Rotation Round" and/or just prior to commissioning.

Name: ______________________ Signature: ______________________ Date: __________

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SCIENTIFIC INSTRUMENTS

SI-7000

Tank Gauging System

Operational Manual Part A

SCIENTIFIC INSTRUMENTS, INC.
4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
090-7000-2A Rev -

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Record of Revisions						
Rev	RFEA #	Description	Rev by	Date	Appr by	Date
-		Initial Release	KN	8/21/19	KN	8/21/19

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1 INTRODUCTION

1.1 GENERAL DESCRIPTION

The Scientific Instruments, Inc. SI-7000 is a state of the art electro-mechanical system designed to measure liquid level, temperature, and density in cryogenic liquids, including liquid natural gas (LNG), propane, and butane. The system has been designed with an emphasis on reliability, modularity, and ease of maintenance and operation. With its computer-controlled operation, the SI-7000 gives operators access to critical data necessary for intelligent and safe operation of the plant.

In addition to providing an accurate measure of the liquid level in a tank, the SI-7000 provides for precise measurement of temperature and density throughout the tank by positioning a multi-sensor probe assembly at any height in the tank. This makes it possible to obtain a profile that gives an accurate representation of the current conditions in the tank. This information is crucial for the safe storage and handling of cryogenic liquids, since they are subject to layering, and over time, possible "rollover" conditions.

1.2 APPLICATION

A significant safety concern in the storage of cryogenic liquids such as LNG is a phenomenon known as "rollover." If this occurs, pressures inside the storage tank may rise to excessive levels. The tanks are equipped with safety vent valves that are designed to keep the pressures from rising to levels that could cause structural damage. However, when these valves operate, gas is vented to the atmosphere at an uncontrolled rate, which is an additional safety concern.

Rollover occurs under certain conditions as stratified LNG comes to equilibrium. Stratification occurs when the product in the tank forms in layers with different densities and different temperatures. Sudden mixing of LNG in any storage tank occurs as two or more layer densities approach equality. Any heat trapped in the system is released rapidly during mixing, generating a vapor which may exceed the venting capability of the tank.

By periodically taking a profile of the tank, the Scientific Instruments SI-7000 can detect stratification and possible rollover conditions and generate an alarm signaling the need for operator intervention. Operators can then take preventative measures (such as mixing the liquid or moving it to another tank).

1.3 SYSTEM CONFIGURATION

The integrated touch screen LCD is the primary interface used for setting all system parameters. The system configuration may also be changed remotely via a Modbus serial communication interface.

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2 SYSTEM SPECIFICATIONS

2.1 CERTIFICATION STANDARDS

IEC 60079-0:2011, IEC 60079-1:2014, IEC 60079-11:2011, EN 60079-0:2012/A11:2013, EN 60079-1:2014, EN 60079-11:2012, ISO 80079-36, ISO 80079-37, IEC/EN61010-1:2010

2.2 EQUIPMENT MARKINGS

Electrical Compartment: II (1) 2 G Ex db [ia IIC Ga] IIB T4 Gb (-20°C ≤ Ta ≤ + 50°C)
Mechanical Compartment: II 1 G Ex h IIC T4 Ga (-20°C ≤ Ta ≤ + 50°C)
Probe Assembly: II 1 G Ex ia IIC T4 Ga (-200°C ≤ Ta ≤ +50°C)

2.3 MEASUREMENT CAPABILITY

Level:
Range 56 m
Resolution: 1 mm
Accuracy: ±2 mm

Temperature:
Range -200° C to +50° C
Resolution: 0.01° C
Accuracy: ± 0.1° C

Density:
Range 400 to 1000 kg/m^3
Resolution: 0.01 kg/m^3
Accuracy: ± 0.5 kg/m^3
Profile Points: 250+

Sensor Isolation: Internal I.S. Barriers

2.4 ELECTRICAL

Operational Voltage: 85-240 VAC
Certified Operational Voltage: 120-240 VAC
Frequency: 50-60 Hz
Power Draw: I_{max}: 3.15 A
Overvoltage: Category III
Pollution: Degree 1

2.5 COMMUNICATIONS

Standard: (2) Standard RS485 / MODBUS Protocol
Optional: 4-20mA, Fiber Optic, HART & Ethernet

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2.6 MECHANICAL

Explosion Group of Ex Atmosphere: IIC
Housing Mat: Stainless Steel
Dimensions: 125 cm x 76 cm x 37 cm
Weight: 180 kg
Ambient Temp: -40°C to +60°C
Safety Certs: IECEx
Protection class: Designed to meet IP65
Seismic Rating: Designed to withstand 3.0 g
Drive Chain: SS304
Mounting: 6" ANSI 150#RF Flange
Op. Pressure: 400 mbar
Stilling Well: Not Required
Isolation Valve: Required
Pinch Valve: Required

2.7 INTERNAL CONNECTIONS

All internal connectors have a flammability rating of V-0. Skipped designators have been removed from the design. Not all connections are used, and some may not be present on the assembly.

SI-7000 Controller Assembly
J1 Mains Power Input (Universal AC Input)
J2 Field Wiring Input (RS485)
J6 Reel Encoder Input
J8 Motor Power Output
J10 Probe Interface Module (PIM)
J14 JTAG Programing (manufacturer use only)
J22 Alternative 28 VDC – 48 VDC Power Input

Probe Interface Module (PIM)
J1 SI-7000 Controller Assembly
J2 Slip-ring
P1 Primary Earth Ground
P2 Optional Earth Ground

3 WARNINGS, CAUTIONS, AND NOTICES

Scientific Instruments, Inc. is not liable for any injuries, death, or damages caused during the operation of the SI-7000. It shall be the customer's responsibility to ensure all operations and maintenance are performed in a safe manner and complies with all government, local and governing agency regulations

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WARNING: All applicable government, local, and governing agency regulations shall be strictly adhered to. Failure to do so could result in death or serious injury

WARNING: External power supply shall be disconnected via switch or circuit breaker prior to opening the electronics enclosure lid or removing the mechanical enclosure cover. Failure to do so could result in death, serious injury, or extensive property damage

WARNING: Care must be taken to ensure adequate clearances are maintained between structures, other equipment, and personnel when lifting and moving the SI-7000 into position. Failure to do so could result in death, serious injury, or extensive property damage

WARNING: Possibility of cryogenic burns! If the SI-7000 is installed on tanks with cryogenic liquids (such as LNG), when the probe is lifted for maintenance it must be given time to reach ambient temperatures to avoid cryogenic burns when handling the probe. It usually takes more than 4 hours to reach ambient temperature. When the temperature indicated on the touch screen main display reaches ambient temperature, it should be safe to open the mechanical unit.

WARNING: The SI-7000 is not for intended oxygen service

WARNING: Do not open when an explosive atmosphere is present.
Maximum experimental safe gap (MESG) of gas/air mixture ≥ 0,05 mm

CAUTION - PINCH POINT: Care must be taken when closing the SI-7000 electronics enclosure lid and all personnel shall keep their hands clear. Failure to do so may result in injuries as fingers may be pinched.

SPECIAL CONDITIONS OF USE: Flameproof joints are not intended to be repaired.

NOTICE: There are no consumable materials associated with using the SI-7000.

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NOTICE: During normal operation there are no surfaces which can cause burns due to high temperatures. If the equipment fails in a manner not covered in the service manual, please consult with Scientific Instruments before touching the equipment.

NOTICE: Instructions for cleaning the probe are given in the service manual. This is the only whetted part that may require periodic cleaning. The drive chain does not need to be cleaned, and no un-whetted parts require periodic cleaning. In the range of applications, the instrument is designed for, no decontamination is required.

NOTICE: All operation instructions in this manual shall be followed. Failure to do so may result in damage to the SI-7000 and void warranty.

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SCIENTIFIC INSTRUMENTS

SI-7000

Tank Gauging System

Operational Manual
Part B

SCIENTIFIC INSTRUMENTS, INC.
4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
090-7000-2B Rev A

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Rev	RFEA #	Description	Rev by	Date	Appr by	Date
-		Initial Release	KN	8/21/19	KN	8/21/19
A		Updated Photos	JR	11/16/20	KN	11/16/20

SCIENTIFIC INSTRUMENTS, INC.
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090-7000-2B Rev A

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1 THEORY OF OPERATION

1.1 OVERVIEW

The SI-7000 controller receives input from the sensors in the mechanical unit and sends signals to control the motor depending on the current mode of operation. This controls the movement of a multi-sensor probe assembly suspended within the tank. Two level sensors inside the probe indicate to the controller whether the probe is in liquid, in vapor or at the interface between liquid and vapor. The controller monitors the signals from the temperature sensor and the density meter, and continuously calculates updated information for transmittal to a host computer.

The system has several modes of operation to accommodate maintenance activities and daily operation. The system may be programmed to perform a profile at regular intervals (as determined by customer requirements) to provide updated information about the conditions in the tank. When not performing a profile, the system is usually set to track liquid level, providing the most up-to-date reading of liquid level.

1.2 LIQUID LEVEL

The liquid level sensors produce different voltages depending on whether they are in liquid or vapor. The controller uses these differences in the sensor voltages to determine their state. By vertically spacing the level sensors a short distance apart, the controller can "recognize" liquid level at the point where the lower level sensor is "in liquid" and the upper level sensor is "in vapor".

1.3 TEMPERATURE

The temperature sensor is a four-leaded platinum element with a resistance of 100 Ω at 0 degrees centigrade. Using a one-milliamp excitation current, a sensor voltage is developed across the sensor proportional to its resistance. This voltage is digitized by an analog to digital converter and compared to data tables in the controller. The appropriate temperature is calculated and constantly updated as the temperature changes.

1.4 DENSITY

The density meter and its associated maintaining amplifier output to the controller a signal whose frequency is proportional to density. The controller determines the frequency of this signal and using a density conversion formula along with calibration parameters for the particular density meter, calculates the density of the medium into which the densitometer is submerged.

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1.5 BOTTOM REFERENCE SWITCH

A mechanical switch assembly is employed to signal the controller when the probe is in contact with the tank floor. When the controller senses this "Probe at Reference" condition, it ceases downward movement of the probe and sets the probe position to the pre-programmed bottom reference value. The bottom reference value accounts for the distance of the level sensors from the bottom of the probe.

1.6 TOP REFERENCE SWITCH

A second mechanical switch assembly is employed to signal the controller when the probe has reached the top of the probe chamber. When the controller senses this "Probe at Top" condition, it ceases upward movement of the probe. This provides an easy method of parking the probe at the top in preparation for maintenance activities.

1.7 OPERATIONAL MODES

The system can be operated in any one of five principal modes using the touch screen interface pictured below. The available modes are Automatic (Auto), Calibration, Profile, Zoom Scan, and Manual. The most commonly used modes in daily operation are Auto and Profile. Manual is used normally only for maintenance activities. The Calibration mode can be used after maintenance activities. When power has been removed. It will quickly find bottom reference again and then return to Auto mode at the surface, or whenever it is desirable to re-establish the liquid level.

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1.7.1 AUTOMATIC MODE (AUTO)

When the system is placed into Auto, the controller causes the probe to locate and track the liquid/vapor interface. The system is at the interface when the lower sensor is in liquid and the upper sensor is in vapor. The Automatic mode is the normal mode of operation; all alarms are reported, and profile runs can start automatically if they are programmed. The system returns to Auto upon completion of a profile run or a calibration run (described below).

1.7.2 CALIBRATION MODE (CAL)

When Cal mode is selected, the controller drives the probe to the bottom of the tank to establish bottom reference and then back to the liquid level interface, upon which the system is returned to Auto. A calibration is most often used after a power loss or after certain maintenance activities. When power is lost, the last probe position is stored, but it is assumed that the probe may have been moved manually while power was off (such as during maintenance activities), so on power-up the system is marked as not being calibrated.

1.7.3 PROFILE MODE

In the Profile mode, temperature and density readings are taken from the bottom of the tank up to liquid level, providing operators with an accurate representation of the current conditions in the tank.

When the Profile mode is started, the probe is driven to the tank floor where it re-establishes bottom reference. At the bottom, the probe pauses for a programmed delay time to allow conditions in the tank around the probe to stabilize. After the prescribed delay, position, temperature, and density readings are taken, and then the probe is driven up, stopping at programmed increments to collect temperature and density data. The same delay occurs at each point to allow readings to stabilize. The profile is terminated when the maximum number of points is taken or liquid level is found.

The parameters that control a profile may be entered via the LCD interface or over the Modbus serial interface.

After all readings are taken, the data is analyzed and alarms are generated as applicable. The system is returned to Auto after completion of the profile. In this manner, a complete profile of the tank is collected and stored for output to a host computer.

NOTE: The SI-7000 was designed to perform profiles when liquid is fully at rest and stabilized. Do NOT attempt to perform a profile while loading or unloading liquid in the tank. Loading LNG either top or bottom creates turbulence, swirls, and insatiability in liquid. High power compressors are used during unloading, this process also creates low level turbulence and instability in the liquid. For best

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accuracy and results profiling must be scheduled a day before or after the loading or unloading process. Consult factory for more information.

1.7.4 ZOOM SCAN MODE

This feature is not implemented at this time.

1.7.5 MANUAL MODE

The Manual mode is used whenever it is desirable to take control of the probe and give specific commands to drive up or down. This would occur most often during maintenance activities. The probe may be stopped so that it will not track liquid level, or it may be driven up or down in fast, medium, or slow.

NOTE: No alarms are reported in the Manual mode and the liquid level reading is not updated. It is important to realize that the system will remain in the Manual mode indefinitely until it is returned to Auto by an operator or by maintenance personnel.

1.8 POWER UP CONDITION

The SI-7000 takes its reference from the floor of the tank; therefore, it is important to re-establish this bottom reference after power up. If this reference has not been established, the current probe position will be marked as "Uncal".

The system can be set so that it will automatically reestablish its bottom reference after a power loss, or it can be set so that it will power up in Manual with the motor stopped. This setting ("Change to Start in Auto" or "Change to Start in Manual") is found in the Configuration section accessed via the LCD interface. If it is set to start in Auto mode, it will first find the tank bottom and then return to liquid level.

If the system is not set to reestablish bottom reference automatically, it is only necessary to start a Cal run (enter Calibration Mode). The probe will travel to the bottom of the tank, re-establish bottom reference and then return to liquid level, where it will return to the Auto mode.

2 OPERATOR INTERFACE

Information about the system status is usually transmitted to a host computer (or DCS) in the Control Room using the Modbus protocol. The information available includes the current mode of operation, direction and speed of travel of the probe, current values for the position, liquid level, temperature, and density, profile data, and various status indications and alarms, but the programming and implementation of an optional DCS operator interface for the SI-7000 may be different at each site, therefore some of the features described below may not appear. The sections below describe what is available on the SI-7000 LCD screen.

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2.1 HOME SCREEN

The home screen is the default display when the gauge is powered on. This screen displays the level, temperature, and density, along with other key parameters.

The current values for position, temperature, and density are the values measured where the probe is located at the moment. If the density is at a very low value (at or equal to zero), it is possible that the probe is in vapor instead of liquid.

The current value for liquid level should also be displayed if it has been established after power-up. If it has not yet been found, it will be zero or the last recorded value. Note that the liquid level can only be guaranteed to be correct when the probe is actually at liquid level. This is indicated by the "At Liquid Level" indicator as described below. In addition, to guarantee an accurate liquid level there should be no "Probe Uncal" status indication (also described below).

There are a number of status indicators that are very important to system operation and must be understood to get a complete picture of the current conditions.

The most basic indicators are the upper and lower level sensors, along with the bottom reference switch and top limit switch. The bottom switch is wired across the lower level sensor, and the top limit switch is wired across the upper sensor. Thus each sensor has 3 possible states:

- Closed: The corresponding switch is closed and the sensor is shorted, so it is not possible to indicate liquid or vapor.
- Liquid: The corresponding switch is open and the sensor is in liquid. (This will be indicated by L-LIQ for the lower sensor, and U-LIQ for the upper sensor.)

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- Vapor: The corresponding switch is open and the sensor is in vapor. (This will be indicated by L-VAP for the lower sensor, and U-VAP for the upper sensor.)

When the probe if fully submersed, both switches will be in liquid. If the probe is completely out of liquid, both sensors should indicate vapor. If the probe is at the interface (liquid level), the lower sensor should be in liquid, and the upper sensor should be in vapor.

The SI-7000 measures level with respect to the bottom of the tank, so to report an accurate liquid level, it must first establish the location of tank bottom. If this reference has not been established, the current probe position will be marked as "Uncal". The system saves its last position when power is lost. However, since it is possible that maintenance personnel have physically moved the probe and its cable during a power loss, the position of the probe is always marked as "Uncal" when the system is first powered up. Bottom reference must be reestablished to clear this condition.

The "Probe Uncal" indicator will also come on if the probe position decreases below where bottom reference was found previously without finding bottom reference there again.

The SI-7000 provides a number of alarms to alert operators to abnormal or potentially dangerous conditions. Some of the alarms relate to current conditions and some relate to conditions found in the tank during a profile run.

2.1.1 REEL ALARM

If the Reel Alarm indicator is on, it indicates that the mechanical unit reel is not turning, and it could be indicative of a mechanical failure in the drive mechanism or an obstruction in the tank. Maintenance personnel should investigate.

2.1.2 PROBE AT LIQUID LEVEL

"Probe at Liquid Level" should be on when the system is in its Auto mode of operation and is at liquid level. If this indicator is not on, the operator cannot be assured that the current liquid level reading is correct. If the probe is beneath the surface or it is up in the vapor, there is no way of knowing if liquid level is changing.

2.1.3 LEVEL ALARMS

Probably the most important alarmed parameter relating to current conditions is level. There are several different level alarms: a low level alarm, a low low level alarm, a high level alarm, and a high high level alarm. Each alarm level setting should be determined by operations personnel and programmed accordingly.

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2.1.4 TEMPERATURE AND DENSITY ALARMS

The other alarms relating to current conditions are the high and low temperature alarms and the high and low density alarms. In most environments, these alarms are not critical, since it is the deviation alarms that are normally used to detect abnormal variations of temperature and density in the profile of the tank, not the high and low alarms.

2.1.5 DEVIATION ALARMS

During a profile run, temperature and density data are collected at each programmed stop point. At the end of the run, the data points are analyzed. If the temperature deviation between one point and the next point is greater than the programmed alarm value, the temperature deviation alarm is set. Similarly, if the density variation from one point to the next is greater than the alarm threshold value, the density deviation alarm is set. These alarms are not cleared until another profile run is made, and it is verified that the alarmed condition no longer exists.

The deviation alarms are very important, since a variation in density or temperature could indicate layering or stratification, which if left unchecked could lead to a potential rollover condition.

2.1.6 ALARM FUNCTIONALITY IN MANUAL MODE

NOTE: It is very important to understand that when the system is in its Manual Mode of operation, no alarms will be reported, even if alarm conditions exist. For the system to report alarms, it must be in the Auto mode. The system will immediately switch to Auto mode after a Profile or Cal run.

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2.2 SYSTEM INFO

This screen displays the software revision and the Tank ID.

2.3 SYSTEM CONFIGURATION MENU

Pressing CONFIG on the main screen brings up the System Configuration Menu. This allows access to parameters that can be changed via the LCD touch screen display.

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2.3.1 LEVEL

One of the primary functions of the LTD is to measure the level of the liquid in a storage tank. If the level falls below the low limits or rises above the high limits, alarms are generated. The set points for these alarms are set in this menu.

To change the values, push on the "Edit Parameter" button. A new screen will appear allowing the values to be changed.

To change the levels, touch the blue squares and then enter the desire reference. When all the levels are entered, click enter and save.

Use the right arrow to access the fourth parameter.

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2.3.2 TEMPERATURE

The temperature sensor in the SI-7000 is mounted inside the density meter. Each sensor is theoretically a 100 Ω platinum resistance thermometer (PRT), but the resistance can vary slightly from the theoretical value. Before each one is shipped, a calibration process is performed, and the exact value is measure in ice and nitrogen. The resulting values are included on the calibration certificate for each density meter.

The SI-7000 electronic circuits are calibrated to measure resistance accurately, so once the correct values are entered for the sensor being used, the instrument can display temperature within the tolerance stated in the system specification.

Temperature Measurement Range: -200°C to +50°C

Error (over-range): 999.99°C

Error (under-range): -555.55°C

Low Temperature and High Temperature alarms can be triggered at set points determined by the customer and are entered in Kelvin.

The Deviation alarm is a very important alarm that is used to determine if the tank is stratified or not. In a profile, if there is a difference from one point to the next greater than the Deviation alarm limit, a Temperature Deviation (TD) alarm is triggered. This alarm will remain active until another profile is performed and the condition is cleared.

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2.3.3 DENSITY

The density meter is a calibrated device. Before each one is shipped, a calibration process is performed, and three coefficients are determined and included on the calibration certificate for each density meter.

The SI-7000 electronic circuits are designed to measure frequency accurately, so once the correct values are entered for the density meter, the instrument can display density within the tolerance stated in the specification.

Density Measurement Range: 400 kg/m^3 to 600 kg/m^3

Error (over-range): 9999.99 kg/m^3

Low Density and High Density alarms can be triggered at set points determined by the customer and are entered in kg/m^3.

The Deviation alarm is a very important alarm that is used to determine if the tank is stratified or not. In a profile, if there is a difference from one point to the next greater than the Deviation alarm limit, a Density Deviation (DD) alarm is triggered. This alarm will remain active until another profile is performed and the condition is cleared.

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2.3.4 SYSTEM

The System Menu provides access to a number of sub menus.

2.3.4.1 BOTTOM REFERENCE

The SI-7000 measures level with respect to the bottom of the tank, and it touches bottom at the beginning of every Cal run and Profile. When the probe touches tank bottom, the motor pulse count is set to zero.

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The height of the lower level sensor above the bottom of the probe is added to the pulse count to establish an accurate position of the lower level sensor.

To change the bottom reference, touch the blue square and then enter the desire reference. Click enter and save.

2.3.4.2 TANK ID

For installations with multiple tanks, the Tank ID distinguishes one tank from another. This is also the Modbus Address.

To change the Tank ID "Address", touch the blue square and then enter the desire ID "Address". Click enter and save.

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2.3.4.3 TIME/DATE

The SI-7000 maintains a record of the date and time and keeps the time for up to one month after power is turned off. If the controller is powered off for more than one month, it may be necessary to reprogram the current date and time.

To change the date and time, push the button with that label. A new screen opens with an option to set the date.

After specifying the correct date, push Next, and another screen opens with an option to set the current time. Push Done when finished.

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2.3.4.4 LEVEL SENSOR CAL CLOSED

This is for internal use, and a password is required.

2.3.4.5 MICROCONTROLLER CAL

This is for internal use, and a password is required.

2.3.4.6 LEVEL SENSOR CAL LIQUID

This is for internal use, and a password is required.

2.3.4.7 UNITS OF MEASURE (METRIC/ENGLISH)

This button gives the option to change to the other system of measure. If the system is currently in metric, the button will display an option to Change to English. If the system is in English units, the button will display an option to change to metric.

2.3.4.8 100 OHM CAL

When this menu is selected, a verification message is displayed.

"Calibrate at 100 OHM?"

This is for internal use, and should only be performed when instructed to do so by the technical support department. Press "No" to exit the menu without calibrating.

2.3.4.9 NITROGEN TEMPERATURE CAL

When this menu is selected, a verification message is displayed.

"Calibrate at Nitrogen temperature?"

This is for internal use, and should only be performed when instructed to do so by the technical support department. Press "No" to exit the menu without calibrating.

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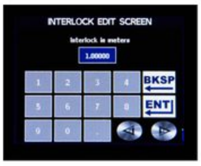
2.3.4.10 LEVEL SCALE

The formula for level is as follows:

Level = Motor Pulses * Level Scale + Bottom Reference

So Level Scale is the calibration value that relates pulses to actual probe position. A calibration is done at the factory, and Level Scale value is determined and then recorded on a site specific Configuration Worksheet. The value shown here should match the value in that Configuration Worksheet.

2.3.4.11 INTERLOCK



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Interlock is the high limit for probe movement. It is normally set to keep the probe down in the cold part of the tank. If the Interlock indicator in the DCS or PC User Interface is on, it means that the probe has reached the highest position allowed. Unless the probe was raised to that point manually, this is an abnormal situation and maintenance personnel should investigate.

To change the interlock, touch the blue square and then enter the desire interlock. Click enter and save.

2.3.5 COM SETTINGS

This menu is not supported at this time.

2.3.6 PROFILE & ZOOM SCAN SETTINGS

This menu has two sub menus.

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2.3.6.1 PROFILE SETUP

One of the most important features of the SI-7000 is the ability to collect data at various points in the tank. A profile starts at the bottom of the tank and points are taken in the upward direction until the maximum number of points is reached or liquid level is found. This menu allows customization of the profile that is taken.

The actual first point of the profile is taken at the tank bottom. The first programmable point is specified in the field, "First Point." As the probe moves off the bottom, it will first stop at this point, and then the Increment will be added to calculate the next stop point. The Dwell Time is the time in milliseconds that the probe will wait at each point before taking readings. The Interval is the amount of time in minutes before the next profile will be taken.

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The time setup screen makes it possible to choose the exact time when the profile will start. Note that the profile run will not start if the system is in Manual when the programmed time arrives, but will start if the system is in Auto, Calibrate, or Profile.

2.3.6.2 ZOOM SCAN SETUP

This menu is not implemented at this time.

2.3.7 START MODE (AUTO/MANUAL)

When the system is powered on, the power up state is programmable. It can either remain in Manual Mode, waiting for instructions, or it can go directly into Auto Mode and measure liquid level. During maintenance, it is usually set to stay in Manual. In normal operation it can be changed to start in Auto.

If it is currently set to start in Manual, the button will show the option to change to start in Auto. If it is set to start in Auto, the button will show the option to start in Manual.

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2.4 COMMAND MENU

The Command Menu allows the current LTD mode to be changed.

Pressing **Manual Mode** will cause the probe to go to Manual Stop. It will remain in this mode indefinitely, waiting for operator commands.

If you push the **Level Calibration** button, the probe will move to the tank bottom to re-establish the bottom reference, and then it will return to the liquid level.

The **Auto Mode** button will return the LTD to normal Auto mode, where it continuously tracks liquid level.

Push **Start Profile** to initiate a new profile, moving to the bottom of the tank, and then back to liquid level, collecting data at the programmed stop points.

Zoom Scan is not implemented at this time.

Use the Back button to return to the previous menu.

2.4.1 MANUAL MODE

In Manual mode the probe is under operator control. Pressing the Up arrow one time starts the probe moving up slowly. Press it again to change to medium speed and press it a 3rd time to change to high speed.

Likewise, pressing down one, two, or three times puts it in slow, medium, or fast speed moving down.

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The Stop button should be pressed before changing directions to avoid jerking the chain and probe.

2.4.2 LEVEL CALIBRATION

Pressing Level Calibration starts the probe moving down to the tank bottom. After it reaches the tank bottom it will return to liquid level. After the calibration button is pushed, the display returns to the normal display.

2.4.3 AUTO MODE

Pressing Auto Mode puts the LTD in Auto, and the probe will return to liquid level from wherever it is, and it will track liquid level continuously. After the Auto Mode button is pushed, the display returns to the normal display.

While in Auto, the probe usually drives in slow to track liquid level, but it can accelerate to a faster speed as needed.

2.4.4 START PROFILE

Pressing Start Profile starts the probe moving down to the tank bottom. After it reaches the tank bottom and collects data at the first point, it begins moving up, stopping to collect data at the programmed stop points along the way. When it reaches liquid level, the last point will be taken, and the profile will complete. After the Start Profile button is pushed, the display returns to the normal display.

2.5 MOTOR

Pressing MOTOR brings up a screen that shows detailed diagnostic information related to the motor drive function and other system parameters.

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3 TECHNICAL ASSISTANCE

For technical assistance, please contact Scientific Instruments via telephone, fax, mail, or email.

Scientific Instruments
4400 W. Tiffany Dr.
West Palm Beach, FL 33407
U.S.A.

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Facsimile: +1 (561) 881-8556
Email: support@scientificinstruments.com
Website: www.scientificinstruments.com

4 DISCLAIMER

WARNING: Only personnel **certified by Scientific Instruments, Inc.** are authorized to execute the activity of installation, commissioning, or any type of work/maintenance on the model SI-7000 LTD gauge. Unauthorized activity on the SI-7000 LTD gauge will negate warranty. Scientific Instruments, Inc. assumes no responsibility for injury, loss or damage to person or property caused by unauthorized activity. Any questions please contact us at email fieldservicetechs@scientificinstruments.com or Phone: (800)466-6031.

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SCIENTIFIC INSTRUMENTS

SI-7000

Tank Gauging System

Service Manual

SCIENTIFIC INSTRUMENTS, INC.
4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
090-7000-3 Rev -

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1 INTRODUCTION

The SI-7000 Tank Gauging System monitors the level, temperature, and density of the cryogenic liquid in a large storage tank and is typically controlled and monitored by a PC based operator interface and/or DCS user interface in the control room. The SI-7000 consists of two sections, mechanical and electrical. The mechanical section consists of the chain reel enclosure and probe enclosure. This section contains all of the mechanical components, except the drive shaft gear box, required to lift and lower the probe assembly in the tank and shall be considered a zone 0 environment. For basic service this section will not need to be accessed. The electrical section consists of the electrical enclosure and contains all the electronics and drive shaft gear box. The electrical section housing is an explosion-proof container and shall be considered a Zone 1 environment. This section is where the input power and communications are physically connected to the SI-7000. For a more detailed explanation of the components in these sections and their function please refer to the "SI-7000 Operational Manual" (090-7000-2). The probe enclosure is mated to the bottom of the chain reel enclosure, and the chain reel enclosure and electrical enclosure are mated back to back


Figure 1 Chain reel & probe enclosures


Figure 2 Chain reel enclosure w/o Cover


Figure 3 SI-7000 Side View


Figure 4 Electrical Enclosure


Figure 5 Electrical Enclosure w/ door open


Figure 6 SI-7000 transparent side view

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This manual provides information necessary to service the system at a typical location. A site-specific mechanical installation drawing and field wiring diagram may be provided along with this manual for clarification of details at a typical location.

Please refer to the "SI-7000 Operational Manual" (090-7000-2) for more detailed instructions on other activities.

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2 WARNINGS, CAUTIONS, AND NOTICES

Scientific Instruments, Inc. is not liable for any injuries, death, or damages caused during operation and servicing of the SI-7000. It shall be the customer's responsibility to ensure all operations and maintenance are performed in a safe manner and comply with all government, local and governing agency regulations.

WARNING: All applicable government, local, and governing agency regulations shall be strictly adhered to. Failure to do so could result in serious injury and/or death.

WARNING: External power supply shall be disconnected via switch or circuit breaker prior to opening the electronics enclosure lid or removing the mechanical enclosure cover. Failure to do so could result in serious injury, death, or extensive property damage

WARNING: Possibility of cryogenic burns! If the SI-7000 is installed on tanks with cryogenic liquids (such as LNG), when the probe is lifted for maintenance it must be given time to reach ambient temperatures to avoid cryogenic burns when handling the probe. It usually takes more than 4 hours to reach ambient temperature. When the temperature indicated on the touch screen main display reaches ambient temperature, it should be safe to open the mechanical unit.

WARNING: Do not open when an explosive atmosphere is present. Maximum experimental safe gap (MESG) of gas/air mixture ≥ 0.05 mm

CAUTION - PINCH POINT: Care must be taken when closing the SI-7000 electronics enclosure lid and all personnel shall keep their hands clear. Failure to do so may result in injuries as fingers may be pinched.

NOTICE: There are no consumable materials associated with using the SI-7000.

NOTICE: During normal operation there are no surfaces which can cause burns due to high temperatures. If the equipment fails in a manner not covered in the service manual, please consult with Scientific Instruments before touching the equipment.

NOTICE: Instructions for cleaning the probe are given here. This is the only whetted part that may require periodic cleaning. The drive chain does not need to be cleaned, and no un-whetted parts require periodic cleaning. In the range of applications the instrument is designed for, no decontamination is required.

NOTICE: All operation instructions in this manual shall be followed. Failure to do so may result in damage to the SI-7000 and void warranty.

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NOTICE: All service instructions in this manual shall be followed. Failure to do so may result in damage to the SI-7000 and/or other nearby equipment.

NOTICE: All service shall be performed by qualified personnel.

NOTICE: Fuses shall be changed by qualified personnel.

NOTICE: Lock-out tag-out procedures shall be followed during service.

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3 OVERVIEW

In general, a system inspection and routine maintenance session will begin in the Control Room, evaluating the performance of the system while the probe is still in the liquid. This provides an opportunity to observe any existing problems and performance problems, allowing corrective action to be taken once the probe is out of the liquid.

After that inspection, the probe will be raised to the probe enclosure and allowed to warm. While the system is warming up, the Mechanical Unit cannot be opened, but the Electronic Unit can be checked during that time. When the probe has reached ambient temperature, the mechanical unit may be opened and the internal components can be checked.

4 GUIDELINES FOR MAINTENANCE

Refer to site specific drawings for more information about field wiring and installation.

4.1 PREPARATION

Before beginning the maintenance procedures, it is recommended to check to see if firmware updates are required. If firmware updates are required, these may need to be completed first.

4.2 POSSIBLE MATERIALS NEEDED

- Vacuum grease
- Grease for pinch valve
- Can of compressed air or clean instrument air

4.3 CALIBRATION OF DENSITY METER

Some customers are concerned about absolute accuracy in density measurement. These customers should return the density meters to the manufacturer every three to five years, depending on how concerned they are with absolute accuracy. Absolute accuracy is not required for detection of stratification. To detect stratification, the system only needs to observe a relative density difference in the tank.

Calibration of the density meter is not included in normal routine maintenance procedures. Please contact the factory for details.

4.4 CALIBRATION OF TEMPERATURE MEASUREMENT

Calibration of temperature measurement consists of two parts: 1) Calibrating the electronics, and 2) Calibrating the temperature sensor. Since the temperature sensor is internal to the density meter, calibration of the temperature sensor can be requested when the density meter is returned to Scientific Instruments. It is possible to perform calibration of the electronics for temperature measurement on site only if the correct test equipment is available. (This would include a decade box, precision DVM, connecting cables.) Please contact the factory for details.

Temperature calibration can also be done every three to five years, depending on how concerned the customer is with absolute accuracy of temperature measurement. Normally,

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absolute accuracy is not critical. Stratification is detected by observing relative changes in temperature.

Calibration of temperature measurement is not included in normal routine maintenance procedures. Please contact the factory for details.

4.5 OPENING ELECTRICAL ENCLOSURE DOOR

To perform maintenance activities, it will be necessary to open the electrical enclosure door. Extreme care must be taken not to scratch or mar the flame path surfaces while working inside the electrical enclosure. Any scratches or damage to the flame path void the enclosure's explosion proof certification.

- Loosen all bolts holding the enclosure's door shut with the Allen wrench provided.
- Open door fully.
- Insert pin to hold the door in the open position.

4.6 CLOSING ELECTRICAL ENCLOSURE DOOR

When maintenance activities are complete, the door should be closed. Extreme care must be taken not to scratch or mar the flame path surfaces while working inside the electrical enclosure. Any scratches or damage to the flame path void the enclosure's explosion proof certification.

A basic functionality test should be performed prior to closing the electrical enclosure. If communications with the control room are available these should be tested as well. This allows for discovering any potential problems before the cover is closed.

- Apply a thin bead of silicone vacuum grease all the way around the enclosure side flame path.
- Remove pin holding the door in the open position.
- Close door fully and insert all the bolts.
- Tighten all bolts on the enclosure door with the Allen wrench provided.
- Torque bolts between 18 to 20 lb·ft (24.5 to 27 N·m)
- Wipe off any silicone lubricant that has been squeezed out from the flame path.

4.7 MECHANICAL ENCLOSURE

This section provides guidance for working with the mechanical enclosure.

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WARNING: Shafts and bushings (flameproof joints) are not intended to be repaired. Contact Scientific Instruments for replacements of these parts.

NOTICE: Bearings shall be replaced after 10 years of service (90% of rated life). Expected use is 2 hours of continuous mechanical movement per day.

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5 PERIODIC MAINTENANCE PROCEDURE

ACTIVITY	REMARKS
On Tank	
Compare current programmable parameters values with Configuration Worksheet.	Note and resolve any discrepancies.
With the probe still in the liquid, observe the operation of the system in Auto at liquid level.	Unless there is turbulence in the tank due to pumping action, the probe should be stable at liquid level with the lower sensor in liquid and the upper sensor in vapor. Constant hunting indicates the need to spread the level sensors apart more.
Check Motor screen and verify all parameters are normal. Observe any alarms present.	
Check for proper values for temperature and density on home screen.	For LNG, temperature is typically between -157°C (-251°F) and -163°C (-261°F), and density is usually between 425 kg/m3 (26.5 lb/ft3) and 470 kg/m3 (29.3 lb/ft3). This is only a functional check. Re-calibration of temperature and re-calibration of density are more complex procedures, and are not included in this procedure.
Drive the probe to the bottom of the tank in manual.	
Raise the probe a little and verify that both sensors indicate liquid.	It is only necessary to raise the probe a little. (The bottom reference switch must not be closed, since it shorts the lower level sensor.)
Start a level calibration and check that the system finds level properly.	The probe should stop quickly when it reaches liquid level.
Drive probe up until it stops at the upper limit switch. Disable drive when the probe reaches the top.	Disabling the drive mechanism prevents the probe from being driven back down into liquid by mistake.
Wait for the probe to warm to ambient temperatures before opening the mechanical unit. Proceed with electronic unit checkout procedure.	In warm weather, the probe should reach ambient temperature in about 4 hours. The mechanical unit cannot be opened during this time, but the electronic unit may be opened. It is not necessary to close the pinch valve while the probe is warming.
Pinch Valve	
After verifying that the probe is at the top limit switch, check that the pinch valve operates freely. (Do not close the pinch valve yet.)	Verify that the top of the probe is at the upper limit switch before closing, otherwise the probe may be damaged! Rotate hand wheel 5 turns clockwise, then 5 turns counter-clockwise. If equipped with a grease fitting, lubricate with quality grease.
Electronic Unit Checks	
Check the value recorded for driving to the top. Close the pinch valve firmly.	The top stop value should correspond closely with the value in the Configuration Worksheet.

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ACTIVITY	REMARKS
Drive the probe down to touch the pinch valve.	This verifies bottom reference operation at ambient temperature.
Leave the probe resting on the pinch valve.	This ensures there is enough slack in the cable when opening the system.
Power off.	
Check that terminals within the electronic unit enclosure are free of corrosion and foreign matter, and are secure.	Check that connectors and terminal screws are tight and no corrosion is present.
Inspect for any sign of deteriorating components.	
Check exposed hardware to be sure it is tight.	
Check connection to motor to make sure the connection is tight.	
Mechanical Unit (after warm up period)	
Loosen bolts in the mechanical unit cover and allow system to depressurize.	If the gas flow does not stop after 15 seconds, close the pinch valve a little tighter.
Remove cover and perform visual inspection of interior components.	
Remove applicable hardware and lift probe out of enclosure. Lay it on a suitable working surface.	
Remove the probe shroud, switch assembly, and screen.	
Inspect the density meter to make sure it is clean and dry and fastened securely to probe cone.	If necessary, clean with compressed air. If it is necessary to use a liquid cleaner, any moisture on the probe must be allowed to dry before returning the probe to liquid.
Inspect level sensors to make sure they are free of debris and foreign material.	
Record spool number of density meter.	Check to see that it matches the Configuration Worksheet.
Re-install the probe shroud, switch assembly, and screen.	
Check all for proper operation, especially the bottom switch.	It is very important that the bottom switch function properly. Verify this carefully.
Return the probe to the probe enclosure and reinstall hardware.	
Inspect chain and sprocket for excessive wear.	If any signs of serious abrasion are evident and drive problems have been reported, carry out a full inspection of the drive chain over its entire length. NOTE: The chain must be handled properly, or it can be adversely affected during the inspection.
Clean drive mechanism interior and remove any foreign matter	
Check decoder for reel feedback signal.	

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ACTIVITY	REMARKS
Closing Mechanical Unit	
Apply a small amount of vacuum grease to the gasket, and apply anti-seize compound to cover bolts. Re-install cover.	
If the system is powered on, power off.	The next step involves venting gas. There should be no exposed active electronics during this step.
Follow site procedures to purge the instrument.	There are lower and upper purge ports.
When purging is complete, make sure all purge openings are secure and tight.	Check that no gas leaks are evident around cover joints. If leaks cannot be stopped, it may be necessary to replace the mechanical unit gasket.
When no gas remains in the area, open Electrical Unit (or power on again).	
Open the pinch valve fully and drive the system down in Manual.	Use high speed. There is no need to stop at the liquid interface. Monitor the system as it drives to make sure it drives smoothly.
Raise the probe a little and verify that sensors read liquid.	
Start a level calibration.	This is the quickest way to return the probe to liquid level. Wait for the Cal Run to finish.
After the probe reaches liquid level, verify that it stops quickly.	
Closing Electronic Unit	
Apply a small amount of silicon grease to sealing surfaces of the electronic unit to protect from environmental effects.	
Apply grease or anti-seize compound to socket bolts and close the electronic unit.	
Run a profile. (If the user interface allows, this may be started from the Control Room.)	This will provide density and temperature profile values for the host computer.
Update Configuration Worksheet.	Make sure all values are up-to-date.
Optional PC Interface	(if PC was provided by SI)
Check that communications cable is fastened securely	
Take backup, etc.	
Check past profiles, if available.	While waiting for the probe to finish its movements, it's a good time to check profile history, if available. Look for any abnormalities in profile data.

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6 BASIC FUNCTIONALITY TESTING

6.1 POWER UP

1. Ensure the SI-7000 power switch is in the "OFF" position.
2. Remove AC power lock out/tag out.
3. Move the SI-7000 power switch to the "ON" position.
4. Check the following on the main controller.
   - Red "A/C" power LEDs are on. If not, try the following:
     - Ensure AC power is being supplied to the controller assembly.
     - Ensure power plug is full inserted into socket .
   - Green "FUSE" LEDs are on. If not, try the following:
     - Check fuses in AC plug module (spare fuses are included in kit).
   - Green "PWR" LED is on. If not, it is most likely the following:
     - DC power supply is not functioning, replace controller assembly.
   - LCD is on and displaying the main menu. If not,
     - Remove controller assembly and ensure LCD is fully connected.
     - If still no success, the LCD or CPU is not functioning, replace controller assembly.
5. Continue to the next test or move the SI-7000 power switch to the "OFF" position.

6.2 MOTOR DRIVER TEST

1. Ensure the SI-7000 power switch is in the "ON" position.
2. On the touch screen LCD navigate to the Motor menu.
3. Touch the down arrow once.
   - The motor should begin to lower the probe assembly at its slowest speed.
4. Touch the down arrow a second time.
   - The motor should increase to its medium speed.
5. Touch the down arrow a third time.
   - The motor should increase to its maximum speed.
6. Touch the down arrow a fourth time.
   - The motor should return to its medium speed.
7. Touch the stop button.
   - The motor should stop and hold the probe at its current position.
8. Touch the up arrow once.
   - The motor should begin to lift the probe assembly at its slowest speed.

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9. Touch the up arrow a second time.
   - The motor should increase to its medium speed.
10. Touch the up arrow a third time.
    - The motor should increase to its maximum speed.
11. Touch the up arrow a fourth time.
    - The motor should return to its medium speed.
12. Allow the motor to lift the probe to the parked position.
    - The motor should automatically stop and hold the probe in the parked position.

6.3 CHECKING LEVEL CALIBRATION

1. Ensure the SI-7000 power switch is in the "ON" position.
2. On the touch screen LCD navigate to the Motor menu.
3. Touch the down arrow once.
   - The motor should begin to lower the probe assembly at its slowest speed
4. Observe the chain movement through the probe enclosure window and stop the motor when the designated marking on the chain is aligned with a reference point. Record the step count.
   - Motor should stop and hold the probe at the current location
5. Touch the down arrow three times.
   - The motor should begin to lower the probe assembly
6. Observe the chain movement through the probe enclosure window and stop the motor when the second marking on the chain is aligned with the reference point. Record the second step count.
7. Calculate the level scale. Consult with factory for details.
8. Touch the up arrow three times.
   - The motor should begin to lift the probe assembly.
9. Allow the motor to lift the probe to the parked position.
   - The motor should automatically stop and hold the probe in the parked position.

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6.4 COMMUNICATIONS TEST

1. Ensure the SI-7000 power switch is in the "ON" position.
2. Ensure the green "COM" LED(s) are on.
3. From the control room send a command to lower the probe.
   - The motor should begin to lower the probe assembly .
4. From the control room send a command to stop the motor.
   - The motor should stop and hold the probe at its current position.
5. From the control room send a command to lift the probe.
   - The motor should begin to lift the probe assembly.
6. Allow the motor to lift the probe to the parked position.
   - The motor should automatically stop and hold the probe in the parked position.

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SCIENTIFIC INSTRUMENTS

Systems Warranty Policy

Scientific Instruments, Inc. (SII) warrants its LTD tank gauging system against all defects in material and workmanship for a period of (24) months from the date of shipment of goods from SII dock, at 4400 West Tiffany Drive, West Palm Beach, Florida.

SII shall not be liable for damages caused by raw materials, designs of work methods instructed by the Purchaser or incompatibility of the Purchaser's equipment with SII products. SII shall not be liable for damages caused by negligence of a contracting party or a third party and not for damages caused by the use of SII product for other purposes than it is intended.

Sub-assemblies not of SII manufacture, but as part of an SII system, such as software, printers, terminals, modems, etc., are warranted in accordance with the terms of sales as extended by the product supplier.

SII shall not be liable for indirect losses and pure financial losses, lost profit or other consequential economic losses.

Should a defect become apparent within this warranty period, SII or properly trained and certified personnel will perform field repair at the customer's facility. Transportation costs and living expenses for the required field service personnel shall be borne by the customer, at cost.

In the event of failure of sub-assemblies or components which can be physically returned to SII, all warranty repairs/replacements will be made at no charge, provided that the customer pays all freight costs, foreign duties, etc., incurred in returning such sub-assembly or components to SII. Any U.S.A. duties or fees imposed upon products entering this country will be paid by SII.

This warranty policy is considered null and void if the installation and commissioning of the system is not performed by SII personnel, or personnel formally trained and certified by SII. A Certificate of Training will be issued by SII when independent contractors or consultants have successfully completed such training.

Authorized by: Leigh Ann Hoey

Name/Title: Leigh Ann Hoey/President

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